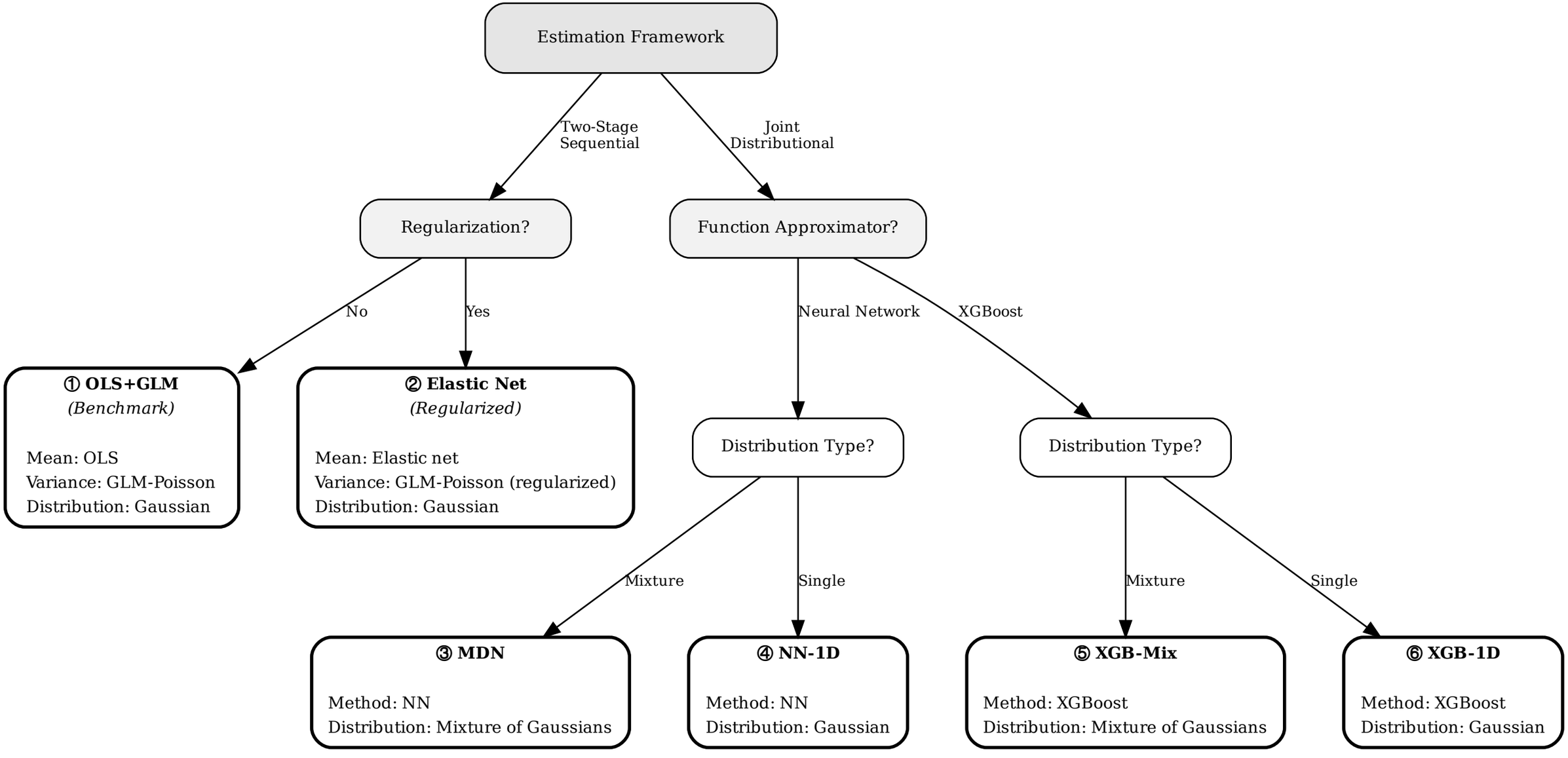

Estimation Framework
Two-Stage
Sequential
Joint
Distributional
Regularization?
Function Approximator?
No
Yes
Neural Network
XGBoost
① OLS+GLM
(Benchmark)
Mean: OLS
Variance: GLM-Poisson
Distribution: Gaussian
② Elastic Net
(Regularized)
Mean: Elastic net
Variance: GLM-Poisson (regularized)
Distribution: Gaussian
Distribution Type?
Distribution Type?
Mixture
Single
Mixture
Single
③ MDN
Method: NN
Distribution: Mixture of Gaussians
④ NN-1D
Method: NN
Distribution: Gaussian
⑤ XGB-Mix
Method: XGBoost
Distribution: Mixture of Gaussians
⑥ XGB-1D
Method: XGBoost
Distribution: Gaussian

OLS + GLM

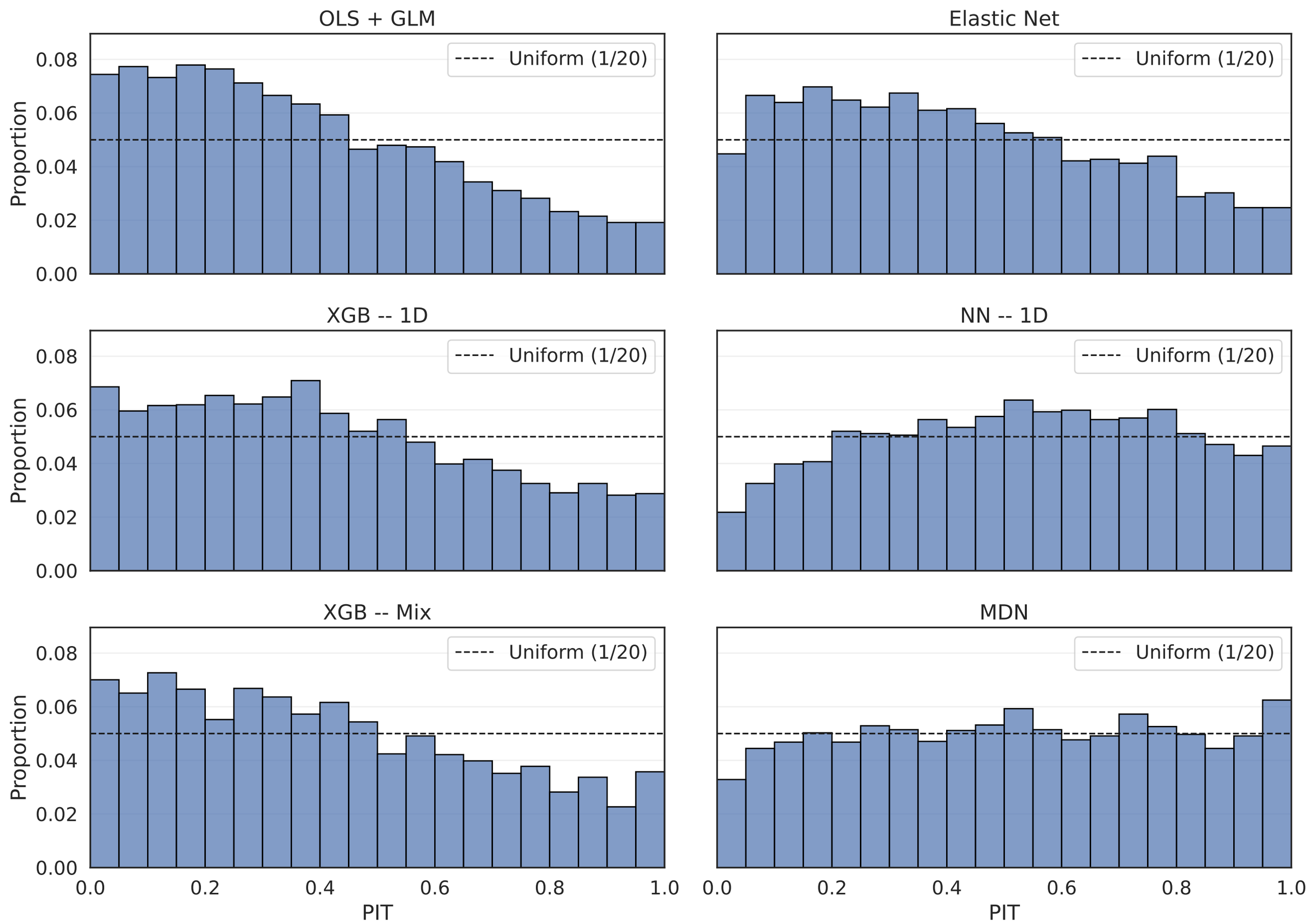
Elastic Net

XGB -- 1D

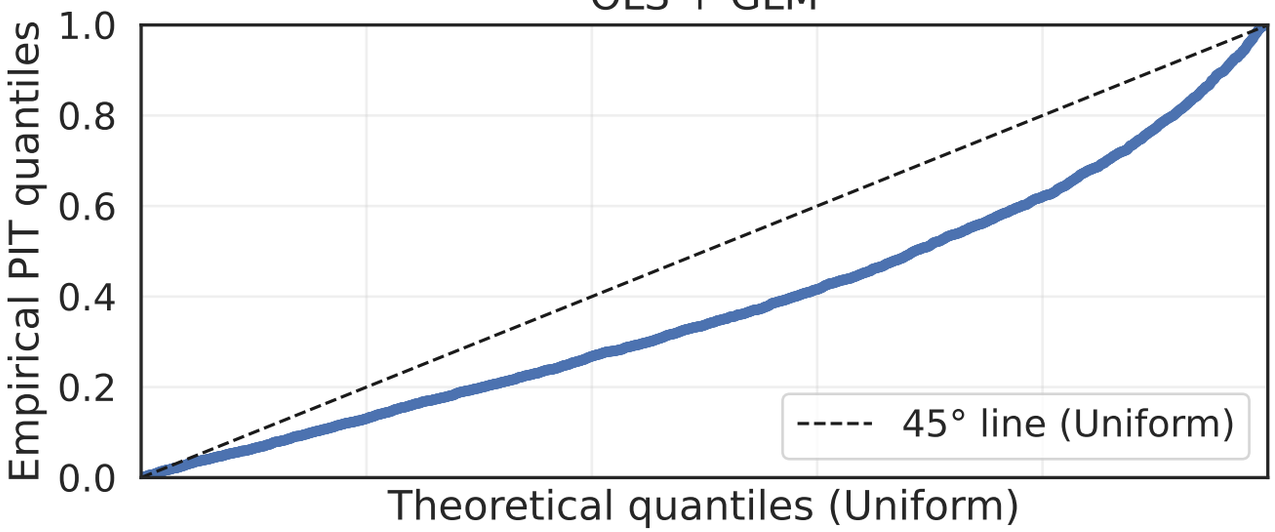
NN -- 1D

XGB -- Mix

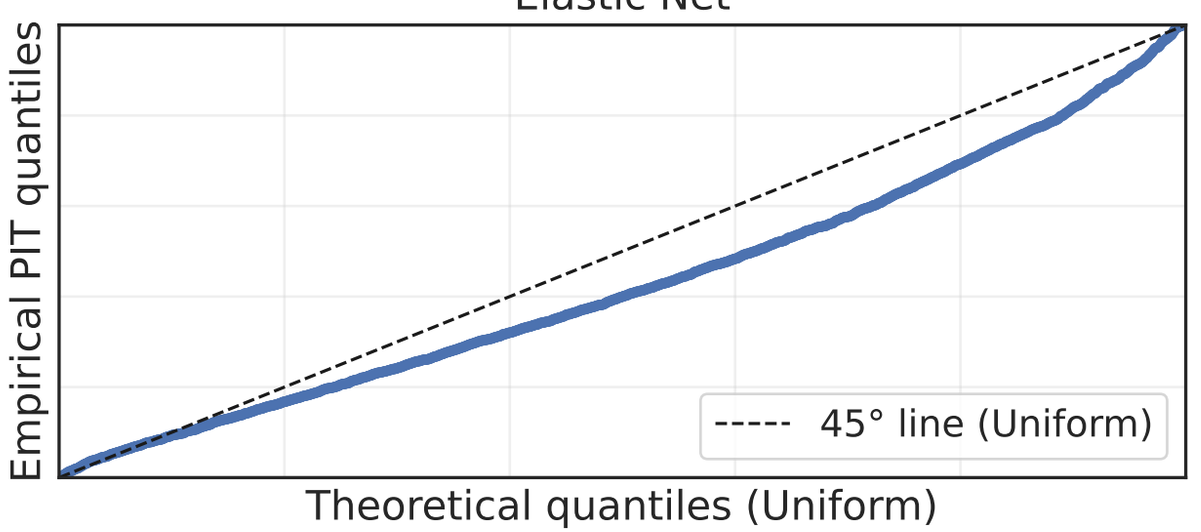
MDN



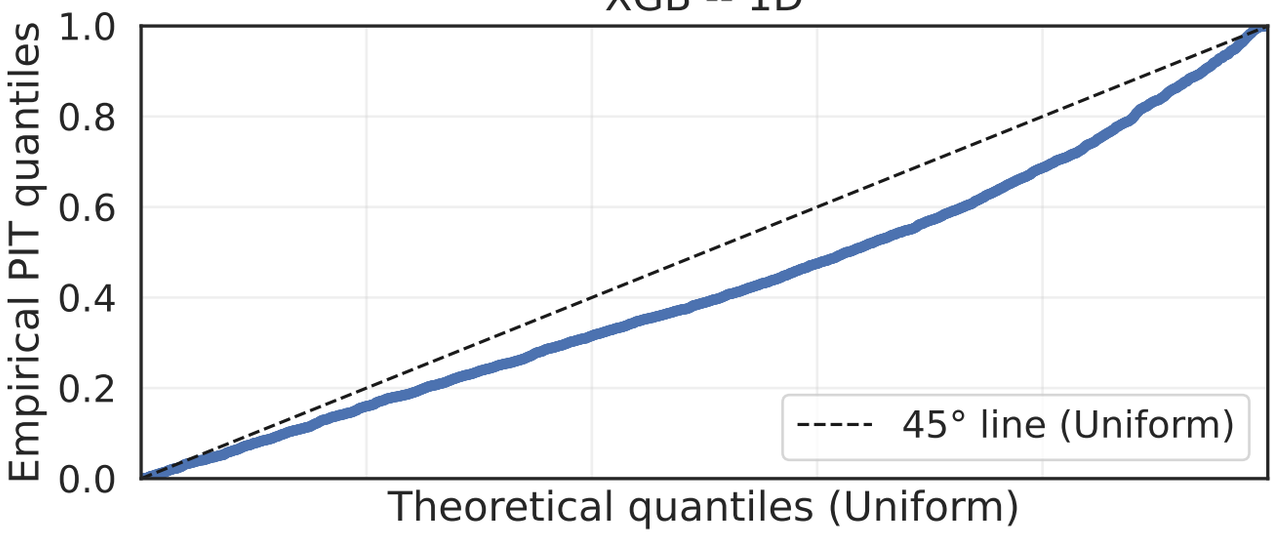
OLS + GLM



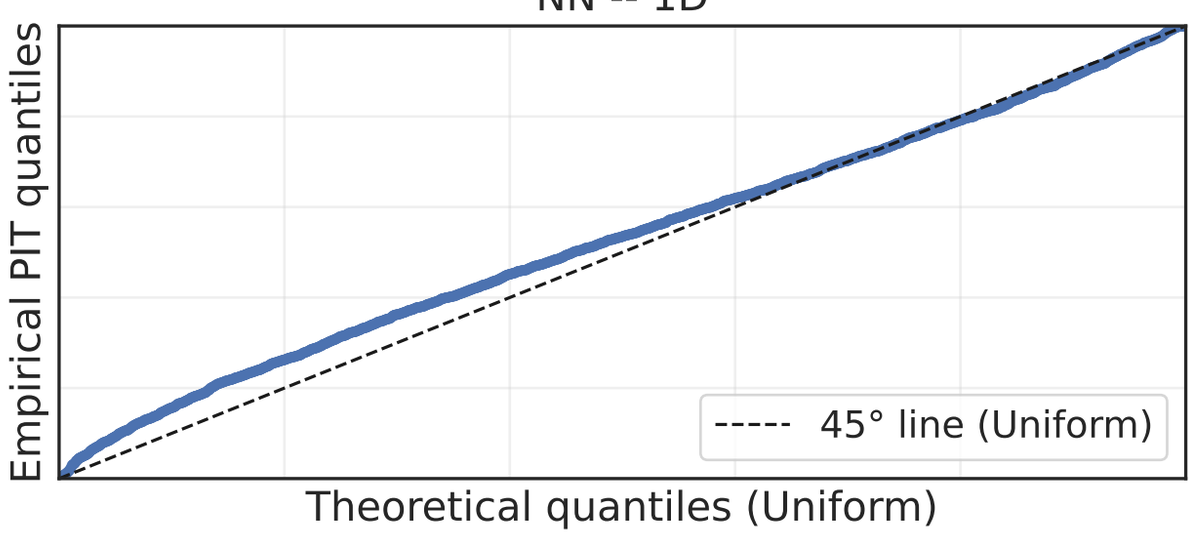
Elastic Net



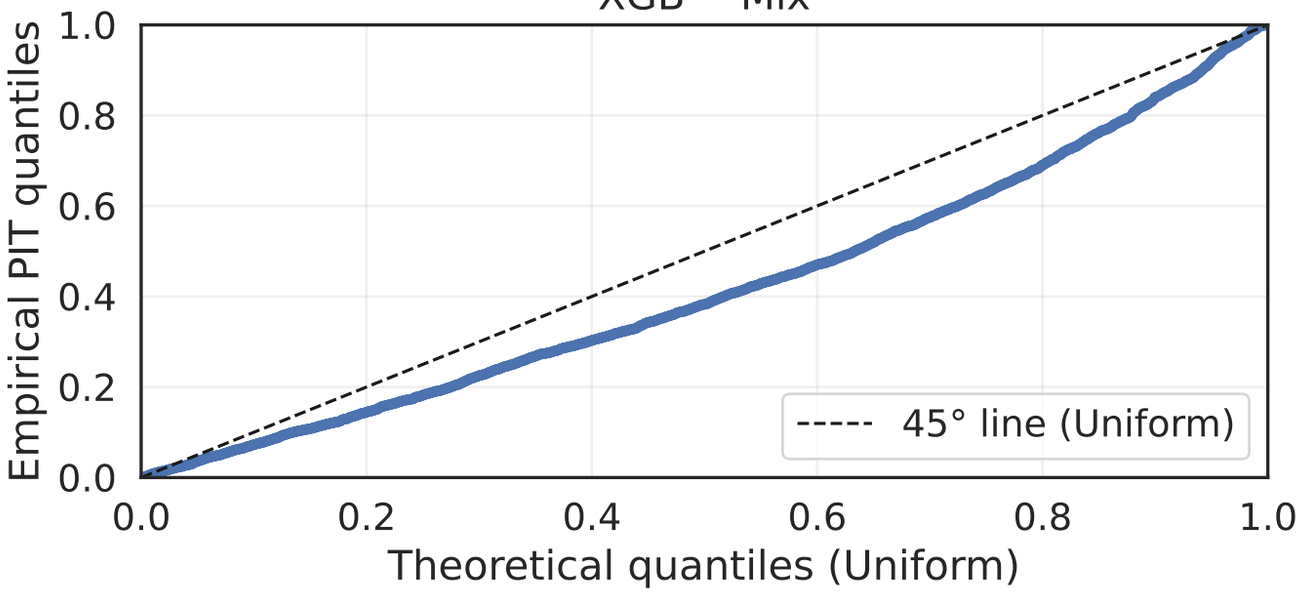
XGB -- 1D



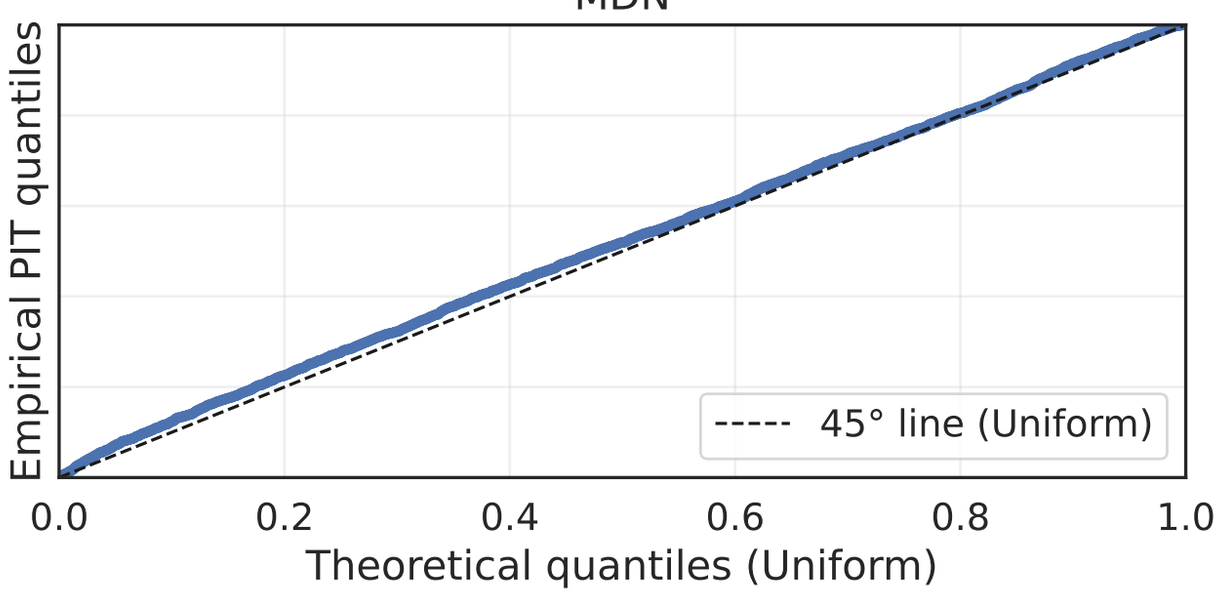
NN -- 1D



XGB -- Mix



MDN



OLS + GLM

Empirical PIT quantiles

1.0
0.8
0.6
0.4
0.2
0.0

----- 45° line (Uniform)

Theoretical quantiles (Uniform)

Elastic Net

Empirical PIT quantiles

----- 45° line (Uniform)

Theoretical quantiles (Uniform)

XGB -- 1D

Empirical PIT quantiles

1.0
0.8
0.6
0.4
0.2
0.0

----- 45° line (Uniform)

Theoretical quantiles (Uniform)

NN -- 1D

Empirical PIT quantiles

----- 45° line (Uniform)

Theoretical quantiles (Uniform)

XGB -- Mix

Empirical PIT quantiles

1.0
0.8
0.6
0.4
0.2
0.0

----- 45° line (Uniform)

0.0
0.2
0.4
0.6
0.8
1.0

Theoretical quantiles (Uniform)

MDN

Empirical PIT quantiles

----- 45° line (Uniform)

0.0
0.2
0.4
0.6
0.8
1.0

Theoretical quantiles (Uniform)

OLS + GLM

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

Elastic Net

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

XGB -- 1D

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

NN -- 1D

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

XGB -- Mix

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

MDN

Empirical PIT quantiles

Theoretical quantiles (Uniform)

----- 45° line (Uniform)

OLS + GLM

Elastic Net

XGB -- 1D

NN -- 1D

XGB -- Mix

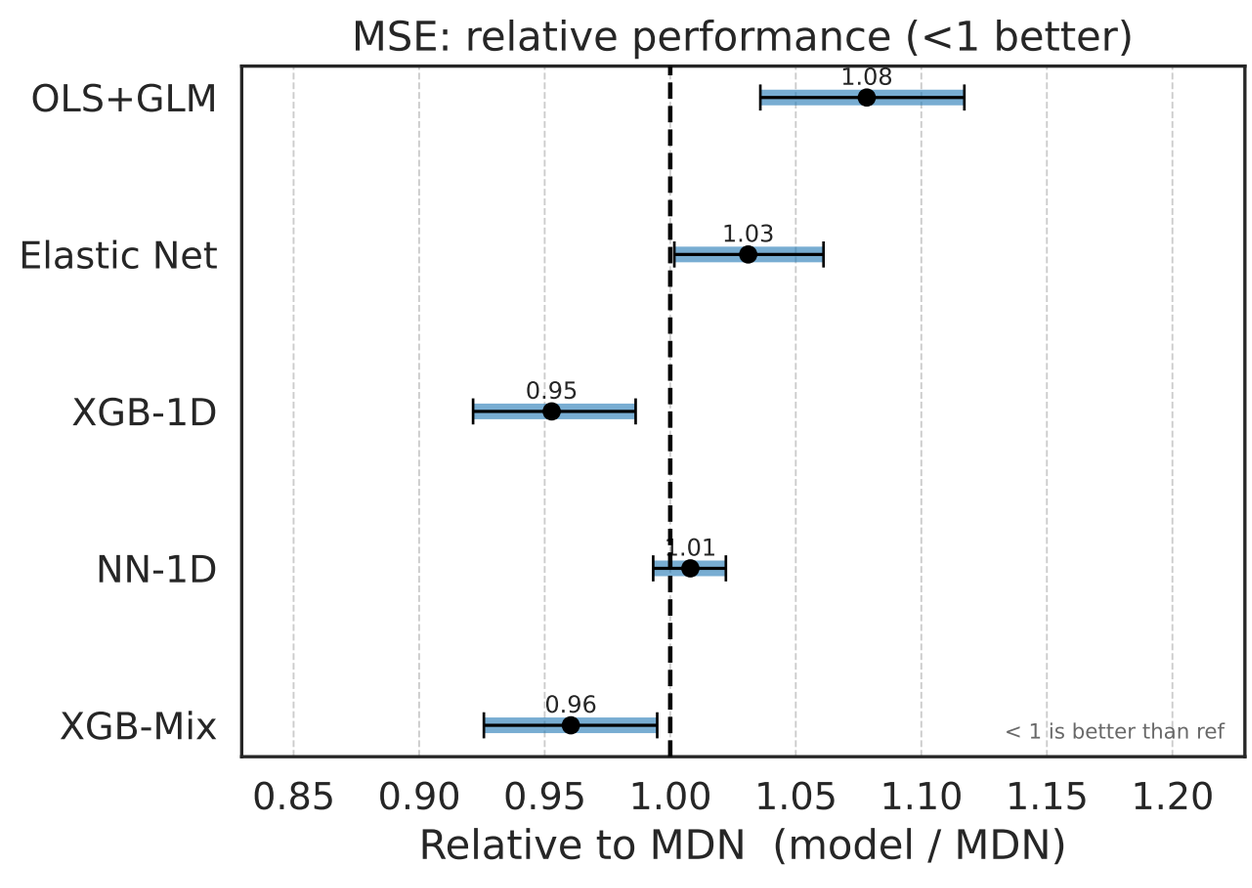
MDN

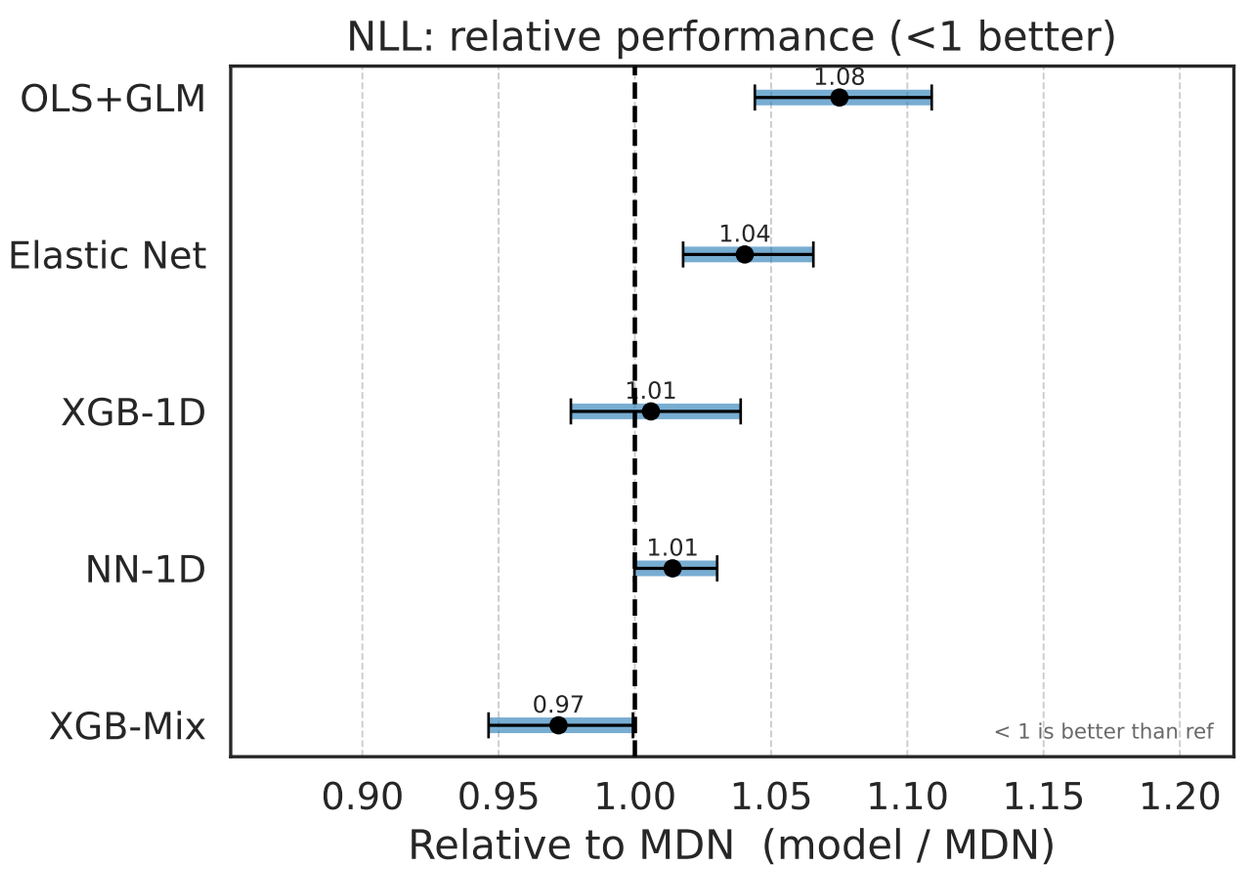
Quantiles

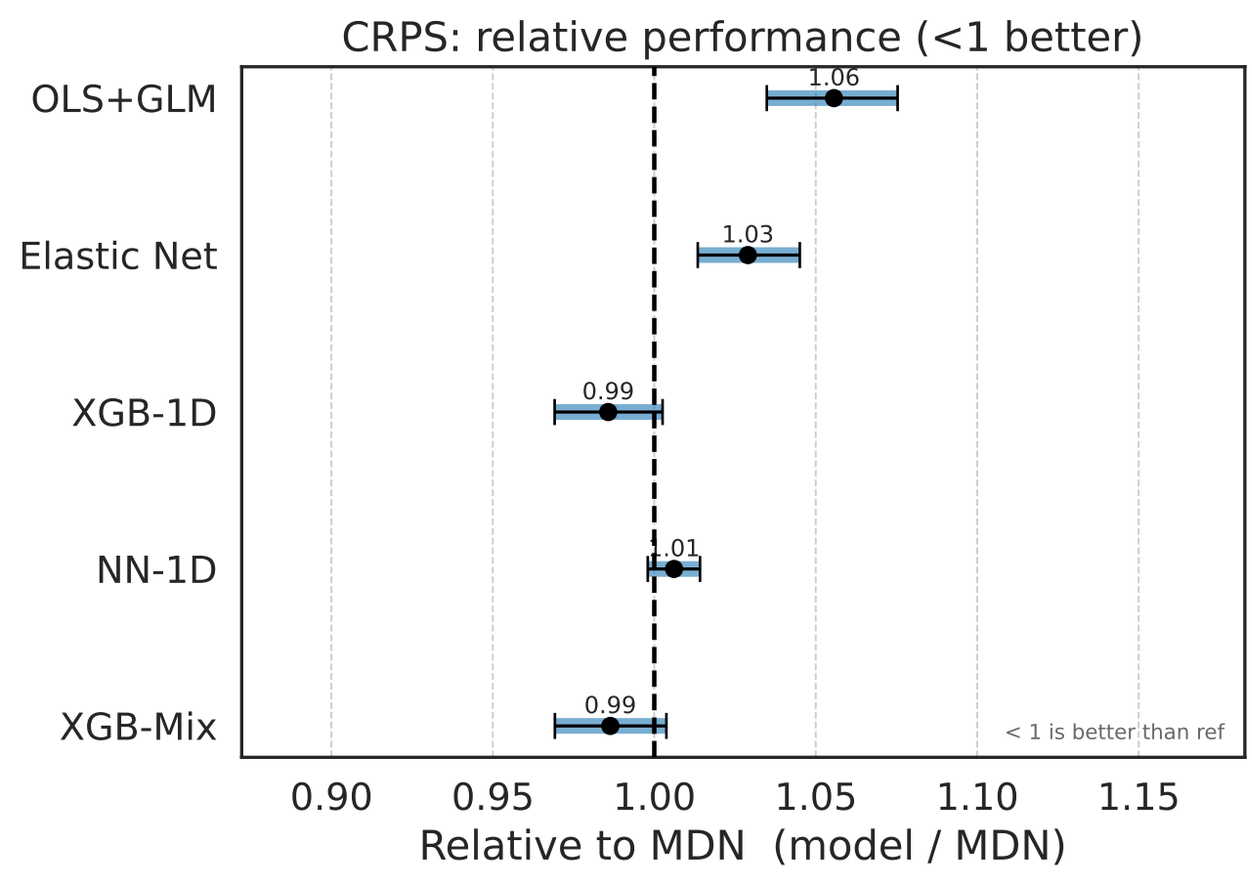
1.0
0.8
0.6
0.4
0.2
0.0

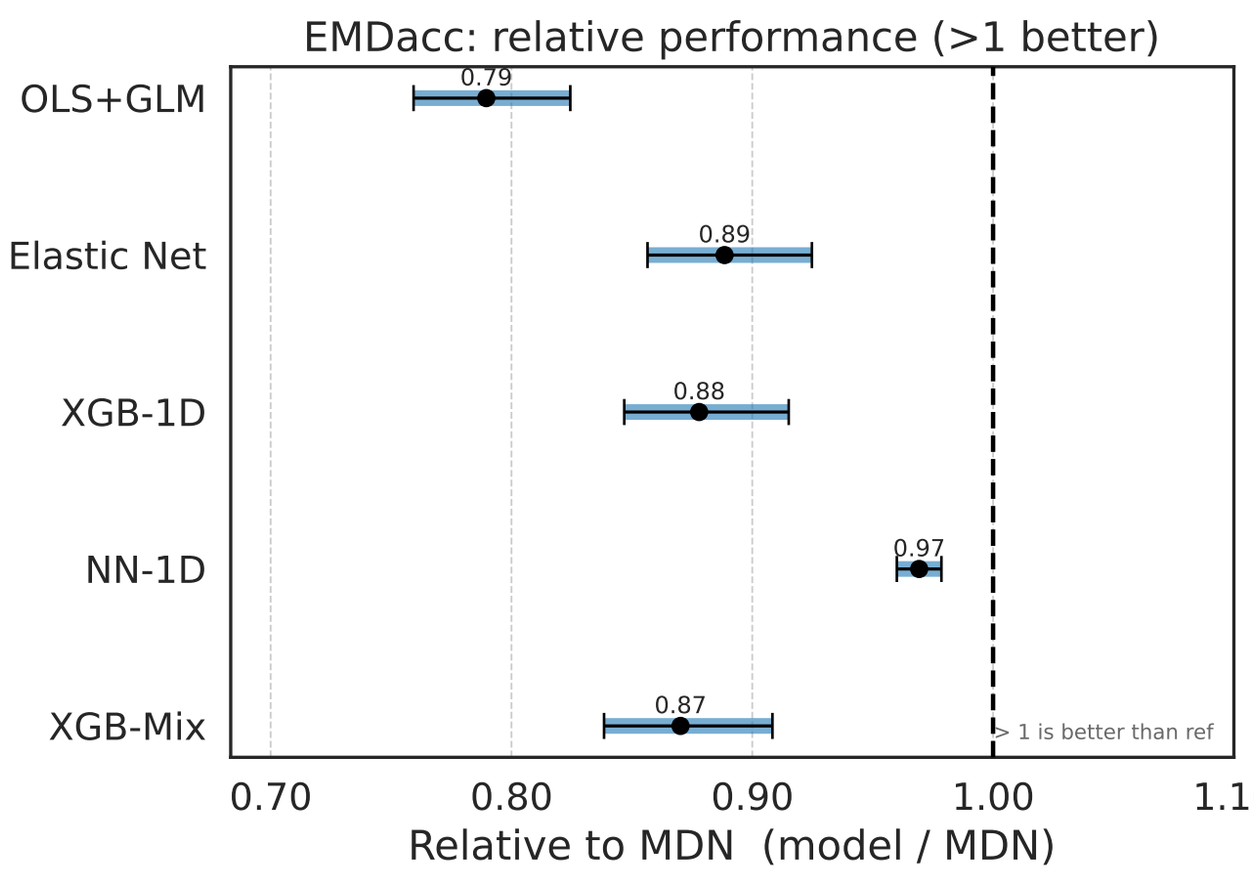
rural_ag
rural_non_ag
urban_non_ag
urban_ag

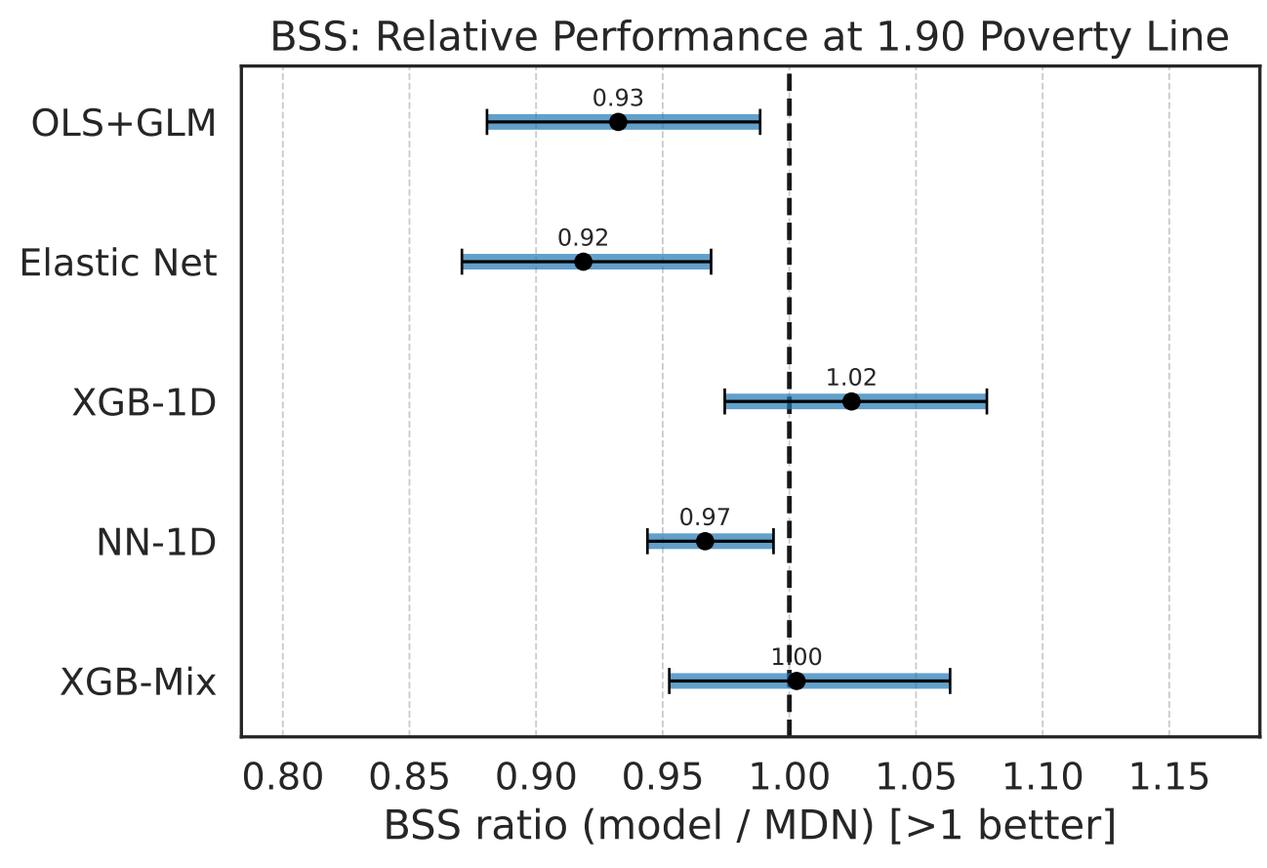
ag_rural_hh

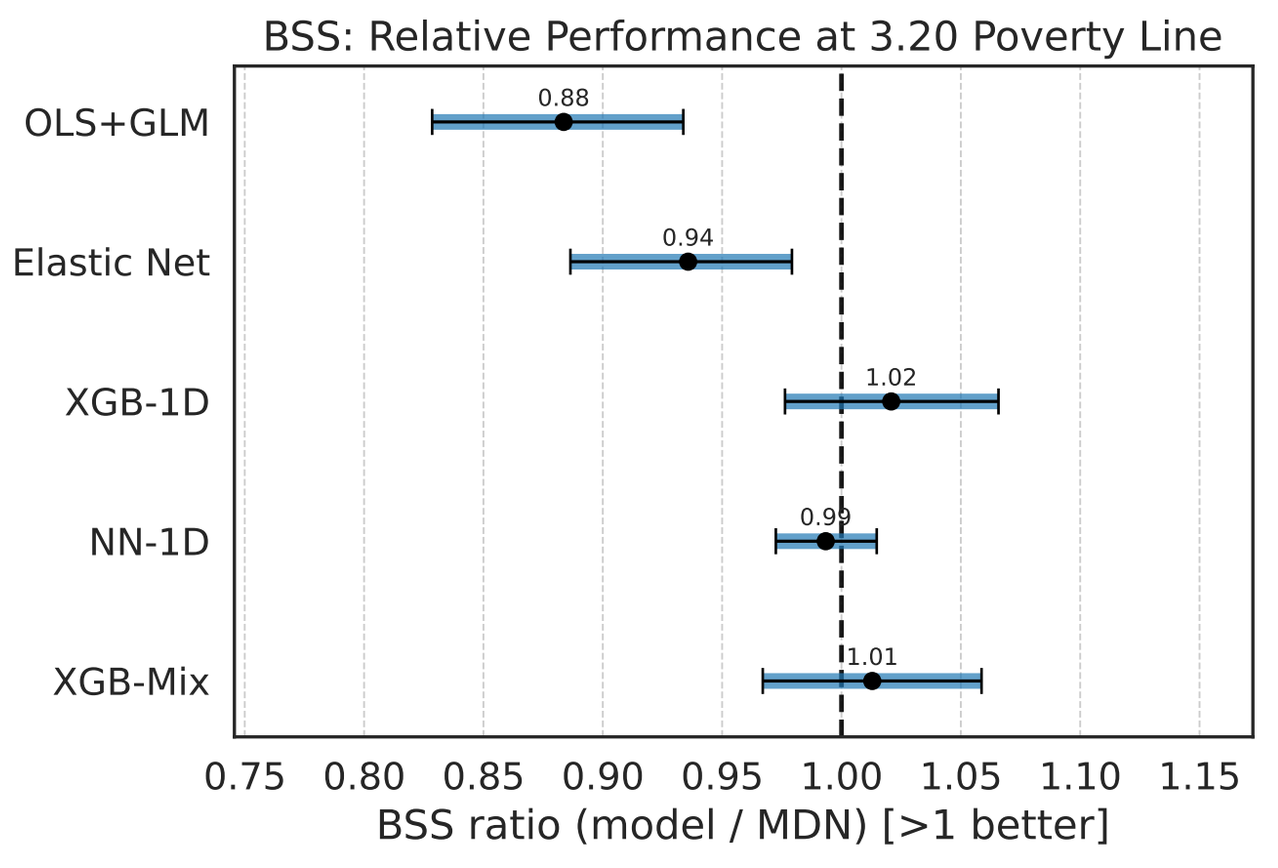

MSE: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.08
1.03
0.95
1.01
0.96
< 1 is better than ref
0.85 0.90 0.95 1.00 1.05 1.10 1.15 1.20
Relative to MDN (model / MDN)

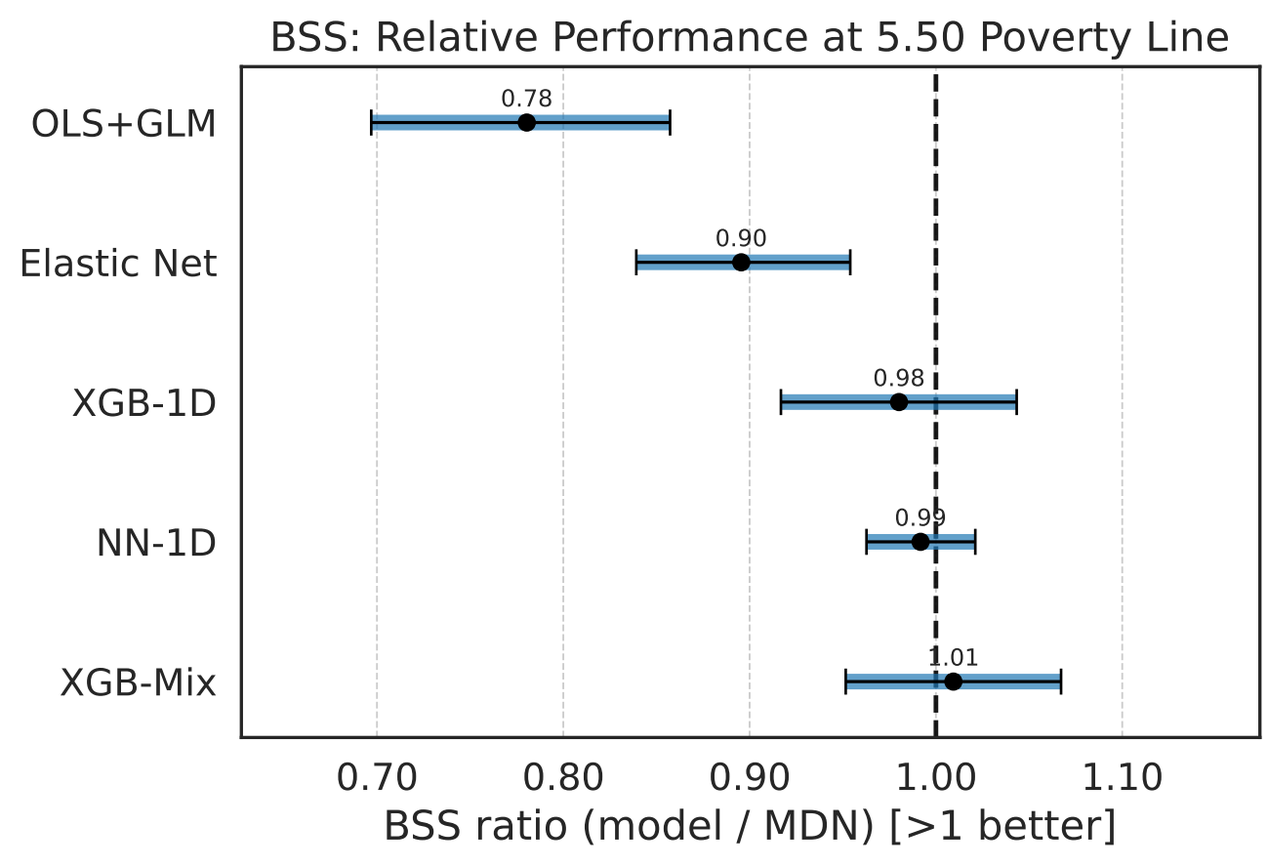

NLL: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.08
1.04
1.01
1.01
0.97
< 1 is better than ref
0.90 0.95 1.00 1.05 1.10 1.15 1.20
Relative to MDN (model / MDN)

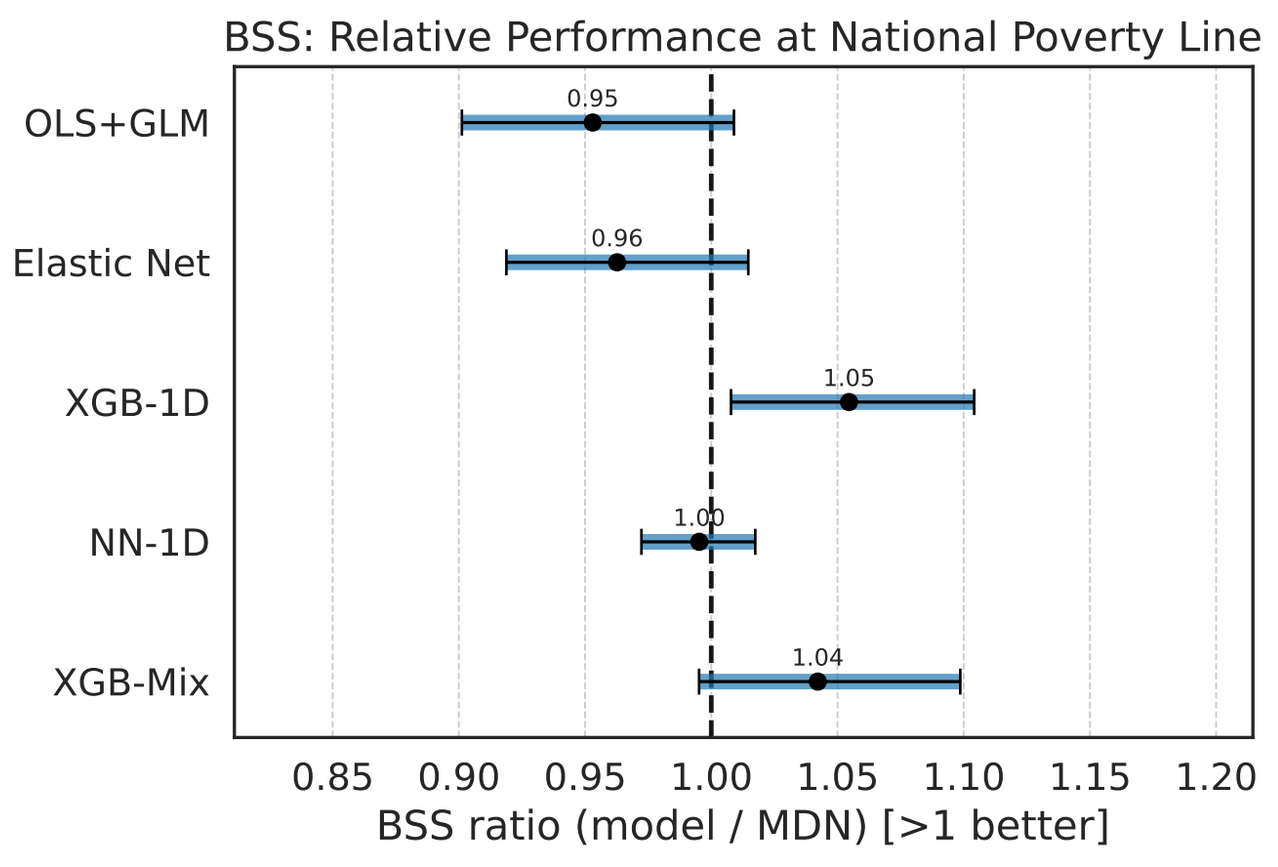

CRPS: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.06
1.03
0.99
1.01
0.99
< 1 is better than ref
0.90 0.95 1.00 1.05 1.10 1.15
Relative to MDN (model / MDN)

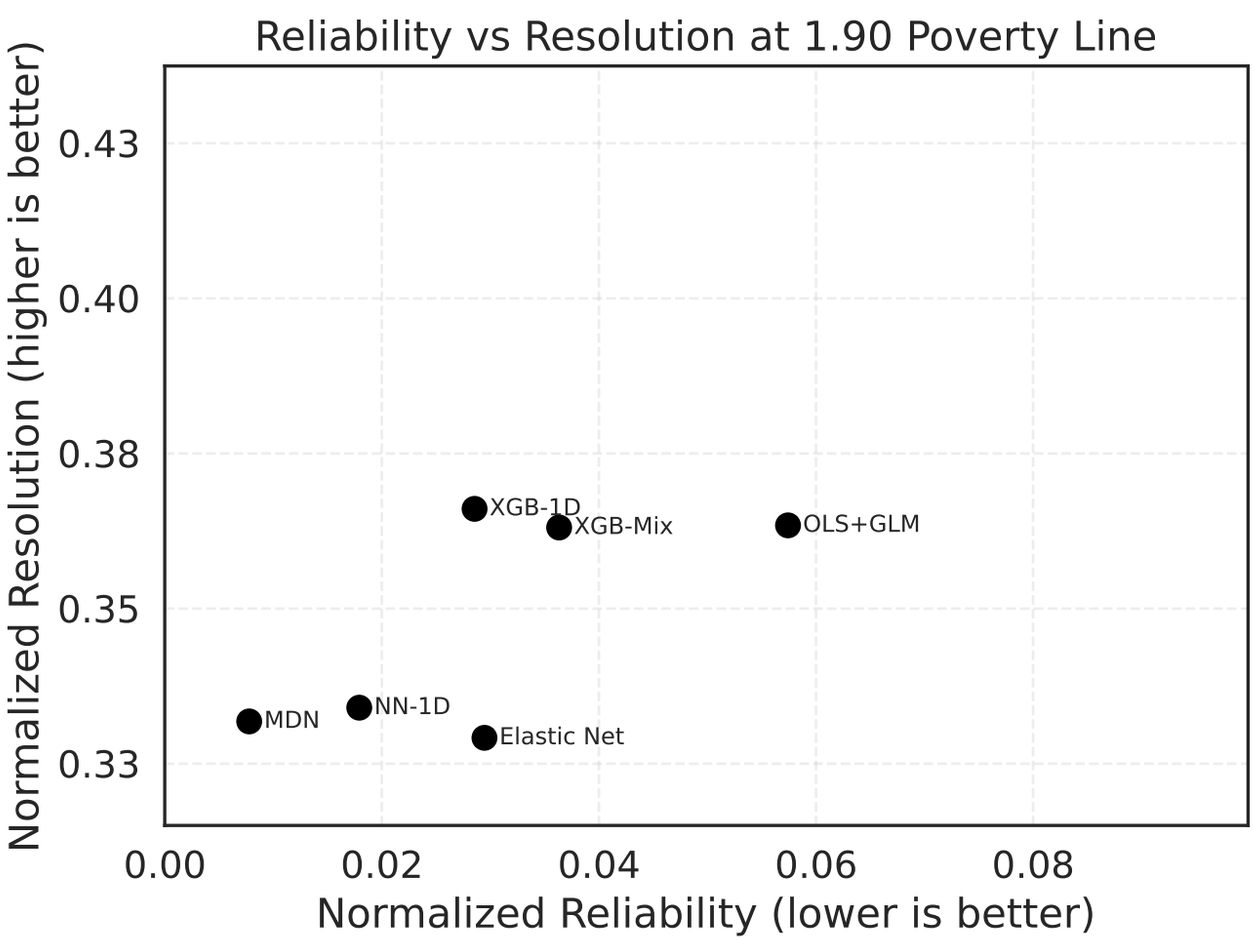

EMDacc: relative performance (>1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.79
0.89
0.88
0.97
0.87
> 1 is better than ref
0.70 0.80 0.90 1.00 1.10
Relative to MDN (model / MDN)

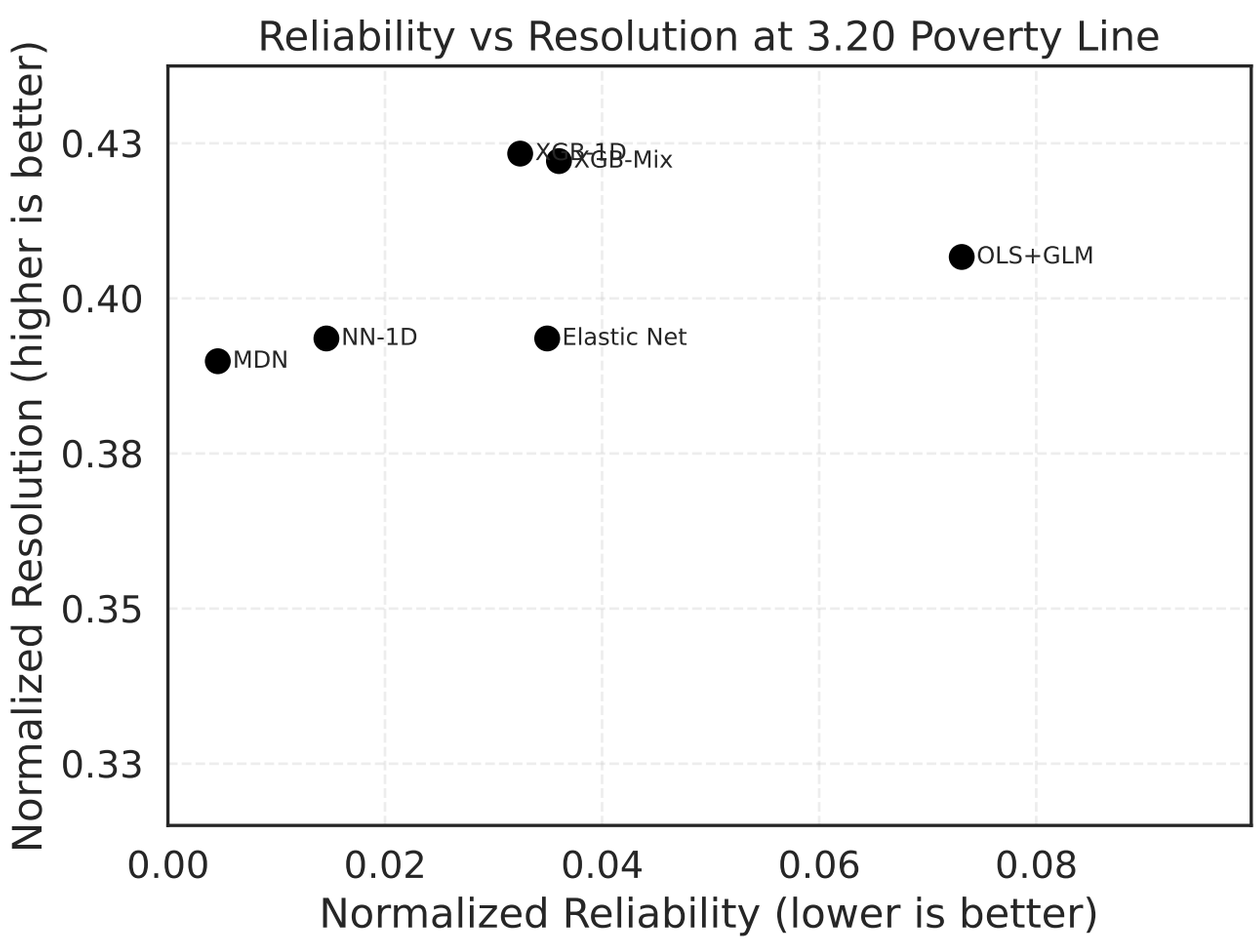

BSS: Relative Performance at 1.90 Poverty Line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.93
0.92
1.02
0.97
1.00
0.80 0.85 0.90 0.95 1.00 1.05 1.10 1.15
BSS ratio (model / MDN) [>1 better]

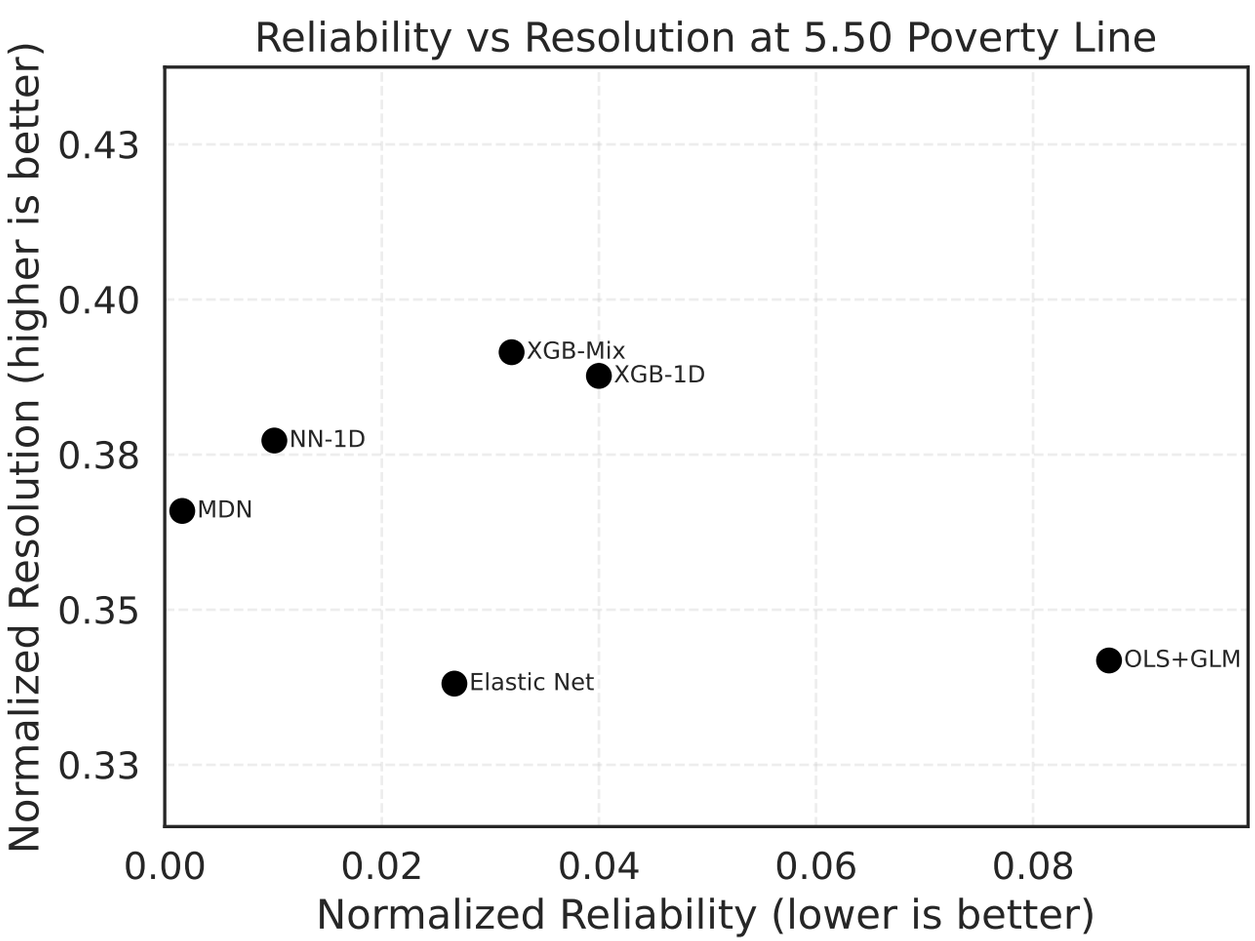

BSS: Relative Performance at 3.20 Poverty Line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.88
0.94
1.02
0.99
1.01
0.75 0.80 0.85 0.90 0.95 1.00 1.05 1.10 1.15
BSS ratio (model / MDN) [>1 better]

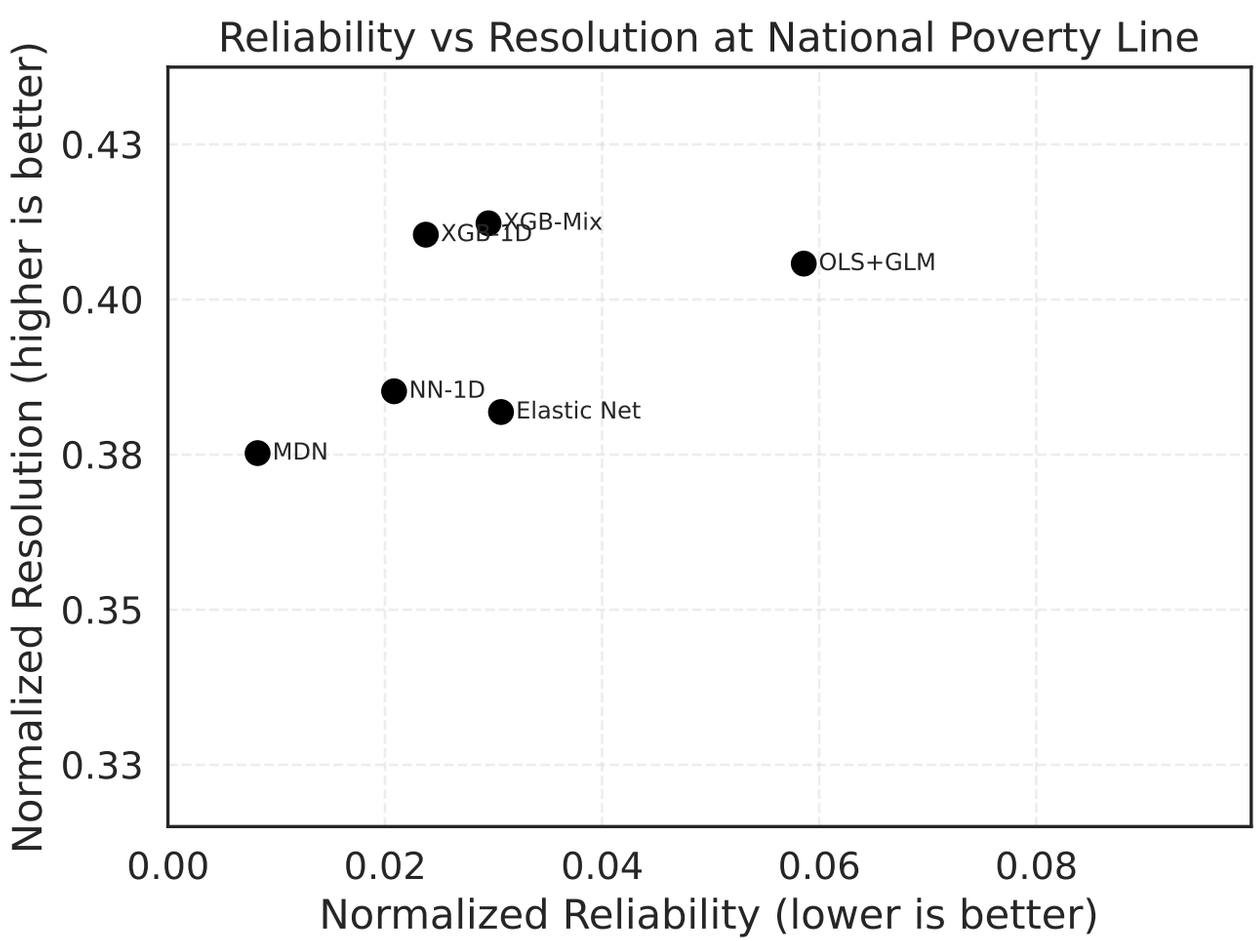

BSS: Relative Performance at 5.50 Poverty Line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.78
0.90
0.98
0.99
1.01
0.70 0.80 0.90 1.00 1.10
BSS ratio (model / MDN) [>1 better]

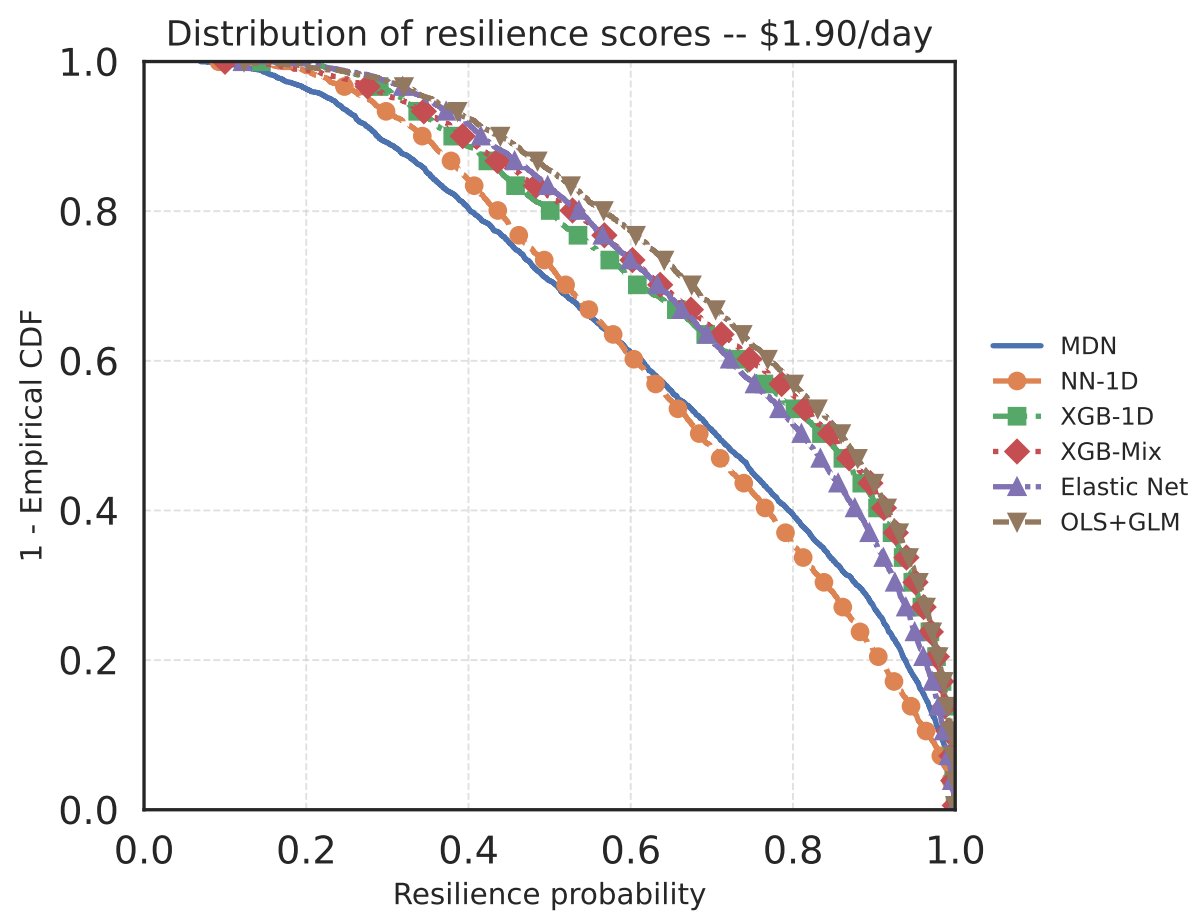

BSS: Relative Performance at National Poverty Line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.95
0.96
1.05
1.00
1.04
0.85 0.90 0.95 1.00 1.05 1.10 1.15 1.20
BSS ratio (model / MDN) [>1 better]

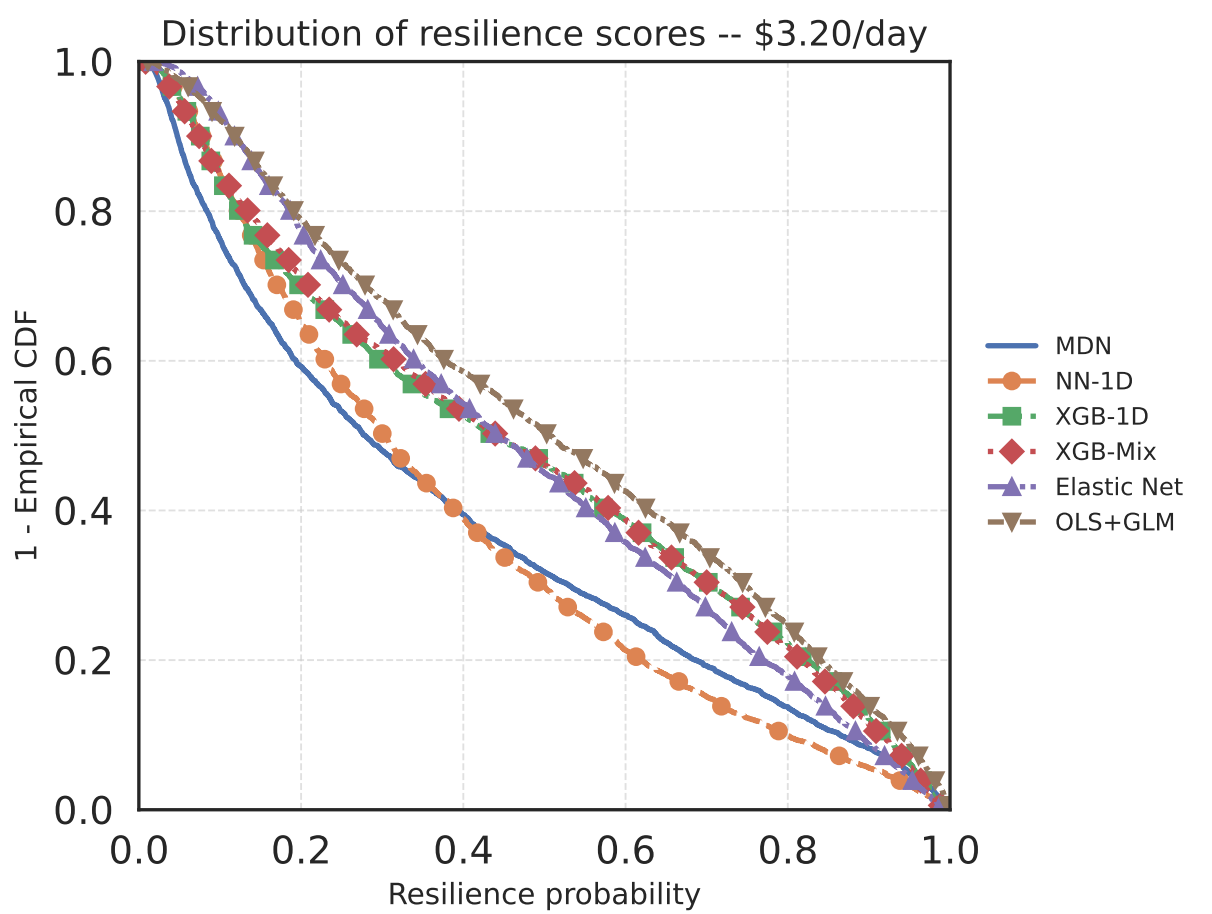

Reliability vs Resolution at 1.90 Poverty Line
Normalized Resolution (higher is better)
Normalized Reliability (lower is better)
0.43
0.40
0.38
0.35
0.33
0.00
0.02
0.04
0.06
0.08
XGB-1D
XGB-Mix
OLS+GLM
MDN
NN-1D
Elastic Net

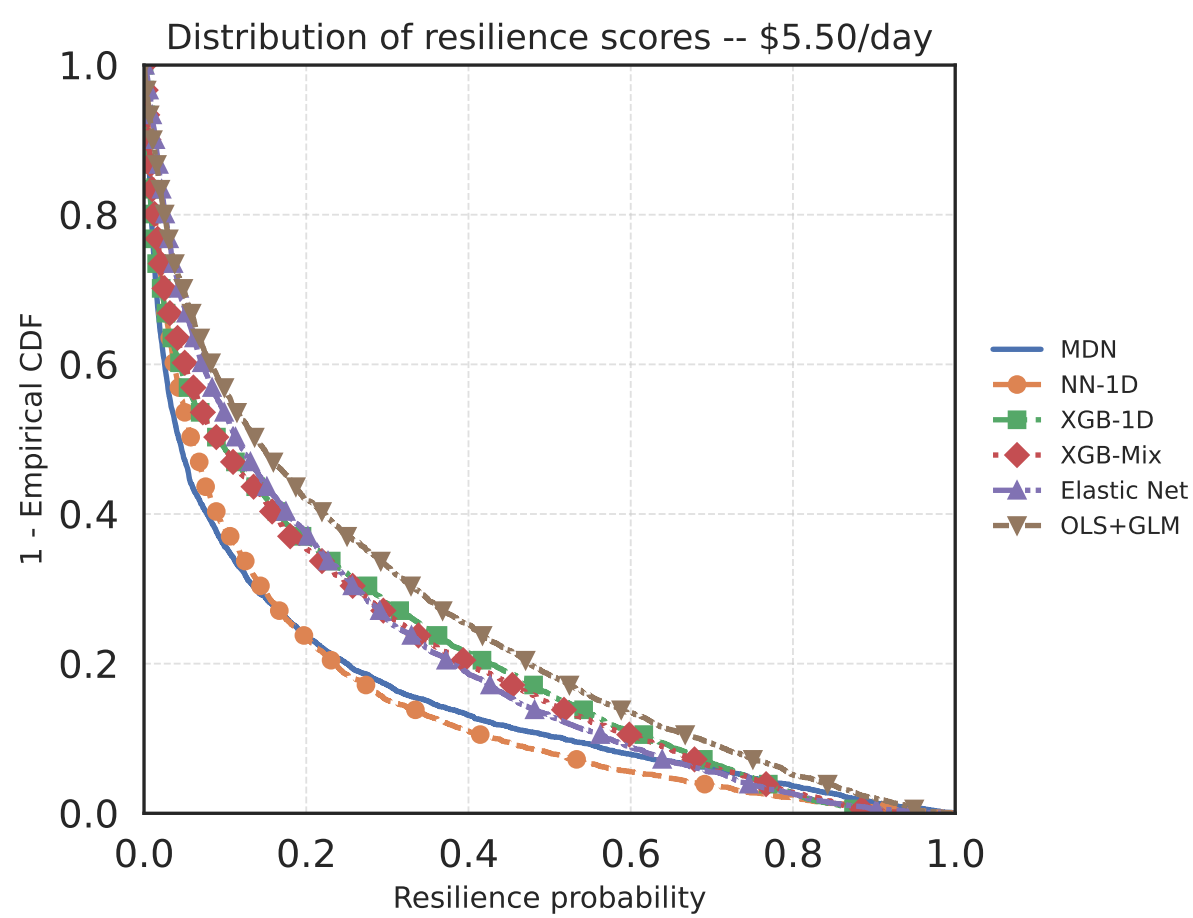

Reliability vs Resolution at 3.20 Poverty Line
Normalized Resolution (higher is better)
Normalized Reliability (lower is better)
0.43
0.40
0.38
0.35
0.33
0.00
0.02
0.04
0.06
0.08
XGB-1D
XGB-Mix
OLS+GLM
MDN
NN-1D
Elastic Net

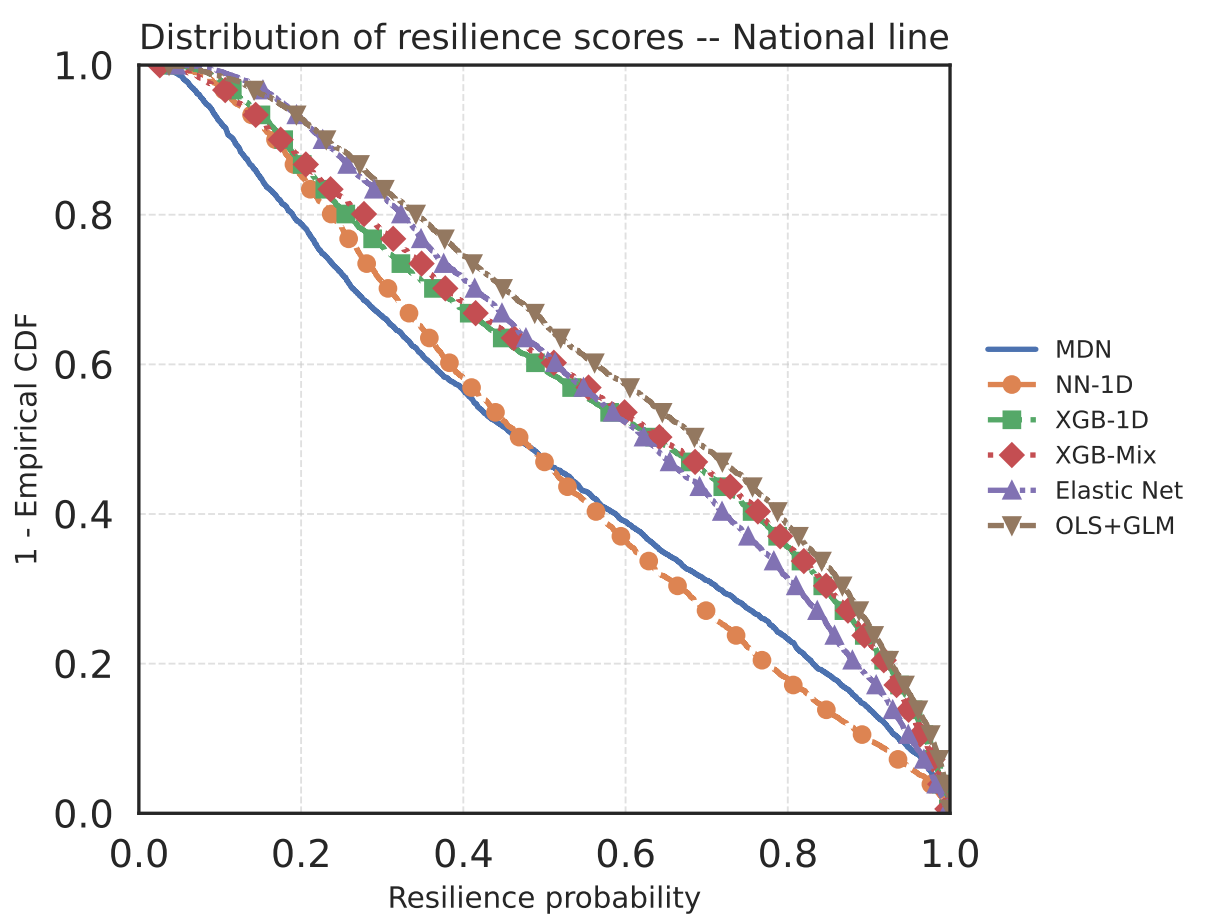

Reliability vs Resolution at 5.50 Poverty Line
Normalized Resolution (higher is better)
Normalized Reliability (lower is better)
0.43
0.40
0.38
0.35
0.33
0.00
0.02
0.04
0.06
0.08
XGB-Mix
XGB-1D
NN-1D
MDN
Elastic Net
OLS+GLM

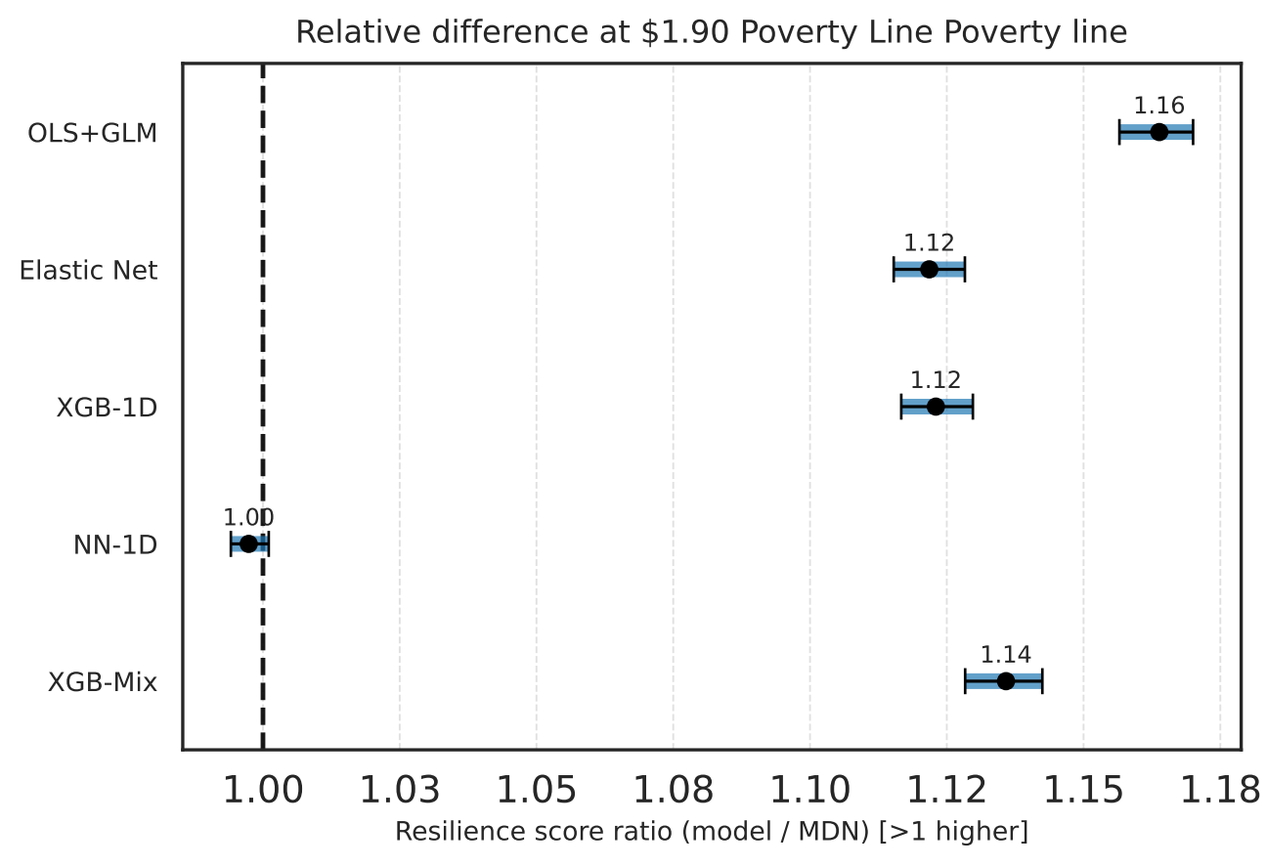

Reliability vs Resolution at National Poverty Line
Normalized Resolution (higher is better)
Normalized Reliability (lower is better)
0.43
0.40
0.38
0.35
0.33
0.00
0.02
0.04
0.06
0.08
XGB-Mix
XGB-1D
OLS+GLM
NN-1D
Elastic Net
MDN

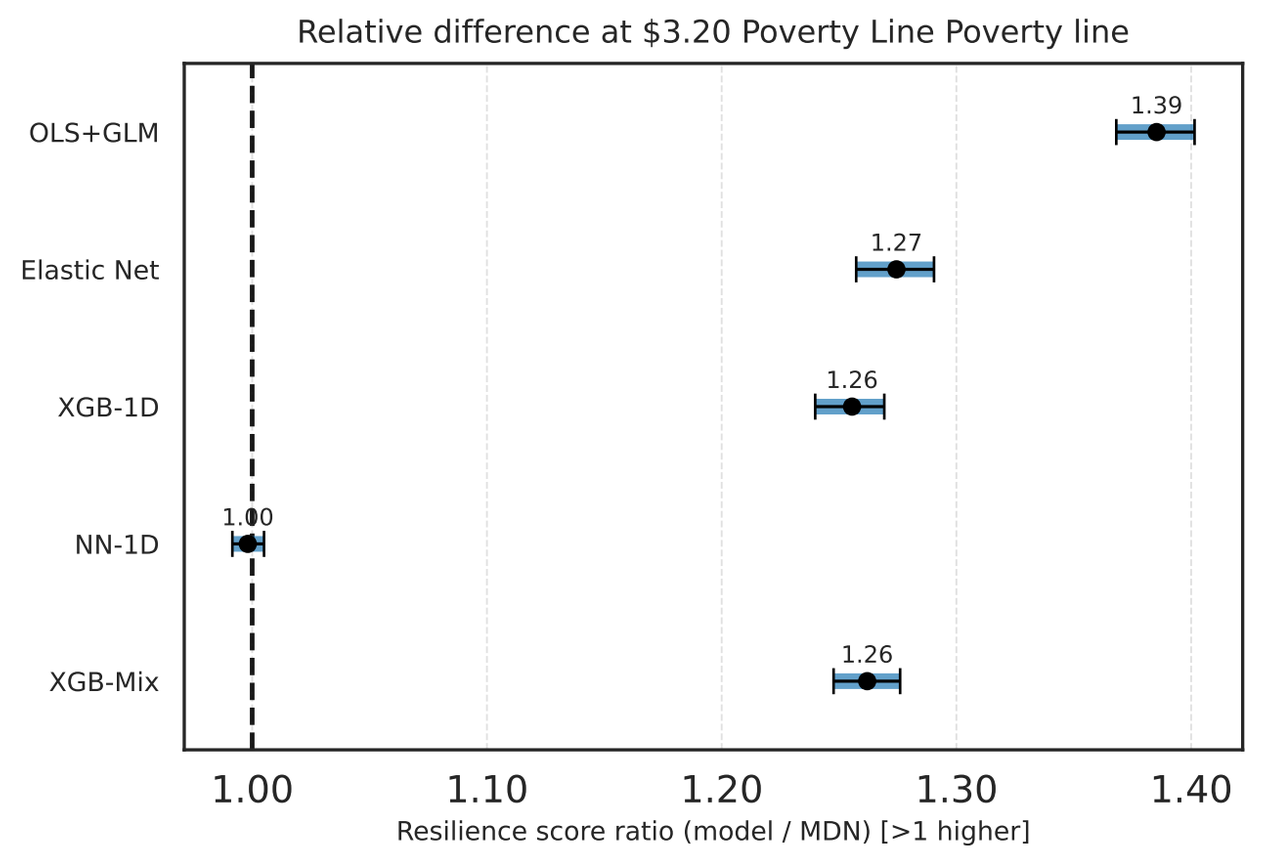

Distribution of resilience scores -- $1.90/day
1 - Empirical CDF
Resilience probability
0.0
0.2
0.4
0.6
0.8
1.0
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM

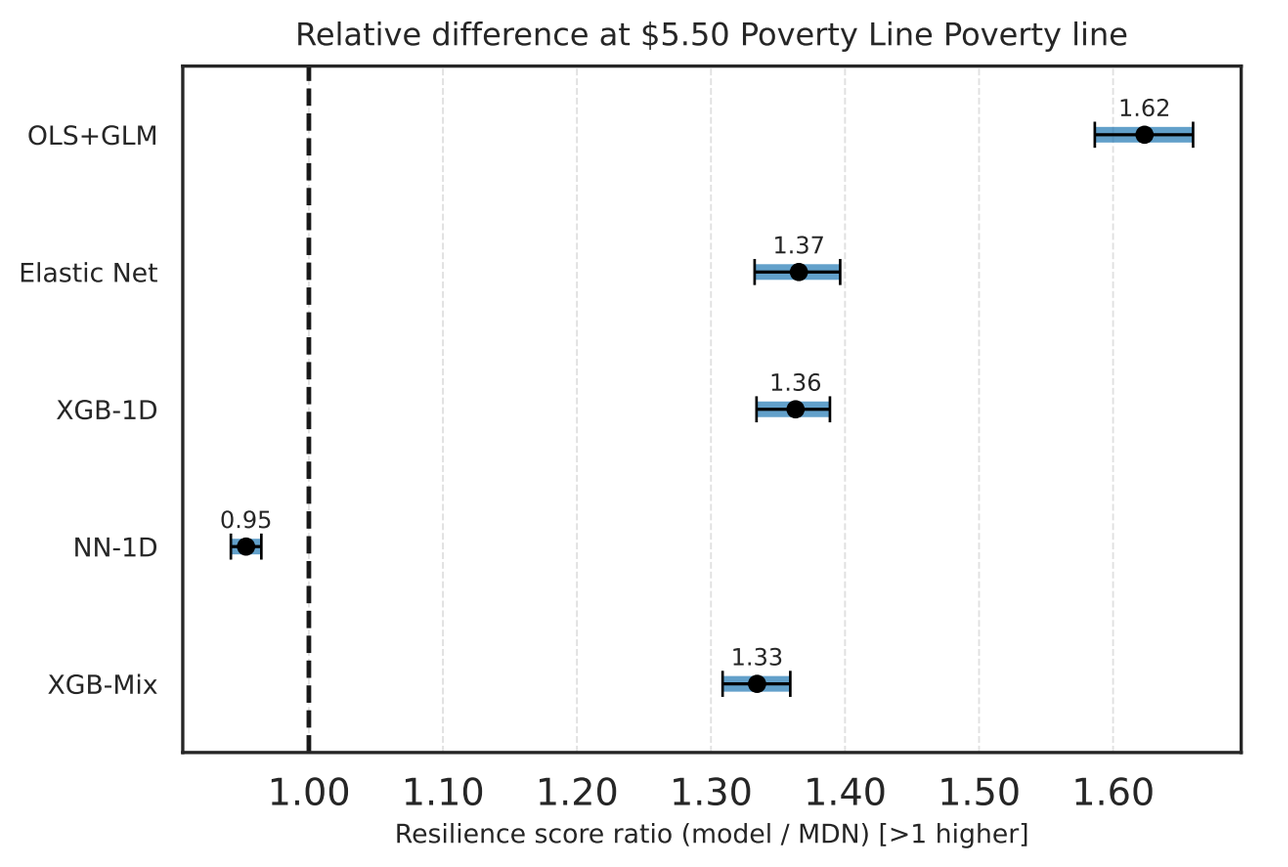

Distribution of resilience scores -- $3.20/day
1 - Empirical CDF
Resilience probability
0.0
0.2
0.4
0.6
0.8
1.0
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM

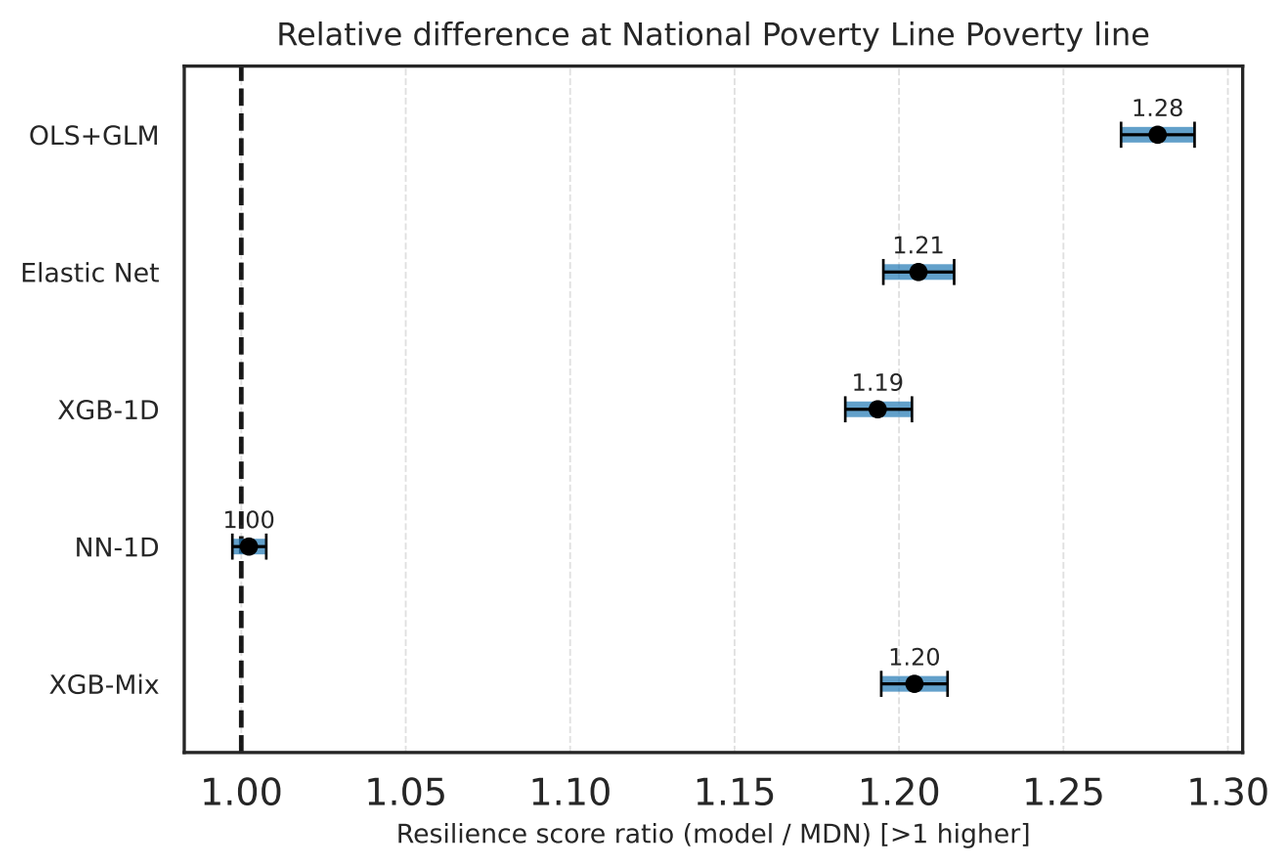

Distribution of resilience scores -- $5.50/day
1 - Empirical CDF
Resilience probability
0.0
0.2
0.4
0.6
0.8
1.0
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM

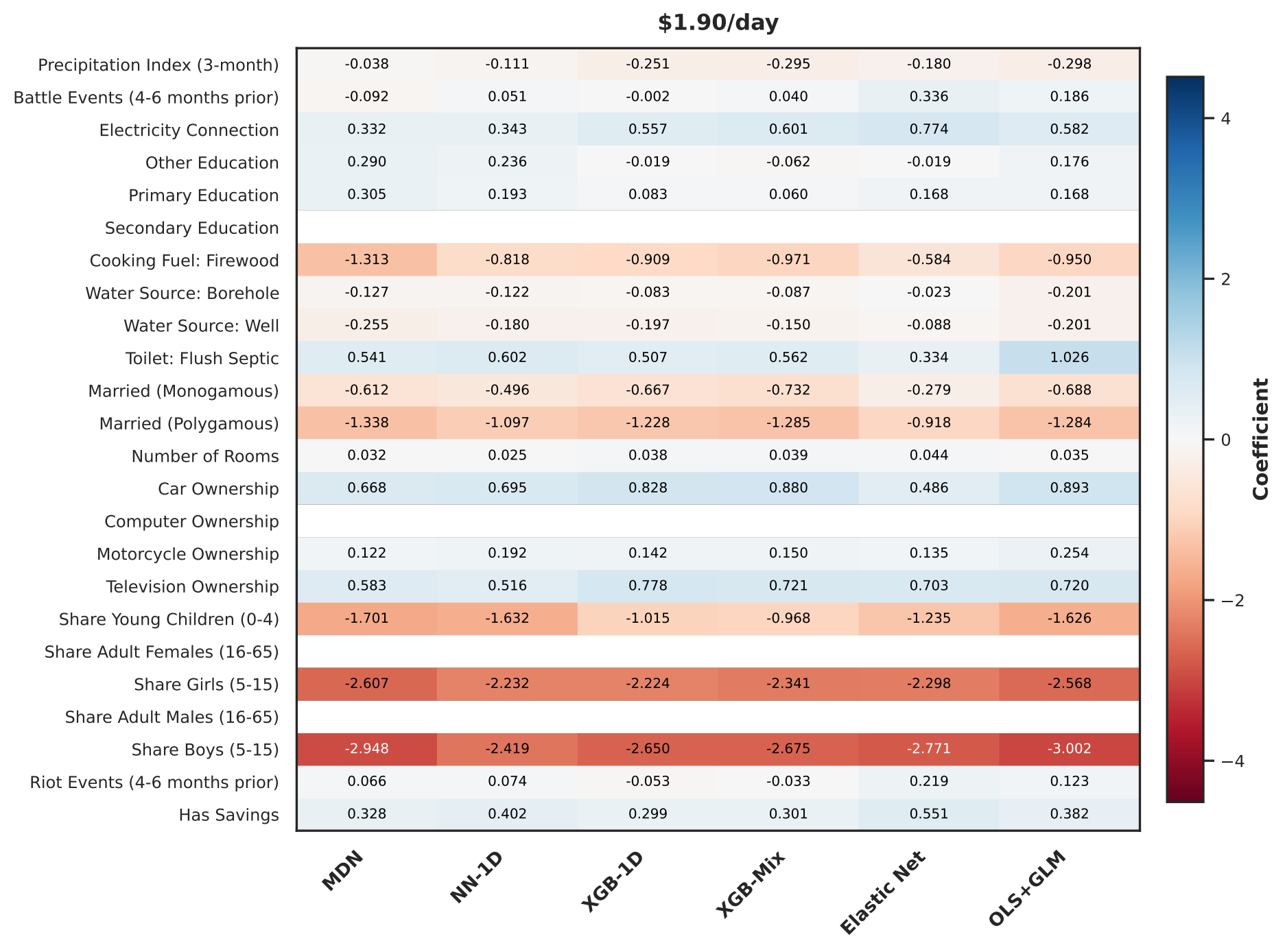

Distribution of resilience scores -- National line
1 - Empirical CDF
Resilience probability
0.0
0.2
0.4
0.6
0.8
1.0
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM

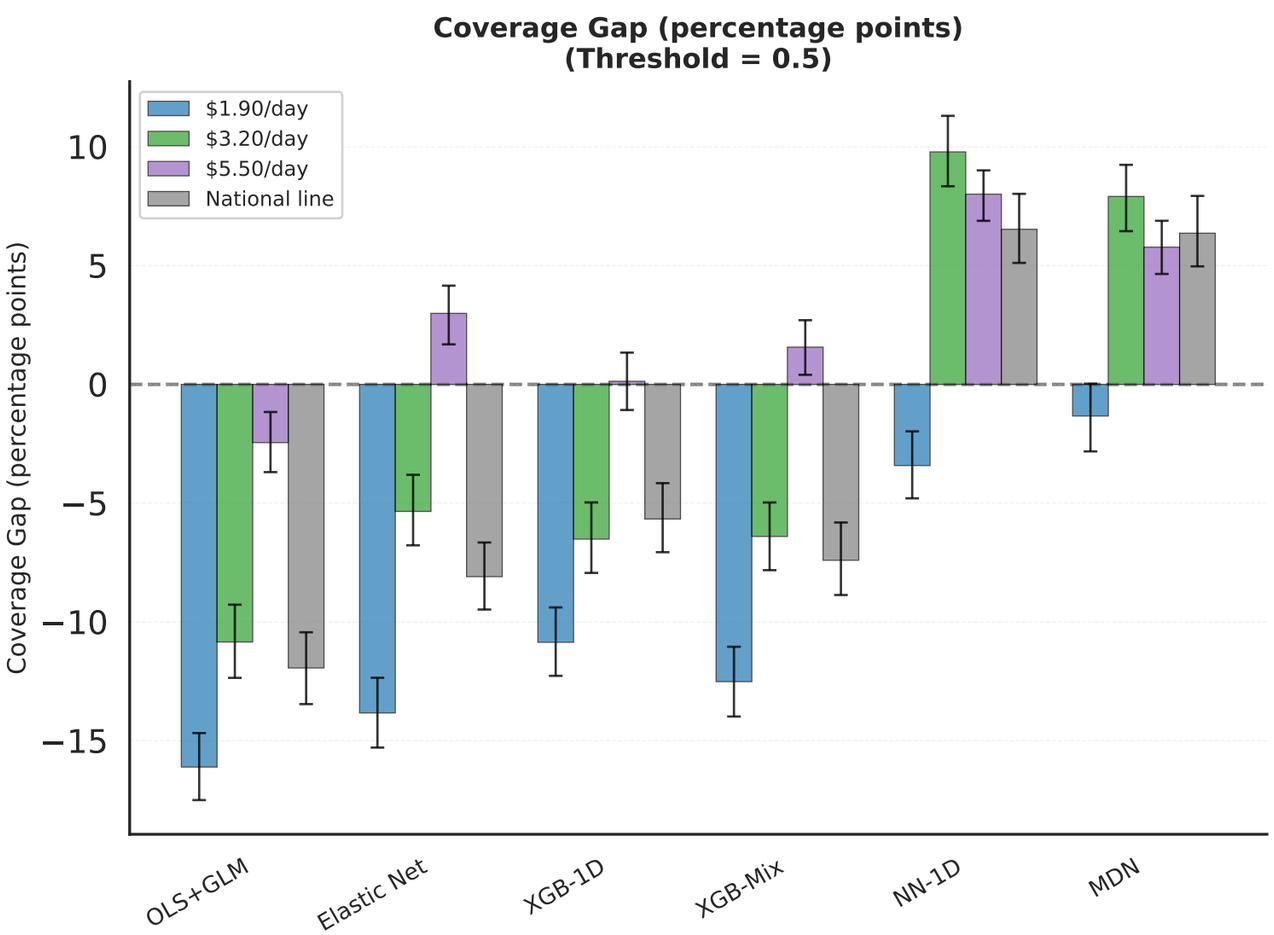

Relative difference at $1.90 Poverty Line Poverty line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.16
1.12
1.12
1.00
1.14
1.00 1.03 1.05 1.08 1.10 1.12 1.15 1.18
Resilience score ratio (model / MDN) [>1 higher]

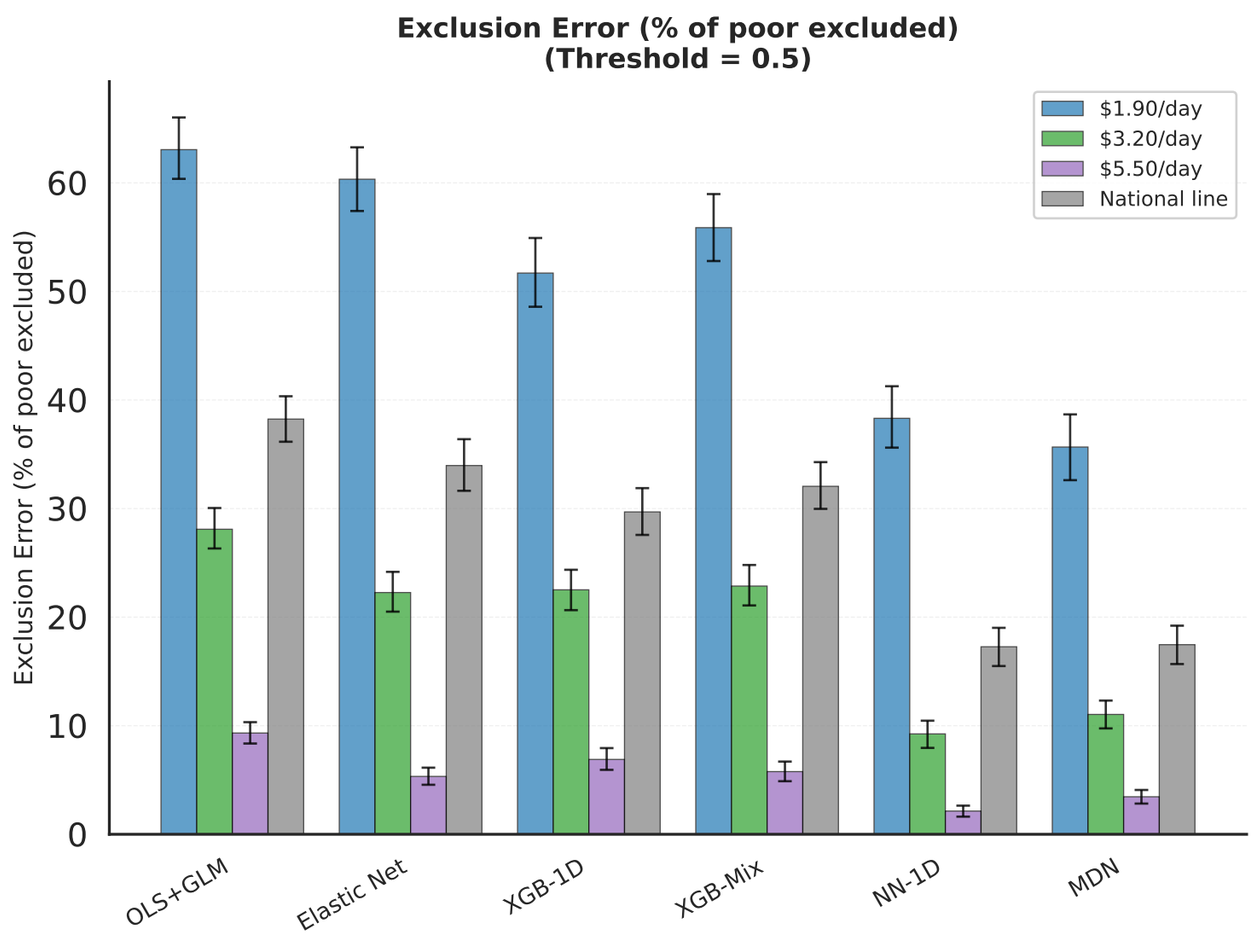

Relative difference at $3.20 Poverty Line Poverty line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.39
1.27
1.26
1.00
1.26
1.00 1.10 1.20 1.30 1.40
Resilience score ratio (model / MDN) [>1 higher]

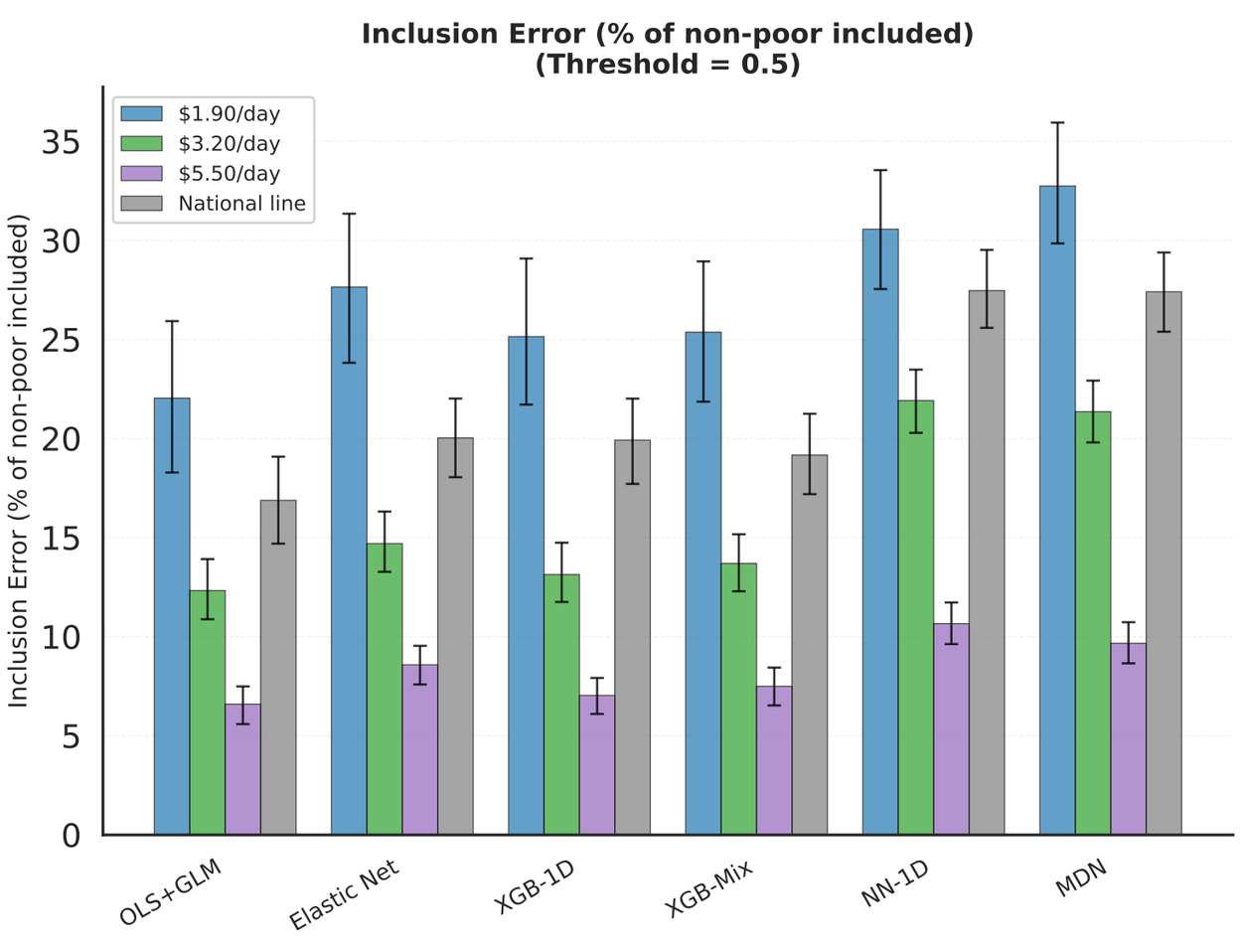

Relative difference at $5.50 Poverty Line Poverty line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.62
1.37
1.36
0.95
1.33
1.00 1.10 1.20 1.30 1.40 1.50 1.60
Resilience score ratio (model / MDN) [>1 higher]

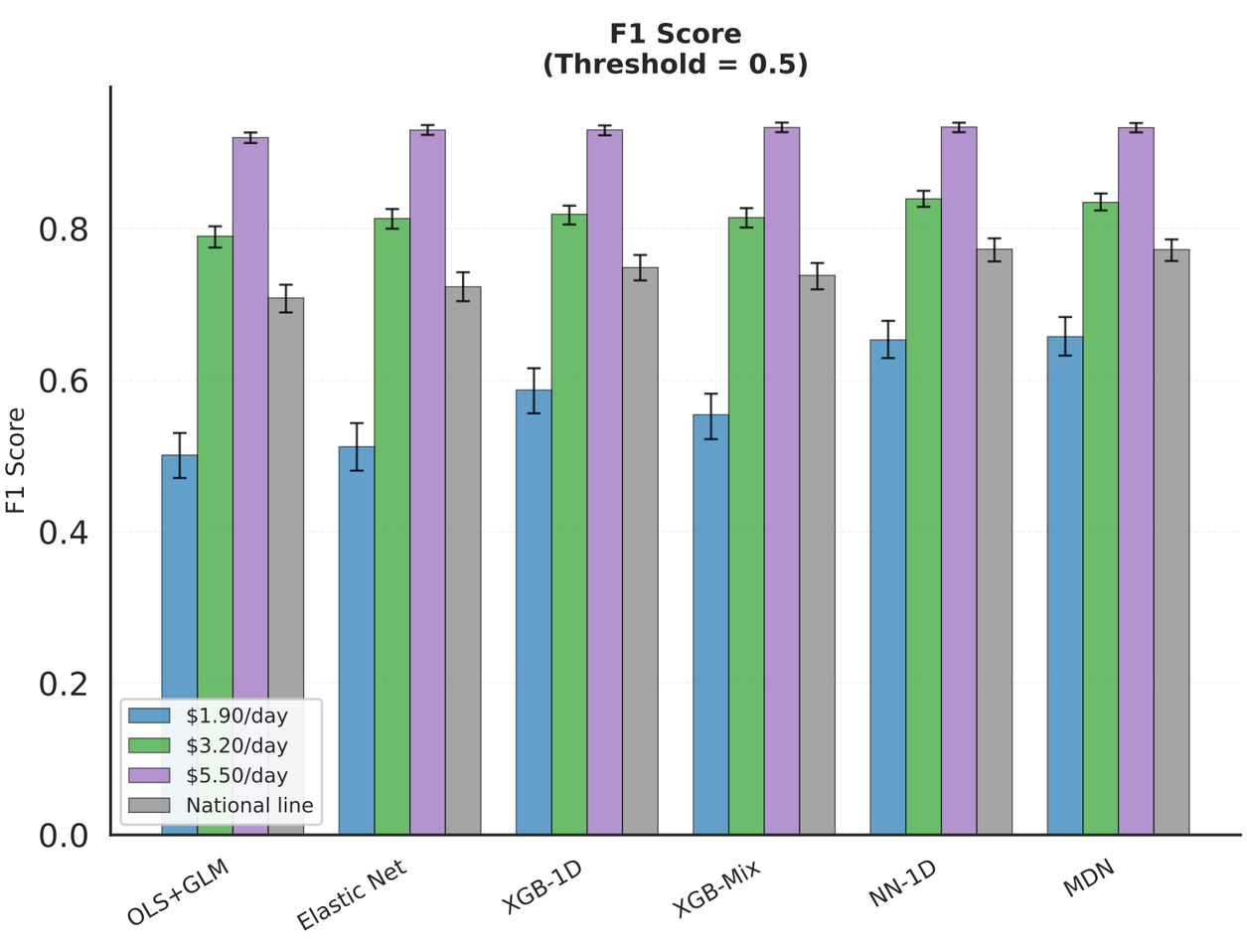

Relative difference at National Poverty Line Poverty line
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.28
1.21
1.19
1.00
1.20
1.00 1.05 1.10 1.15 1.20 1.25 1.30
Resilience score ratio (model / MDN) [>1 higher]


$1.90/day
Precipitation Index (3-month) -0.038 -0.111 -0.251 -0.295 -0.180 -0.298
Battle Events (4-6 months prior) -0.092 0.051 -0.002 0.040 0.336 0.186
Electricity Connection 0.332 0.343 0.557 0.601 0.774 0.582
Other Education 0.290 0.236 -0.019 -0.062 -0.019 0.176
Primary Education 0.305 0.193 0.083 0.060 0.168 0.168
Secondary Education
Cooking Fuel: Firewood -1.313 -0.818 -0.909 -0.971 -0.584 -0.950
Water Source: Borehole -0.127 -0.122 -0.083 -0.087 -0.023 -0.201
Water Source: Well -0.255 -0.180 -0.197 -0.150 -0.088 -0.201
Toilet: Flush Septic 0.541 0.602 0.507 0.562 0.334 1.026
Married (Monogamous) -0.612 -0.496 -0.667 -0.732 -0.279 -0.688
Married (Polygamous) -1.338 -1.097 -1.228 -1.285 -0.918 -1.284
Number of Rooms 0.032 0.025 0.038 0.039 0.044 0.035
Car Ownership 0.668 0.695 0.828 0.880 0.486 0.893
Computer Ownership
Motorcycle Ownership 0.122 0.192 0.142 0.150 0.135 0.254
Television Ownership 0.583 0.516 0.778 0.721 0.703 0.720
Share Young Children (0-4) -1.701 -1.632 -1.015 -0.968 -1.235 -1.626
Share Adult Females (16-65)
Share Girls (5-15) -2.607 -2.232 -2.224 -2.341 -2.298 -2.568
Share Adult Males (16-65)
Share Boys (5-15) -2.948 -2.419 -2.650 -2.675 -2.771 -3.002
Riot Events (4-6 months prior) 0.066 0.074 -0.053 -0.033 0.219 0.123
Has Savings 0.328 0.402 0.299 0.301 0.551 0.382
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
4
2
0
−2
−4
Coefficient


Coverage Gap (percentage points)
(Threshold = 0.5)
Coverage Gap (percentage points)
$1.90/day
$3.20/day
$5.50/day
National line
10
5
0
−5
−10
−15
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN


Exclusion Error (% of poor excluded)
(Threshold = 0.5)
Exclusion Error (% of poor excluded)
$1.90/day
$3.20/day
$5.50/day
National line
60
50
40
30
20
10
0
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN


Inclusion Error (% of non-poor included)
(Threshold = 0.5)
Inclusion Error (% of non-poor included)
$1.90/day
$3.20/day
$5.50/day
National line
35
30
25
20
15
10
5
0
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN


F1 Score
(Threshold = 0.5)
F1 Score
$1.90/day
$3.20/day
$5.50/day
National line
0.8
0.6
0.4
0.2
0.0
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN

OLS + GLM

Elastic Net

XGB -- 1D

NN -- 1D

XGB -- Mix

MDN

Empirical PIT quantiles

Theoretical quantiles (Uniform)

0.0

0.2

0.4

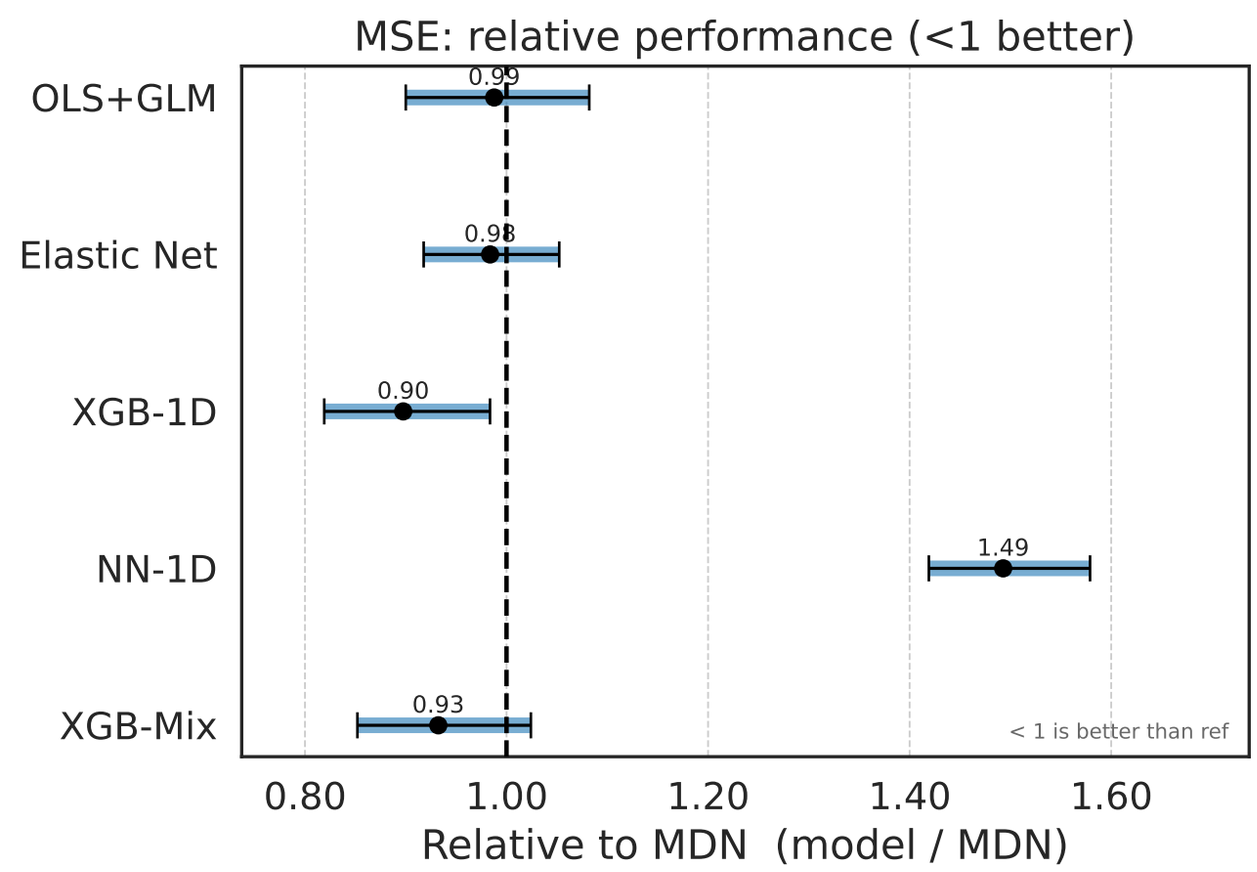
0.6

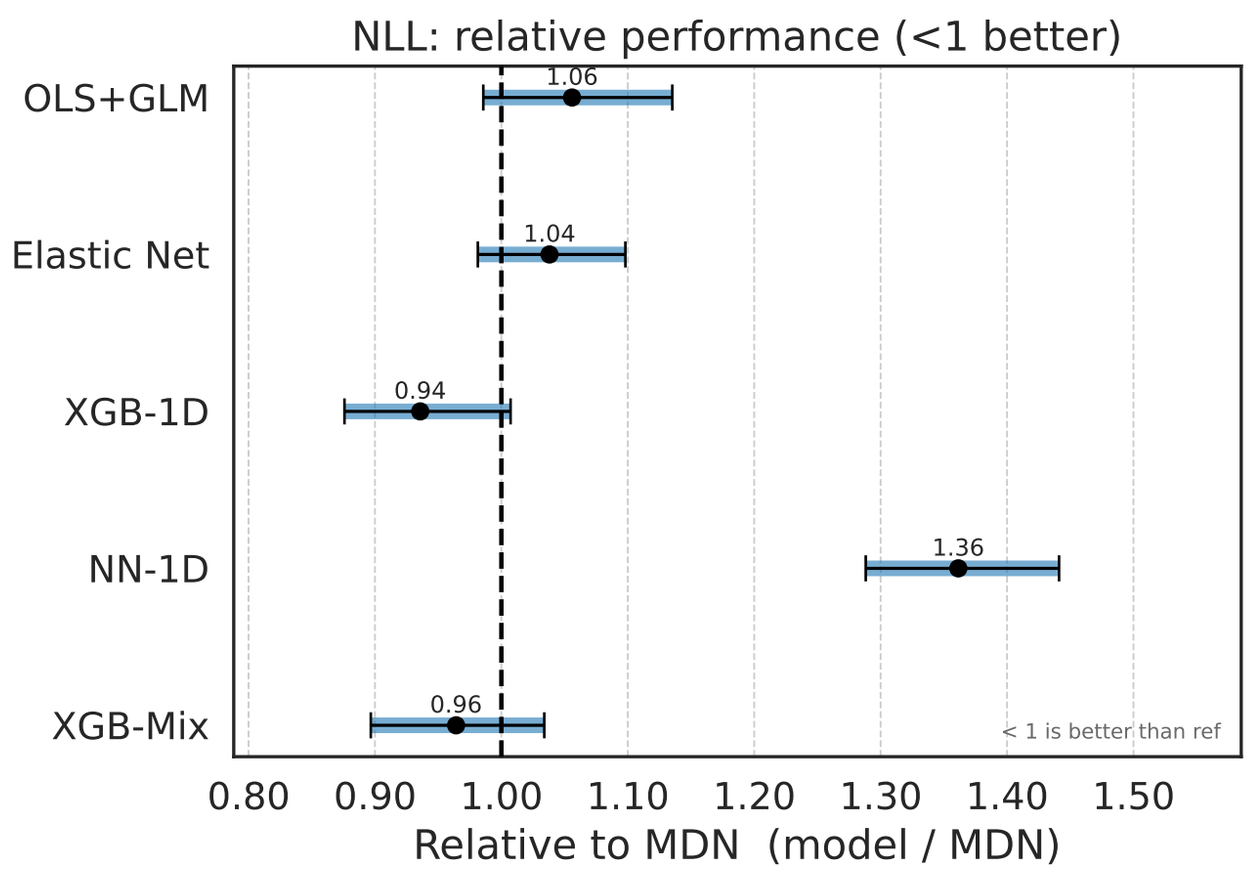
0.8

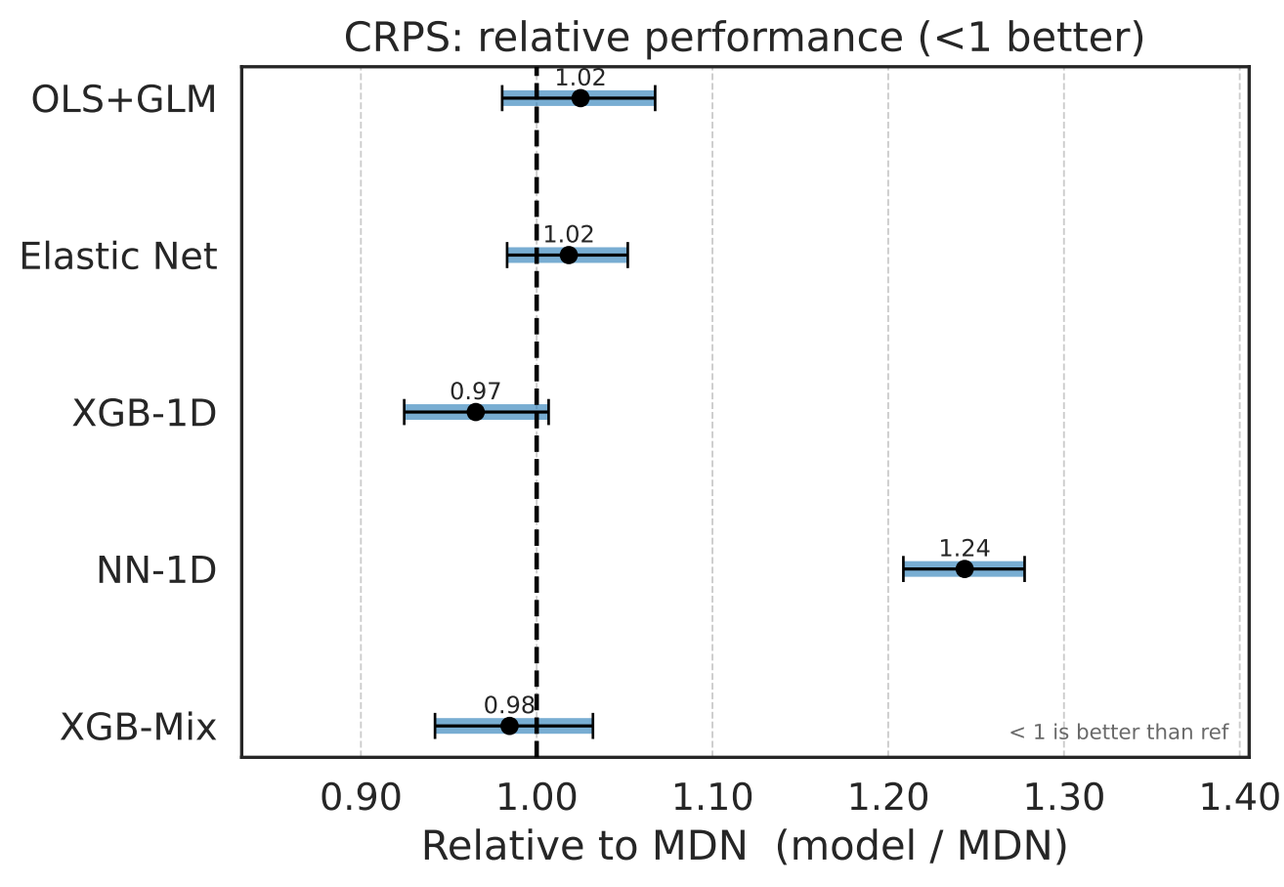
1.0

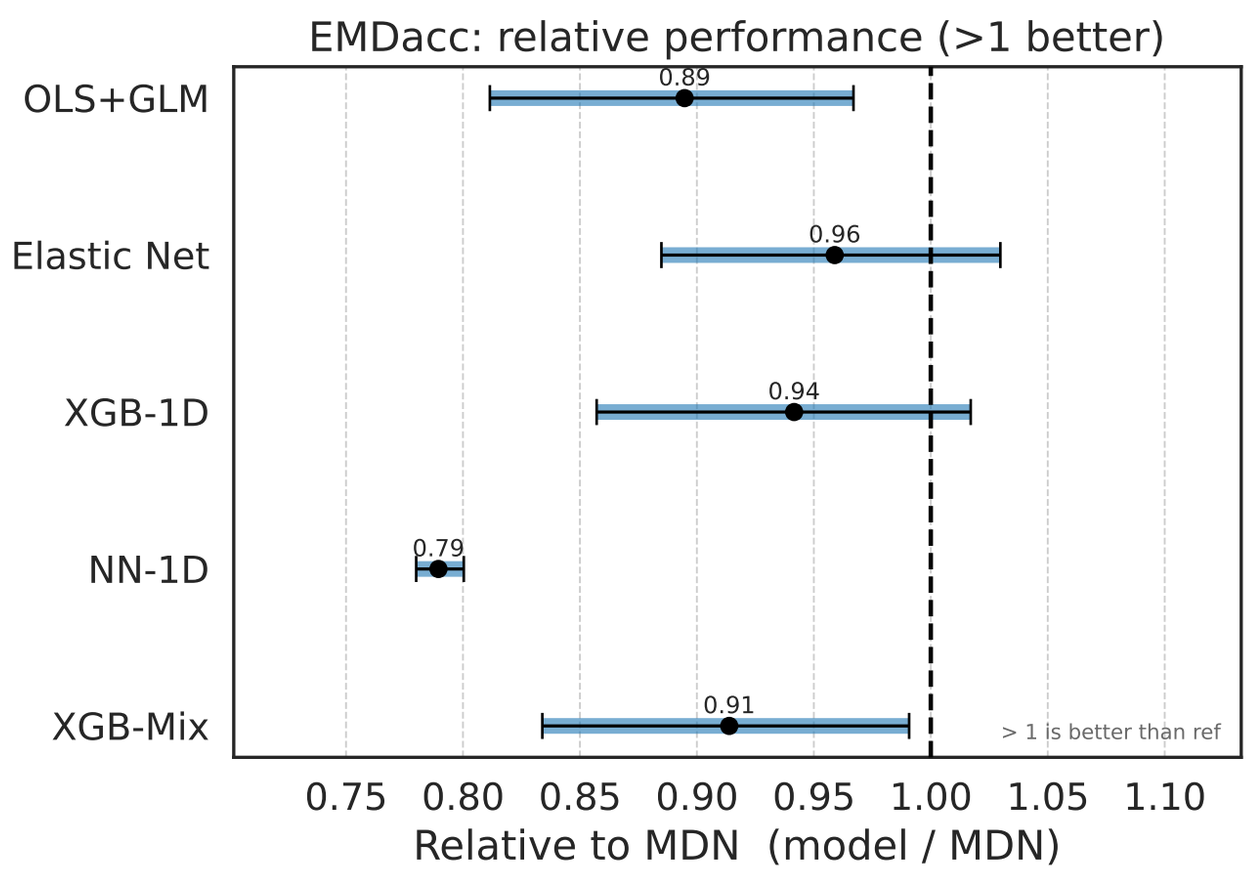
----- 45° line (Uniform)

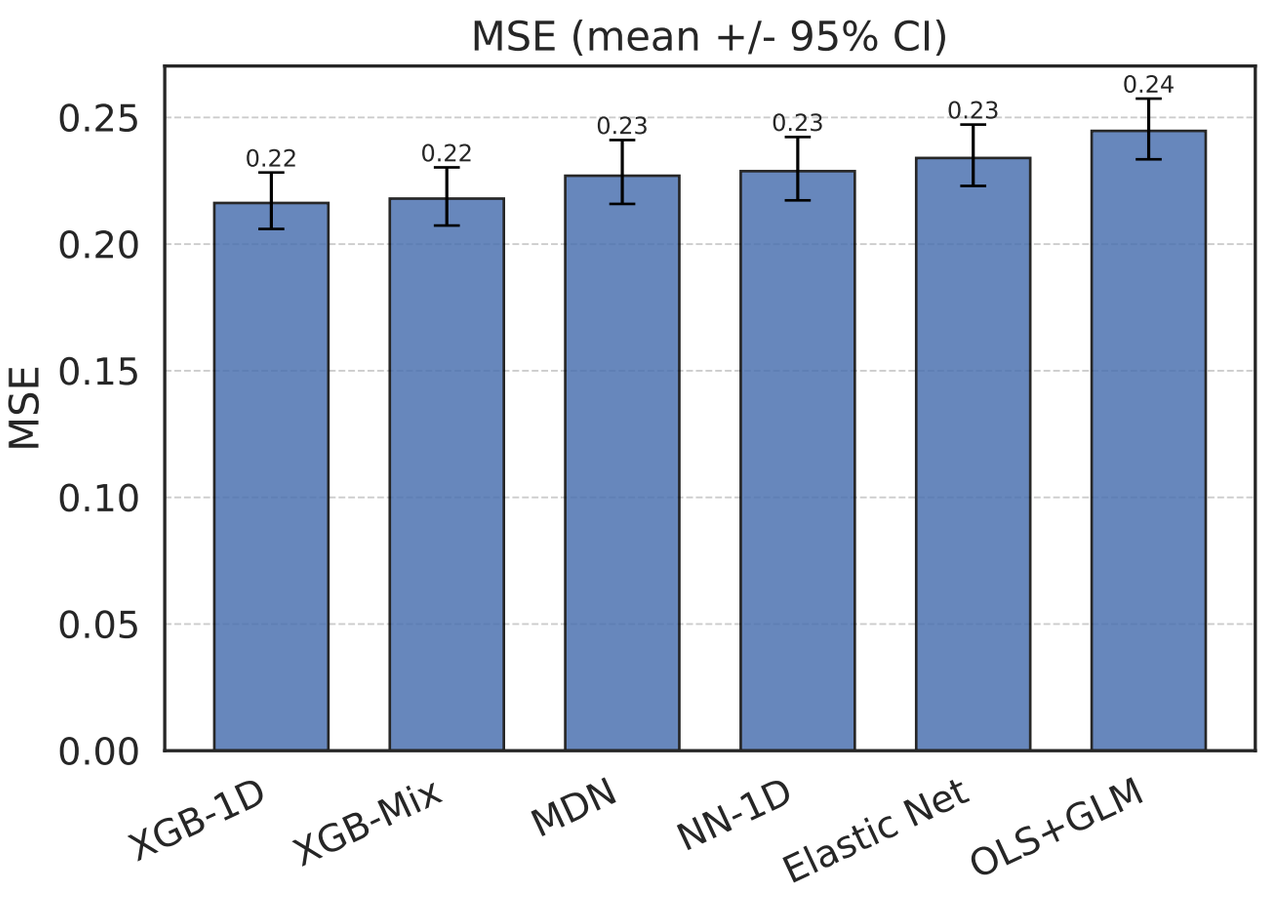
95% band

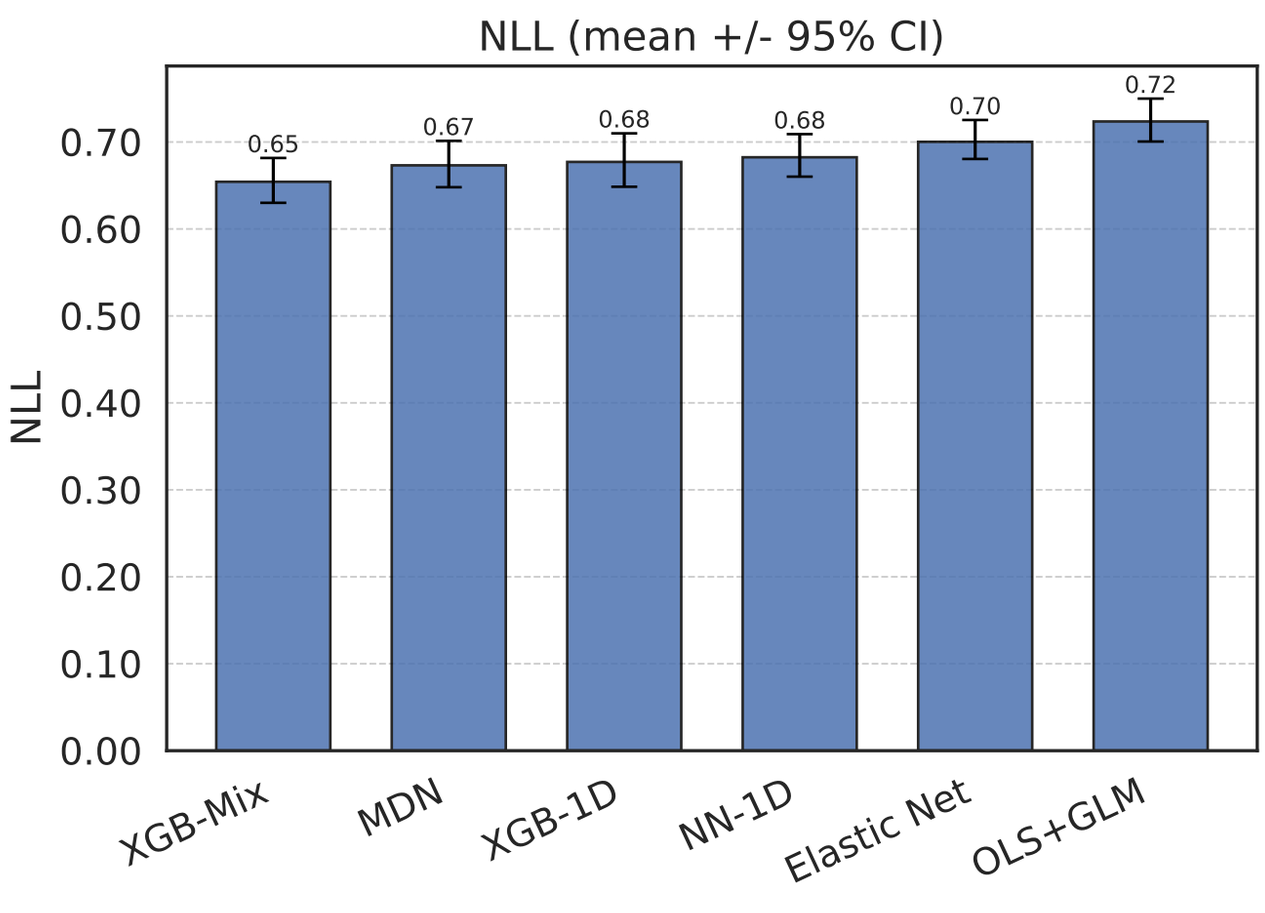

MSE: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.99
0.98
0.90
1.49
0.93
< 1 is better than ref
0.80
1.00
1.20
1.40
1.60
Relative to MDN (model / MDN)

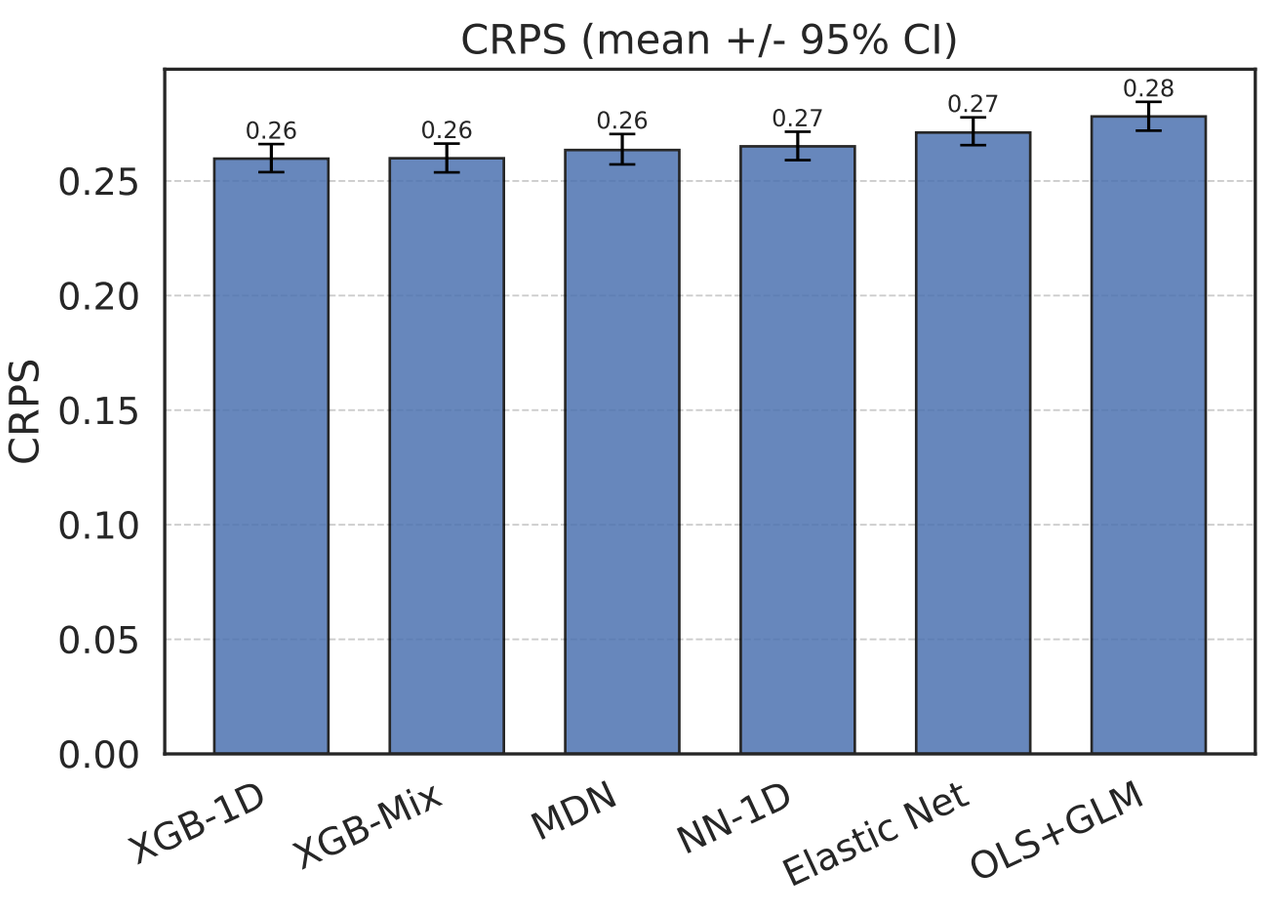

NLL: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.06
1.04
0.94
1.36
0.96
< 1 is better than ref
0.80
0.90
1.00
1.10
1.20
1.30
1.40
1.50
Relative to MDN (model / MDN)

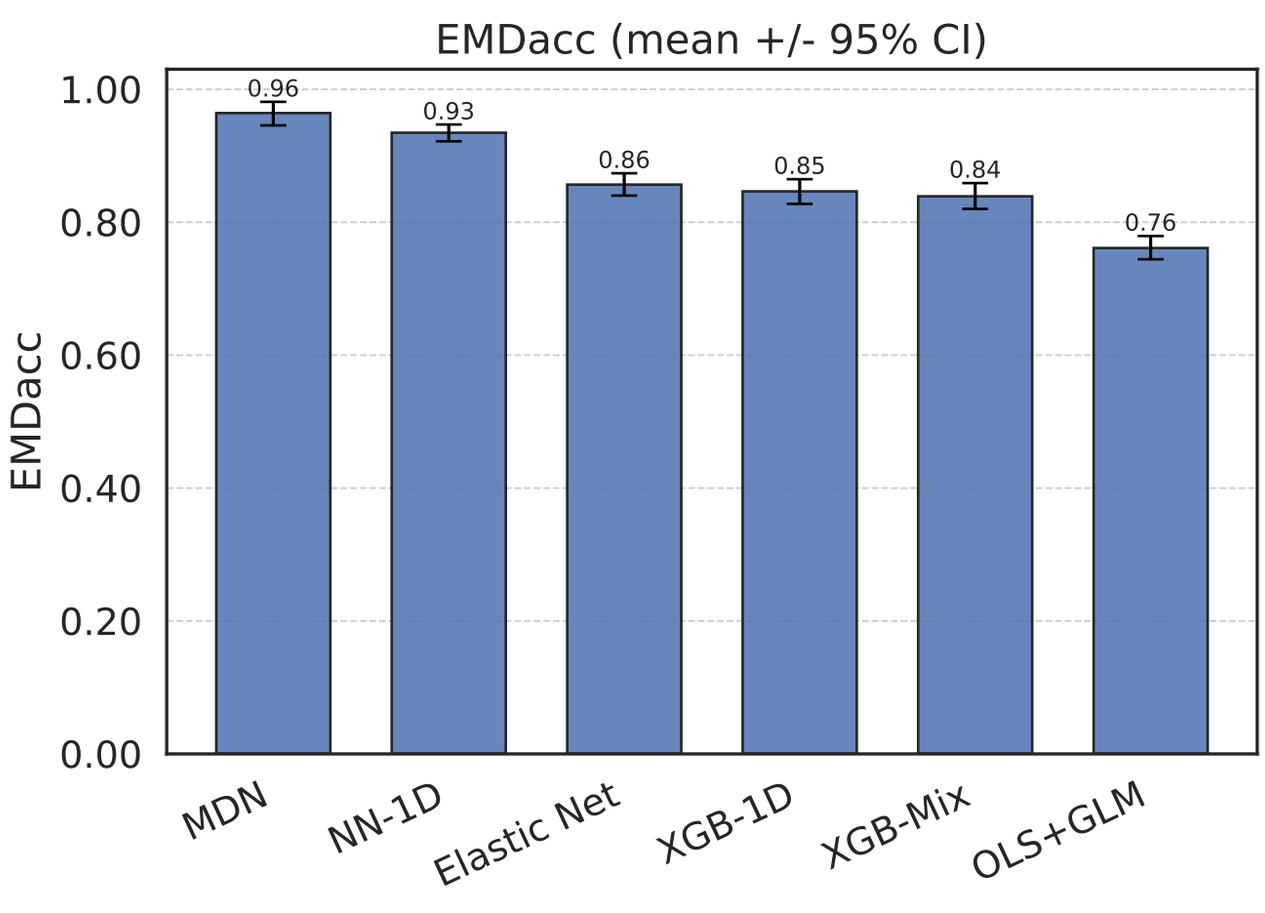

CRPS: relative performance (<1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
1.02
1.02
0.97
1.24
0.98
< 1 is better than ref
0.90
1.00
1.10
1.20
1.30
1.40
Relative to MDN (model / MDN)


EMDacc: relative performance (>1 better)
OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix
0.89
0.96
0.94
0.79
0.91
> 1 is better than ref
0.75
0.80
0.85
0.90
0.95
1.00
1.05
1.10
Relative to MDN (model / MDN)


MSE (mean +/- 95% CI)
MSE
0.00
0.05
0.10
0.15
0.20
0.25
0.22
0.22
0.23
0.23
0.23
0.24
XGB-1D
XGB-Mix
MDN
NN-1D
Elastic Net
OLS+GLM


NLL (mean +/- 95% CI)
NLL
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.65
0.67
0.68
0.68
0.70
0.72
XGB-Mix
MDN
XGB-1D
NN-1D
Elastic Net
OLS+GLM


CRPS (mean +/- 95% CI)
CRPS
0.00
0.05
0.10
0.15
0.20
0.25
0.26
0.26
0.26
0.27
0.27
0.28
XGB-1D
XGB-Mix
MDN
NN-1D
Elastic Net
OLS+GLM


EMDacc (mean +/- 95% CI)
EMDacc
0.00
0.20
0.40
0.60
0.80
1.00
0.96
0.93
0.86
0.85
0.84
0.76
MDN
NN-1D
Elastic Net
XGB-1D
XGB-Mix
OLS+GLM

Within MDN band: Non-resilient (<0.5)

Within MDN band: Resilient (0.5-0.8)

Within MDN band: Most-resilient (>=0.8)

$1.90/day Poverty Line

$3.20/day Poverty Line

Share of household

$5.50/day Poverty Line

National Poverty Line

100%
80%
60%
40%
20%
0%

OLS+GLM
Elastic Net
XGB-1D
NN-1D
XGB-Mix

$1.90/day Poverty Line

Within MDN band: Non-resilient (<0.5)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Resilient (0.5-0.8)	51% [48, 54]	45% [42, 49]	36% [33, 39]	14% [12, 16]	41% [38, 44]
Non-resilient (<0.5)	48% [45, 51]	53% [50, 56]	62% [59, 65]	86% [84, 88]	57% [54, 60]

Within MDN band: Resilient (0.5-0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	55% [52, 58]	42% [39, 45]	47% [44, 50]	12% [8, 16]	51% [48, 54]
Resilient (0.5-0.8)	43% [40, 46]	54% [51, 57]	48% [46, 51]	88% [86, 90]	44% [42, 48]

Within MDN band: Most-resilient (>=0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	100% [99, 100]	96% [95, 97]	98% [98, 99]	86% [84, 88]	98% [98, 99]
Resilient (0.5-0.8)				14% [12, 16]	

$3.20/day Poverty Line

Within MDN band: Non-resilient (<0.5)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Resilient (0.5-0.8)	26% [24, 28]	20% [18, 21]	20% [19, 22]	2% [1, 3]	20% [19, 22]
Non-resilient (<0.5)	72% [71, 74]	79% [78, 81]	78% [76, 80]	98% [97, 98]	78% [77, 80]

Within MDN band: Resilient (0.5-0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	56% [51, 60]	30% [26, 34]	45% [42, 49]	1% [0, 2]	42% [39, 47]
Resilient (0.5-0.8)	44% [39, 48]	66% [62, 69]	52% [47, 56]	79% [75, 82]	55% [51, 59]
Non-resilient (<0.5)		4% [3, 5]	3% [2, 4]	20% [17, 23]	2% [1, 3]

Within MDN band: Most-resilient (>=0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	97% [96, 99]	86% [83, 89]	94% [92, 96]	70% [66, 74]	96% [94, 98]
Resilient (0.5-0.8)	3% [2, 4]	14% [11, 17]	6% [5, 7]	30% [26, 34]	4% [3, 5]

$5.50/day Poverty Line

Within MDN band: Non-resilient (<0.5)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Resilient (0.5-0.8)	9% [8, 10]	5% [4, 6]	7% [6, 8]	0% [0, 0]	5% [4, 6]
Non-resilient (<0.5)	91% [90, 92]	95% [95, 96]	93% [92, 94]	100% [100, 100]	95% [94, 95]

Within MDN band: Resilient (0.5-0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	30% [24, 36]	11% [6, 15]	11% [8, 15]	1% [0, 2]	12% [8, 16]
Resilient (0.5-0.8)	67% [61, 73]	70% [64, 76]	83% [78, 87]	62% [56, 68]	79% [74, 84]
Non-resilient (<0.5)	4% [3, 5]	19% [15, 25]	6% [5, 7]	37% [31, 43]	9% [5, 13]

Within MDN band: Most-resilient (>=0.8)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	81% [73, 88]	52% [43, 61]	50% [42, 59]	56% [48, 64]	54% [47, 64]
Resilient (0.5-0.8)	19% [12, 27]	47% [38, 56]	49% [40, 57]	44% [36, 52]	45% [36, 53]

National Poverty Line

Within MDN band: Non-resilient (<0.5)

	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Resilient (0.5-0.8)	34% [32, 36]	28% [26, 30]	23% [21, 25]	5% [4, 6]	26% [24, 28]
Non-resilient (<0.5)	65% [62, 67]	71% [69, 73]	75% [73, 77]	95% [93, 96]	72% [70, 74]

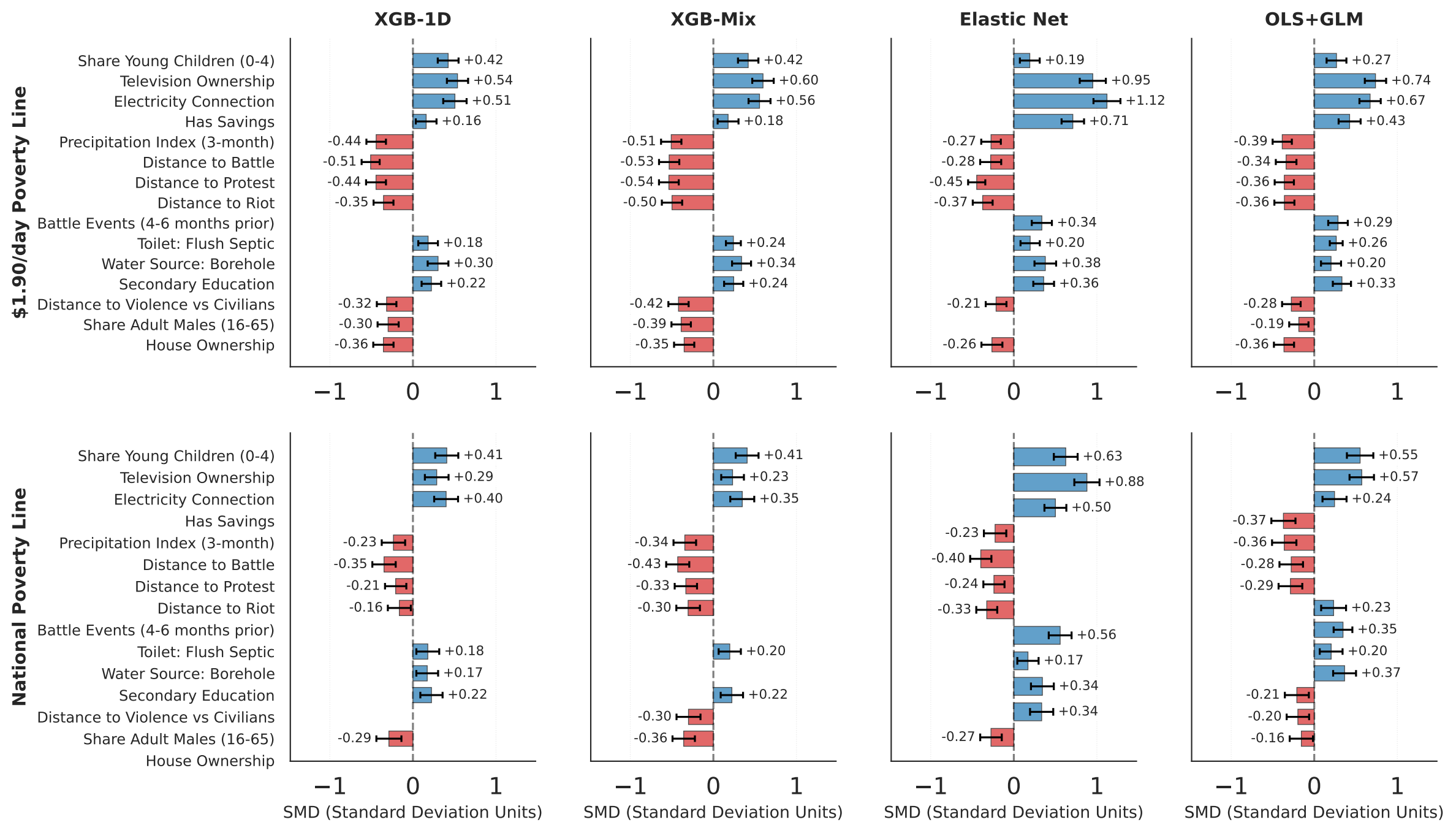
Within MDN band: Resilient (0.5-0.8)

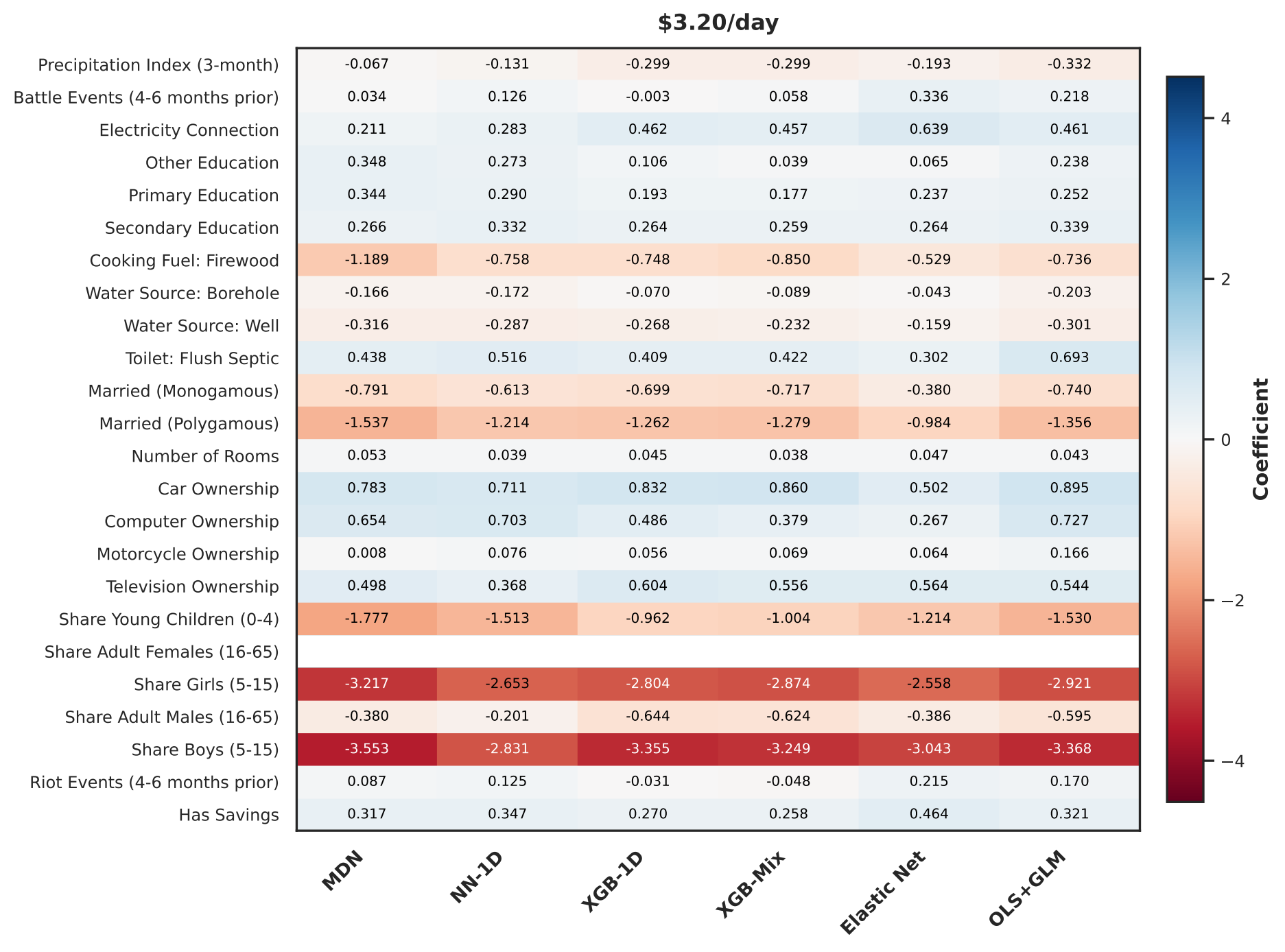
	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	61% [58, 65]	39% [36, 43]	51% [48, 54]	2% [1, 3]	52% [49, 56]
Resilient (0.5-0.8)	38% [34, 41]	57% [53, 60]	46% [42, 49]	85% [82, 87]	45% [41, 49]
Non-resilient (<0.5)	1% [0, 2]	4% [3, 5]	4% [3, 5]	13% [10, 15]	3% [2, 4]

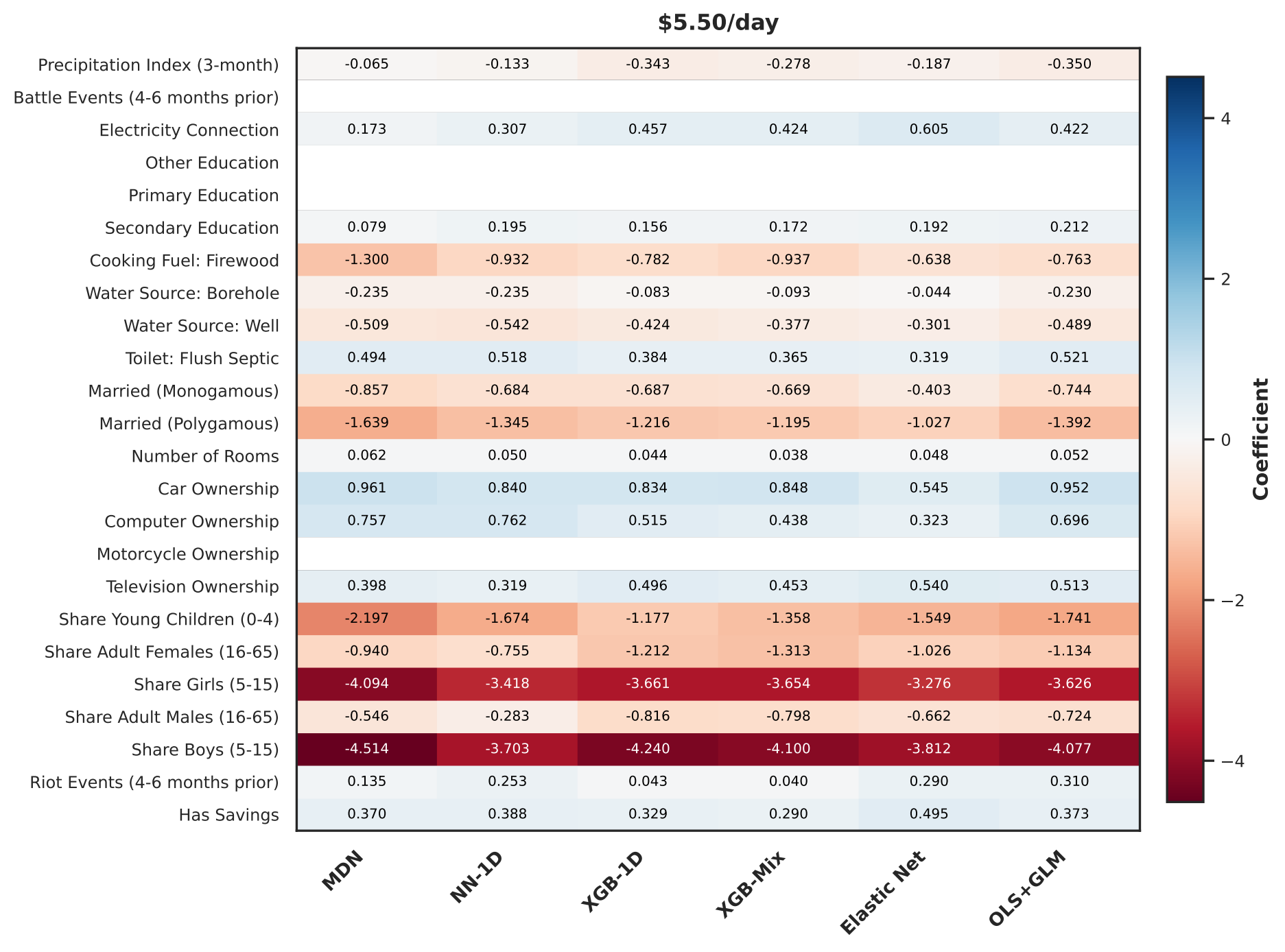
Within MDN band: Most-resilient (>=0.8)

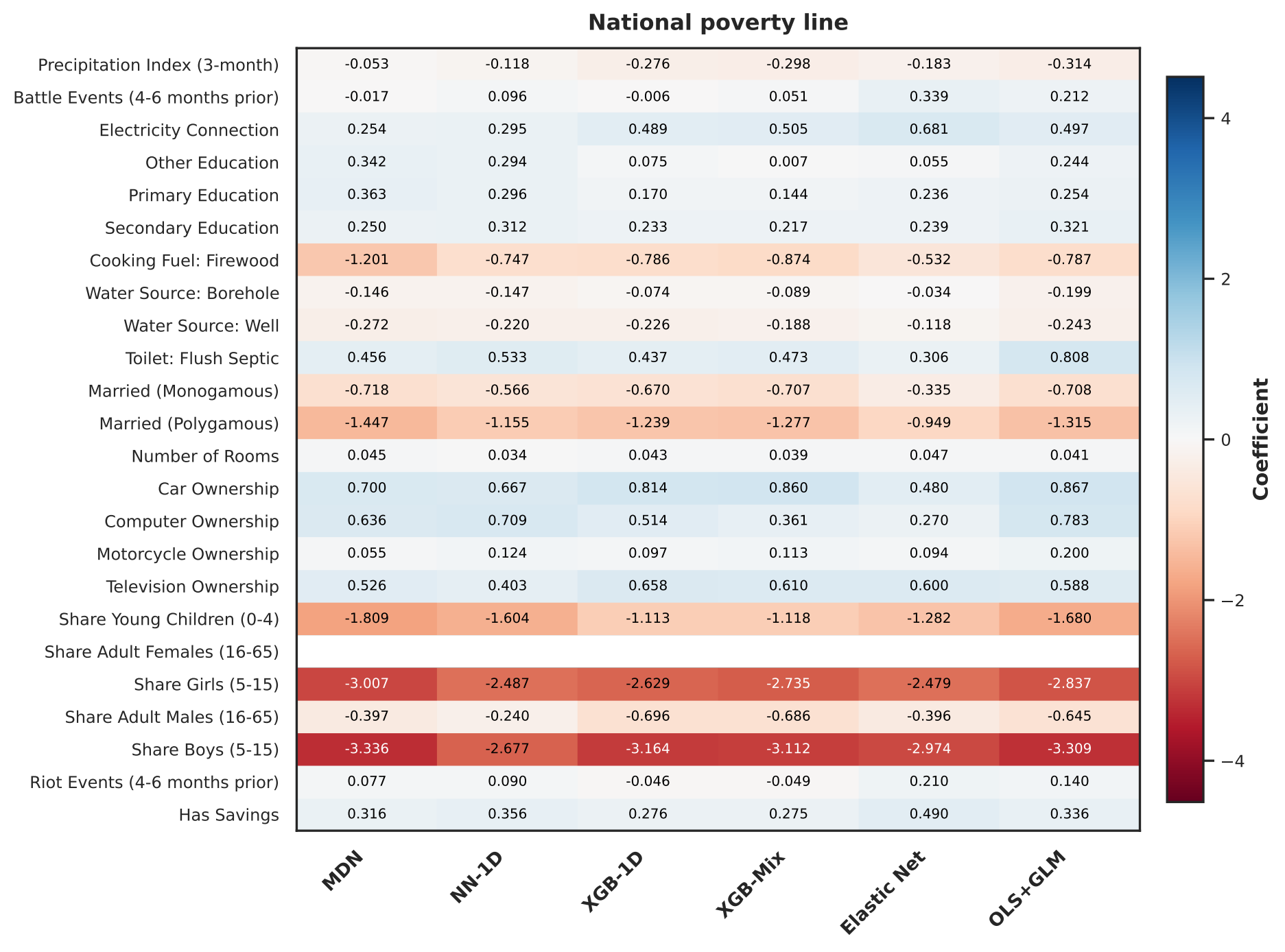
	OLS+GLM	Elastic Net	XGB-1D	NN-1D	XGB-Mix
Most-resilient (>=0.8)	99% [98, 100]	92% [90, 94]	96% [94, 97]	75% [72, 77]	97% [96, 98]
Resilient (0.5-0.8)	1% [0, 2]	8% [6, 10]	4% [3, 5]	25% [23, 28]	3% [2, 4]

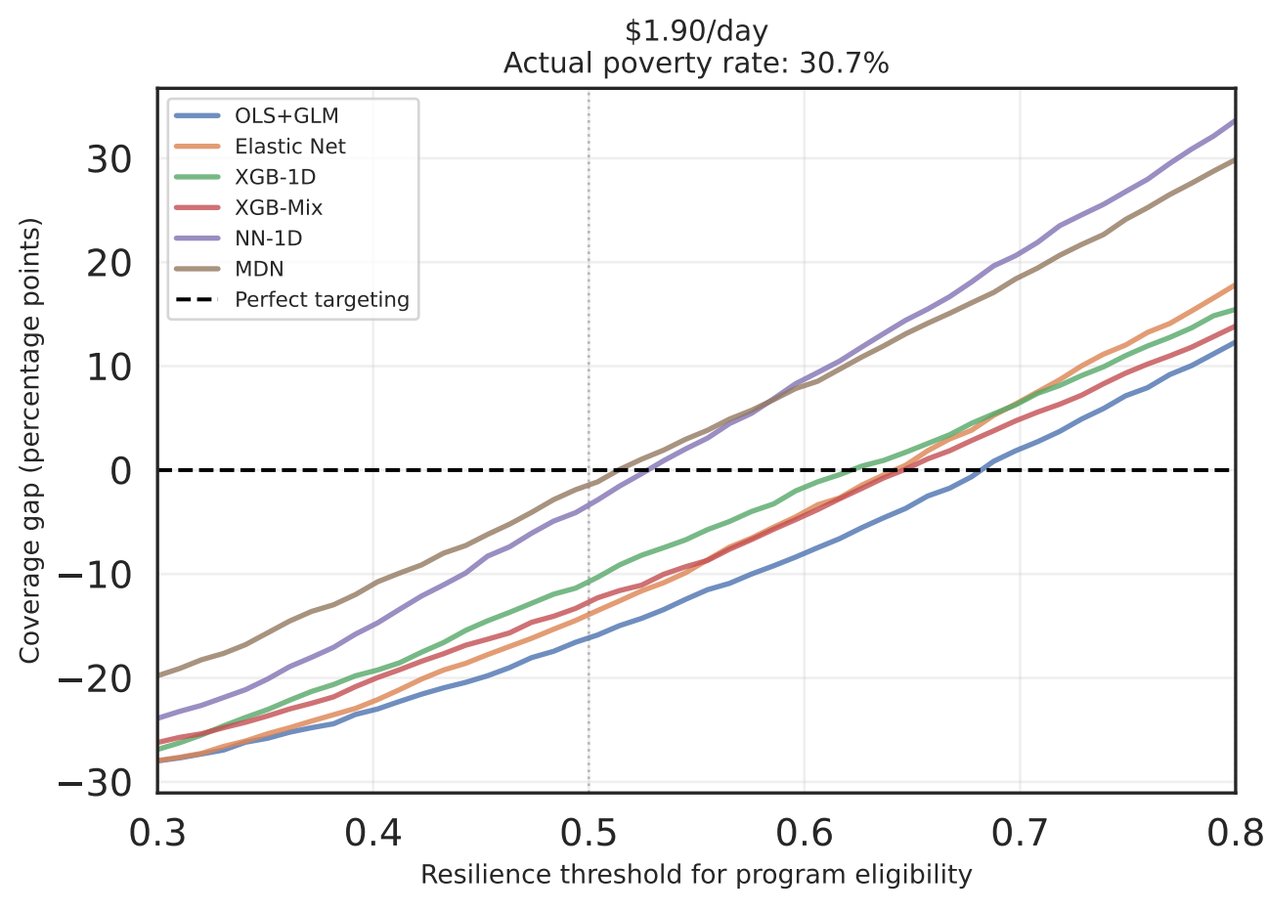
Non-resilient (<0.5)
Resilient (0.5-0.8)
Most-resilient (>=0.8)

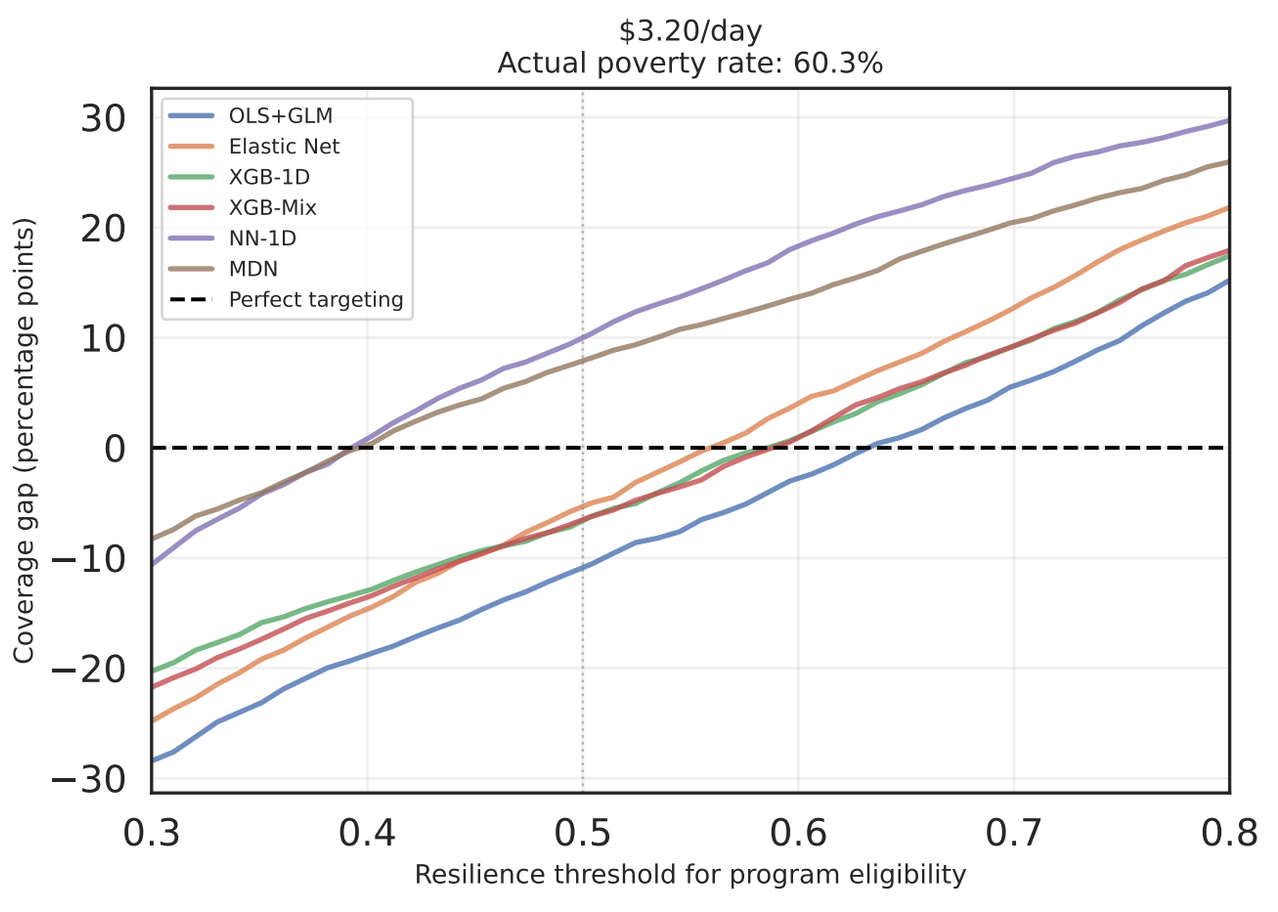

XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
$1.90/day Poverty Line
National Poverty Line
Share Young Children (0-4)
Television Ownership
Electricity Connection
Has Savings
Precipitation Index (3-month)
Distance to Battle
Distance to Protest
Distance to Riot
Battle Events (4-6 months prior)
Toilet: Flush Septic
Water Source: Borehole
Secondary Education
Distance to Violence vs Civilians
Share Adult Males (16-65)
House Ownership
+0.42 +0.54 +0.51 +0.16 -0.44 -0.51 -0.44 -0.35 +0.18 +0.30 +0.22 -0.32 -0.30 -0.36
+0.42 +0.60 +0.56 +0.18 -0.51 -0.53 -0.54 -0.50 +0.24 +0.34 +0.24 -0.42 -0.39 -0.35
+0.19 +0.95 +1.12 +0.71 -0.27 -0.28 -0.45 -0.37 +0.34 +0.20 +0.38 +0.36 -0.21 -0.26
+0.27 +0.74 +0.67 +0.43 -0.39 -0.34 -0.36 -0.36 +0.29 +0.26 +0.20 +0.33 -0.28 -0.19 -0.36
+0.41 +0.29 +0.40 -0.23 -0.35 -0.21 -0.16 +0.18 +0.17 +0.22 -0.29
+0.41 +0.23 +0.35 -0.34 -0.43 -0.33 -0.30 +0.20 +0.22 -0.30 -0.36
+0.63 +0.88 +0.50 -0.23 -0.40 -0.24 -0.33 +0.56 +0.17 +0.34 +0.34 -0.27
+0.55 +0.57 +0.24 -0.37 -0.36 -0.28 -0.29 +0.23 +0.35 +0.20 +0.37 -0.21 -0.20 -0.16
−1 0 1
SMD (Standard Deviation Units)

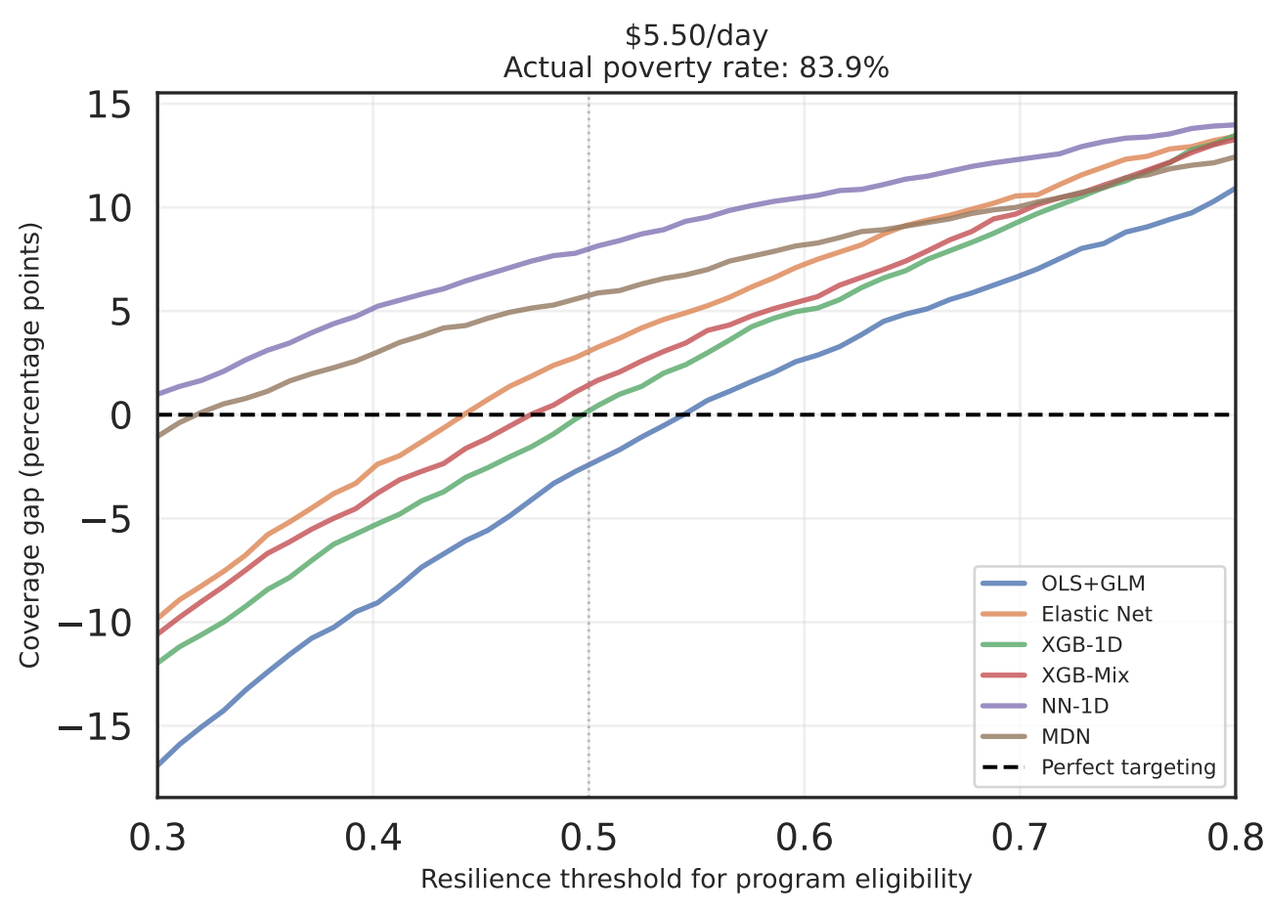

$3.20/day
Precipitation Index (3-month) -0.067 -0.131 -0.299 -0.299 -0.193 -0.332
Battle Events (4-6 months prior) 0.034 0.126 -0.003 0.058 0.336 0.218
Electricity Connection 0.211 0.283 0.462 0.457 0.639 0.461
Other Education 0.348 0.273 0.106 0.039 0.065 0.238
Primary Education 0.344 0.290 0.193 0.177 0.237 0.252
Secondary Education 0.266 0.332 0.264 0.259 0.264 0.339
Cooking Fuel: Firewood -1.189 -0.758 -0.748 -0.850 -0.529 -0.736
Water Source: Borehole -0.166 -0.172 -0.070 -0.089 -0.043 -0.203
Water Source: Well -0.316 -0.287 -0.268 -0.232 -0.159 -0.301
Toilet: Flush Septic 0.438 0.516 0.409 0.422 0.302 0.693
Married (Monogamous) -0.791 -0.613 -0.699 -0.717 -0.380 -0.740
Married (Polygamous) -1.537 -1.214 -1.262 -1.279 -0.984 -1.356
Number of Rooms 0.053 0.039 0.045 0.038 0.047 0.043
Car Ownership 0.783 0.711 0.832 0.860 0.502 0.895
Computer Ownership 0.654 0.703 0.486 0.379 0.267 0.727
Motorcycle Ownership 0.008 0.076 0.056 0.069 0.064 0.166
Television Ownership 0.498 0.368 0.604 0.556 0.564 0.544
Share Young Children (0-4) -1.777 -1.513 -0.962 -1.004 -1.214 -1.530
Share Adult Females (16-65)
Share Girls (5-15) -3.217 -2.653 -2.804 -2.874 -2.558 -2.921
Share Adult Males (16-65) -0.380 -0.201 -0.644 -0.624 -0.386 -0.595
Share Boys (5-15) -3.553 -2.831 -3.355 -3.249 -3.043 -3.368
Riot Events (4-6 months prior) 0.087 0.125 -0.031 -0.048 0.215 0.170
Has Savings 0.317 0.347 0.270 0.258 0.464 0.321
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
Coefficient
4
2
0
−2
−4

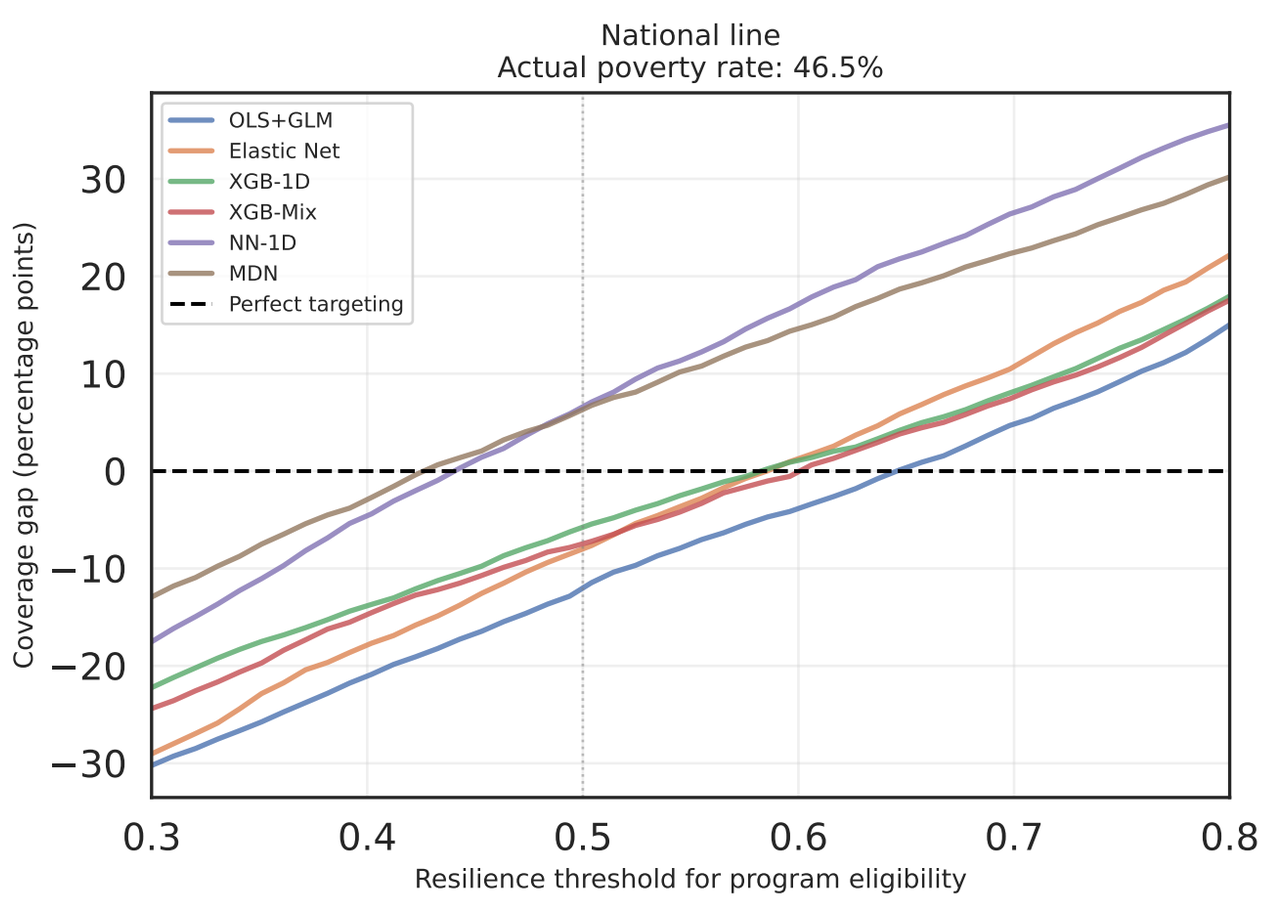

$5.50/day
Precipitation Index (3-month) -0.065 -0.133 -0.343 -0.278 -0.187 -0.350
Battle Events (4-6 months prior)
Electricity Connection 0.173 0.307 0.457 0.424 0.605 0.422
Other Education
Primary Education
Secondary Education 0.079 0.195 0.156 0.172 0.192 0.212
Cooking Fuel: Firewood -1.300 -0.932 -0.782 -0.937 -0.638 -0.763
Water Source: Borehole -0.235 -0.235 -0.083 -0.093 -0.044 -0.230
Water Source: Well -0.509 -0.542 -0.424 -0.377 -0.301 -0.489
Toilet: Flush Septic 0.494 0.518 0.384 0.365 0.319 0.521
Married (Monogamous) -0.857 -0.684 -0.687 -0.669 -0.403 -0.744
Married (Polygamous) -1.639 -1.345 -1.216 -1.195 -1.027 -1.392
Number of Rooms 0.062 0.050 0.044 0.038 0.048 0.052
Car Ownership 0.961 0.840 0.834 0.848 0.545 0.952
Computer Ownership 0.757 0.762 0.515 0.438 0.323 0.696
Motorcycle Ownership
Television Ownership 0.398 0.319 0.496 0.453 0.540 0.513
Share Young Children (0-4) -2.197 -1.674 -1.177 -1.358 -1.549 -1.741
Share Adult Females (16-65) -0.940 -0.755 -1.212 -1.313 -1.026 -1.134
Share Girls (5-15) -4.094 -3.418 -3.661 -3.654 -3.276 -3.626
Share Adult Males (16-65) -0.546 -0.283 -0.816 -0.798 -0.662 -0.724
Share Boys (5-15) -4.514 -3.703 -4.240 -4.100 -3.812 -4.077
Riot Events (4-6 months prior) 0.135 0.253 0.043 0.040 0.290 0.310
Has Savings 0.370 0.388 0.329 0.290 0.495 0.373
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
Coefficient
4
2
0
−2
−4

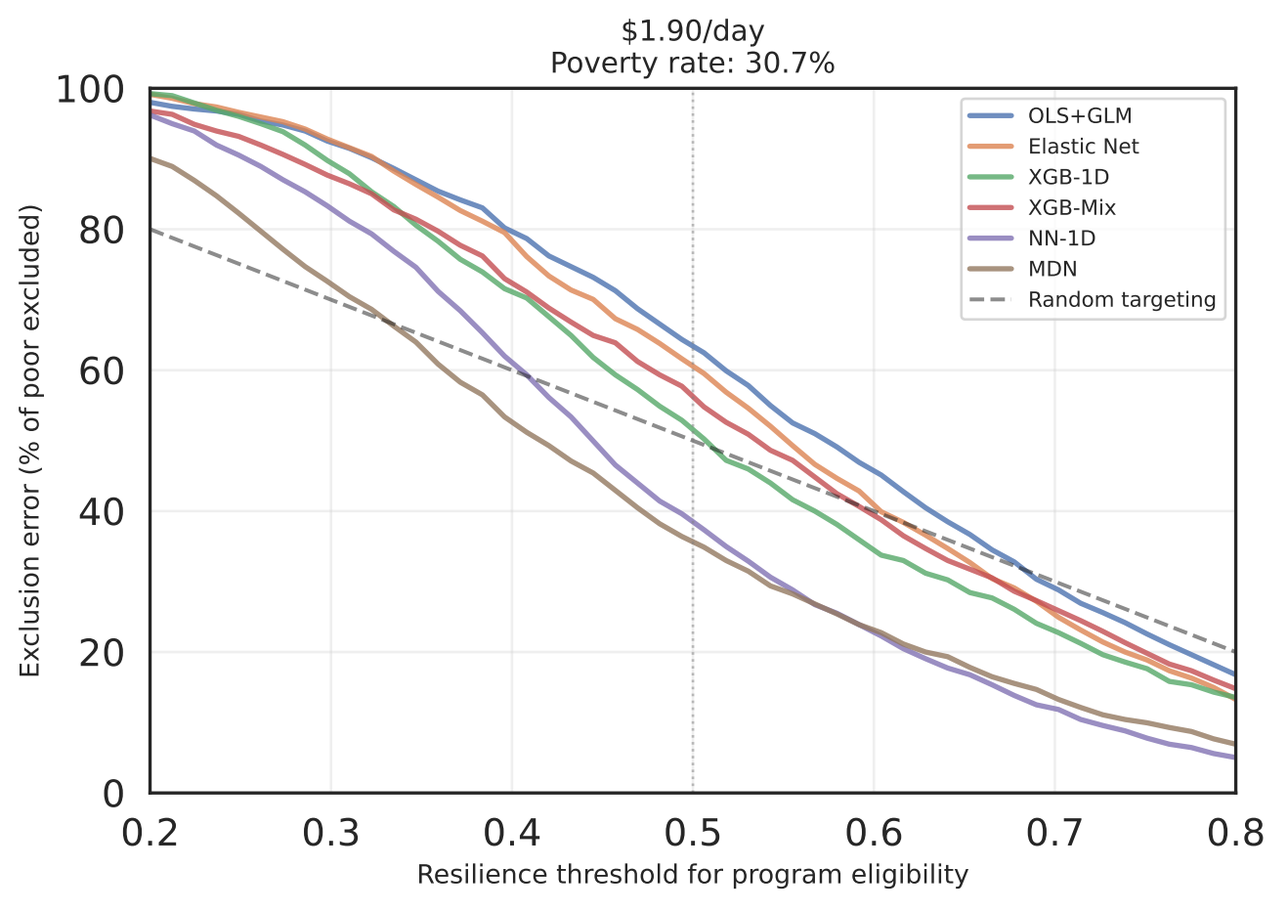

National poverty line
Precipitation Index (3-month)
Battle Events (4-6 months prior)
Electricity Connection
Other Education
Primary Education
Secondary Education
Cooking Fuel: Firewood
Water Source: Borehole
Water Source: Well
Toilet: Flush Septic
Married (Monogamous)
Married (Polygamous)
Number of Rooms
Car Ownership
Computer Ownership
Motorcycle Ownership
Television Ownership
Share Young Children (0-4)
Share Adult Females (16-65)
Share Girls (5-15)
Share Adult Males (16-65)
Share Boys (5-15)
Riot Events (4-6 months prior)
Has Savings
-0.053 -0.118 -0.276 -0.298 -0.183 -0.314
-0.017 0.096 -0.006 0.051 0.339 0.212
0.254 0.295 0.489 0.505 0.681 0.497
0.342 0.294 0.075 0.007 0.055 0.244
0.363 0.296 0.170 0.144 0.236 0.254
0.250 0.312 0.233 0.217 0.239 0.321
-1.201 -0.747 -0.786 -0.874 -0.532 -0.787
-0.146 -0.147 -0.074 -0.089 -0.034 -0.199
-0.272 -0.220 -0.226 -0.188 -0.118 -0.243
0.456 0.533 0.437 0.473 0.306 0.808
-0.718 -0.566 -0.670 -0.707 -0.335 -0.708
-1.447 -1.155 -1.239 -1.277 -0.949 -1.315
0.045 0.034 0.043 0.039 0.047 0.041
0.700 0.667 0.814 0.860 0.480 0.867
0.636 0.709 0.514 0.361 0.270 0.783
0.055 0.124 0.097 0.113 0.094 0.200
0.526 0.403 0.658 0.610 0.600 0.588
-1.809 -1.604 -1.113 -1.118 -1.282 -1.680
-3.007 -2.487 -2.629 -2.735 -2.479 -2.837
-0.397 -0.240 -0.696 -0.686 -0.396 -0.645
-3.336 -2.677 -3.164 -3.112 -2.974 -3.309
0.077 0.090 -0.046 -0.049 0.210 0.140
0.316 0.356 0.276 0.275 0.490 0.336
MDN
NN-1D
XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
Coefficient
4
2
0
−2
−4

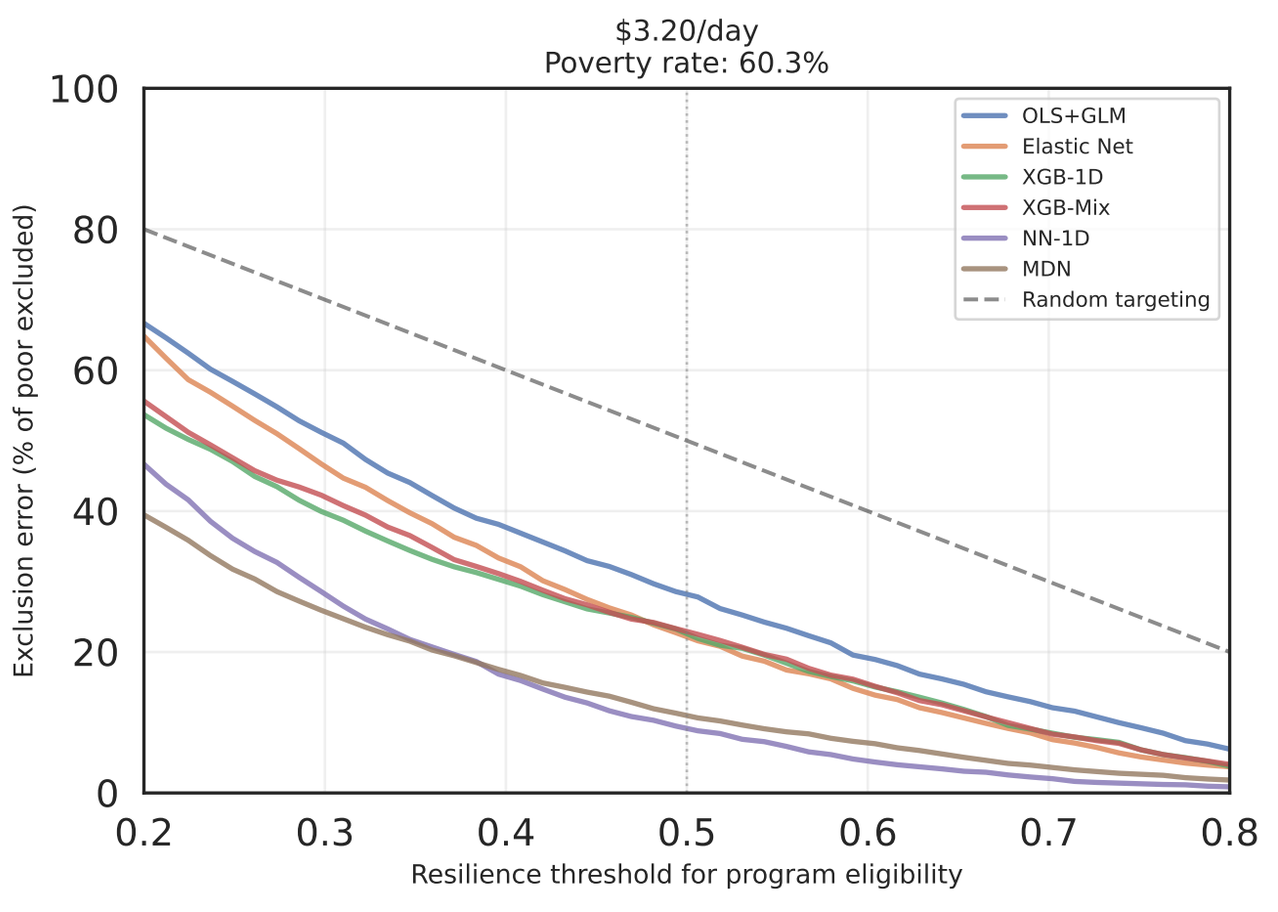

$1.90/day
Actual poverty rate: 30.7%
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Coverage gap (percentage points)
Resilience threshold for program eligibility

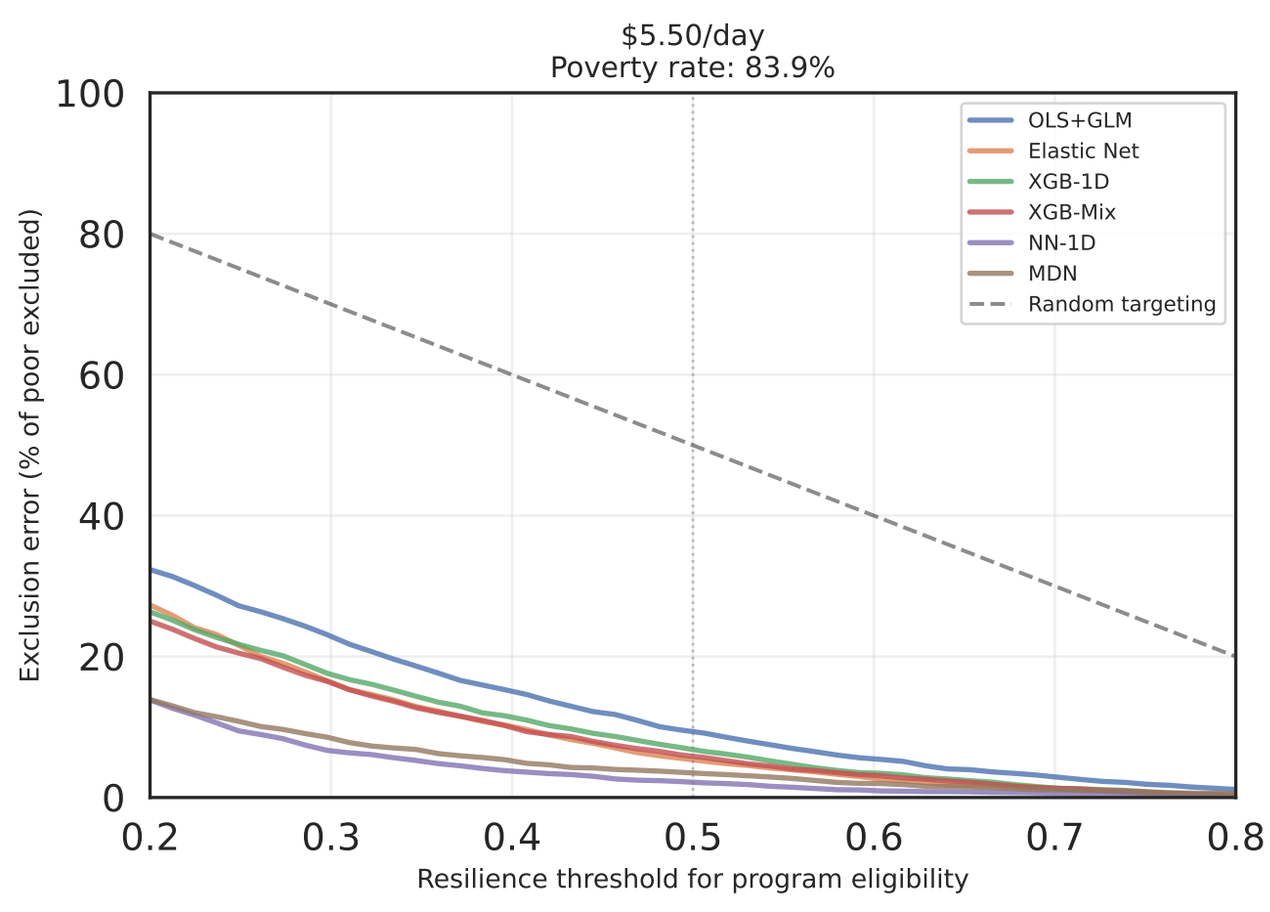

$3.20/day
Actual poverty rate: 60.3%
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Coverage gap (percentage points)
Resilience threshold for program eligibility

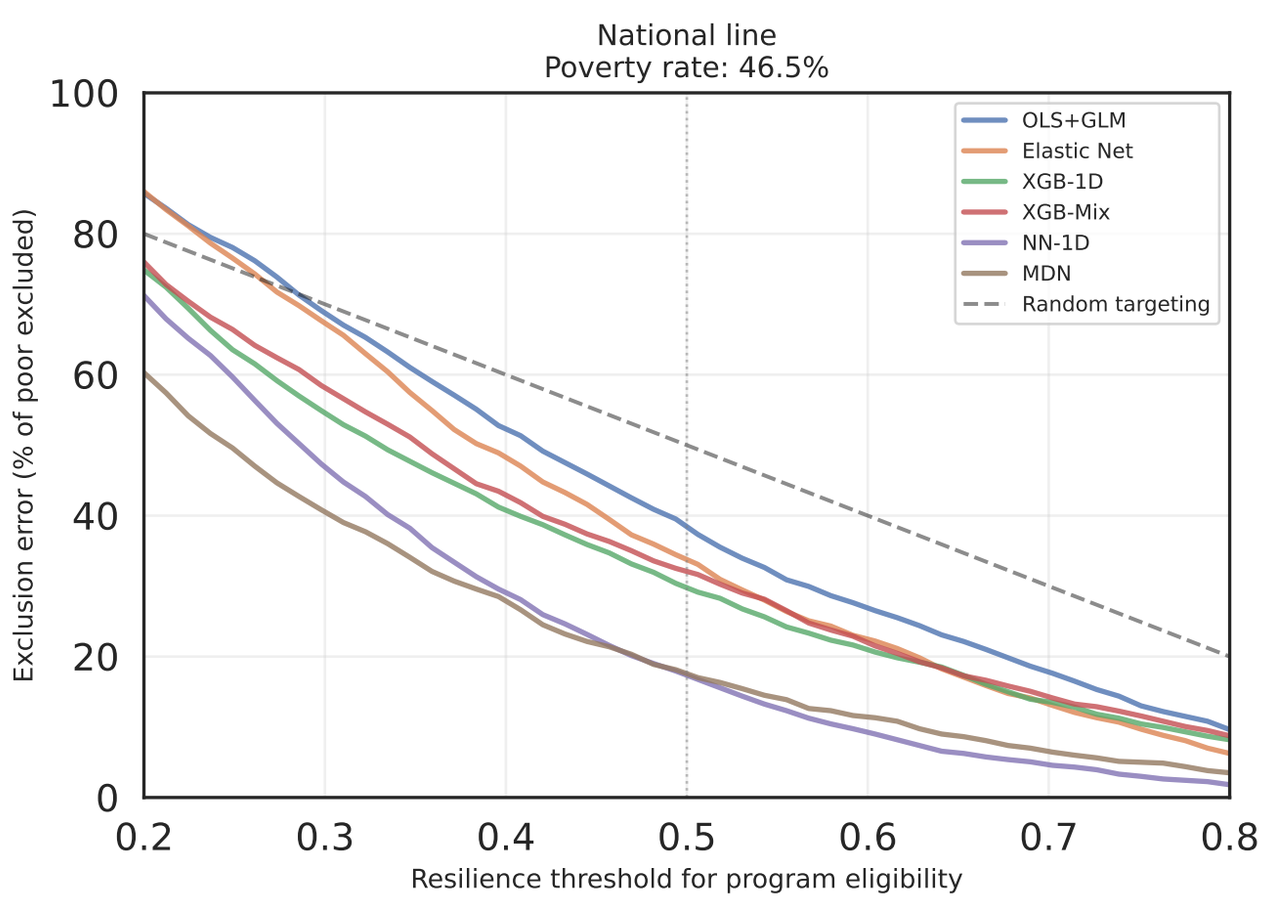

$5.50/day
Actual poverty rate: 83.9%
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Coverage gap (percentage points)
Resilience threshold for program eligibility

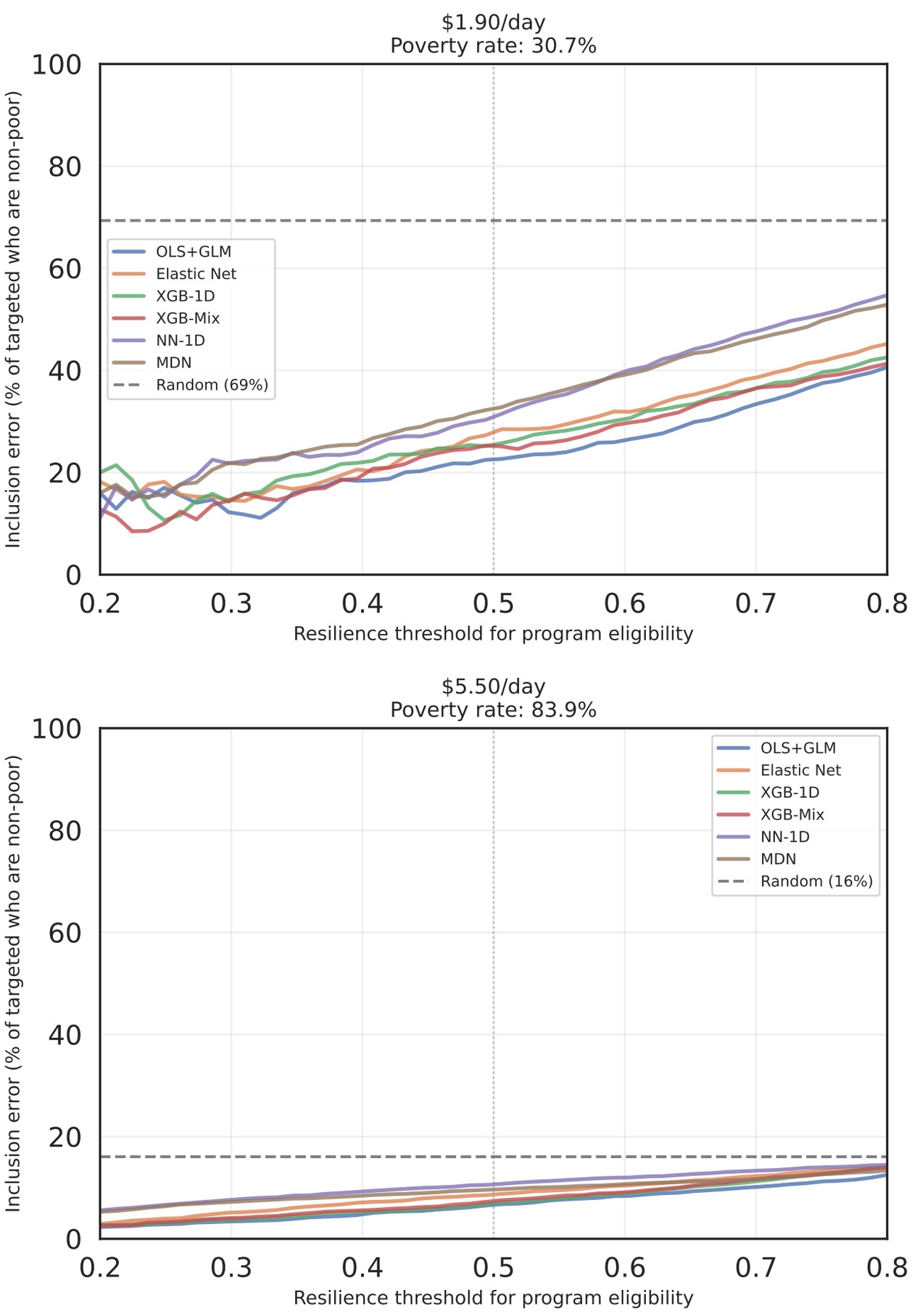

National line
Actual poverty rate: 46.5%
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Coverage gap (percentage points)
Resilience threshold for program eligibility

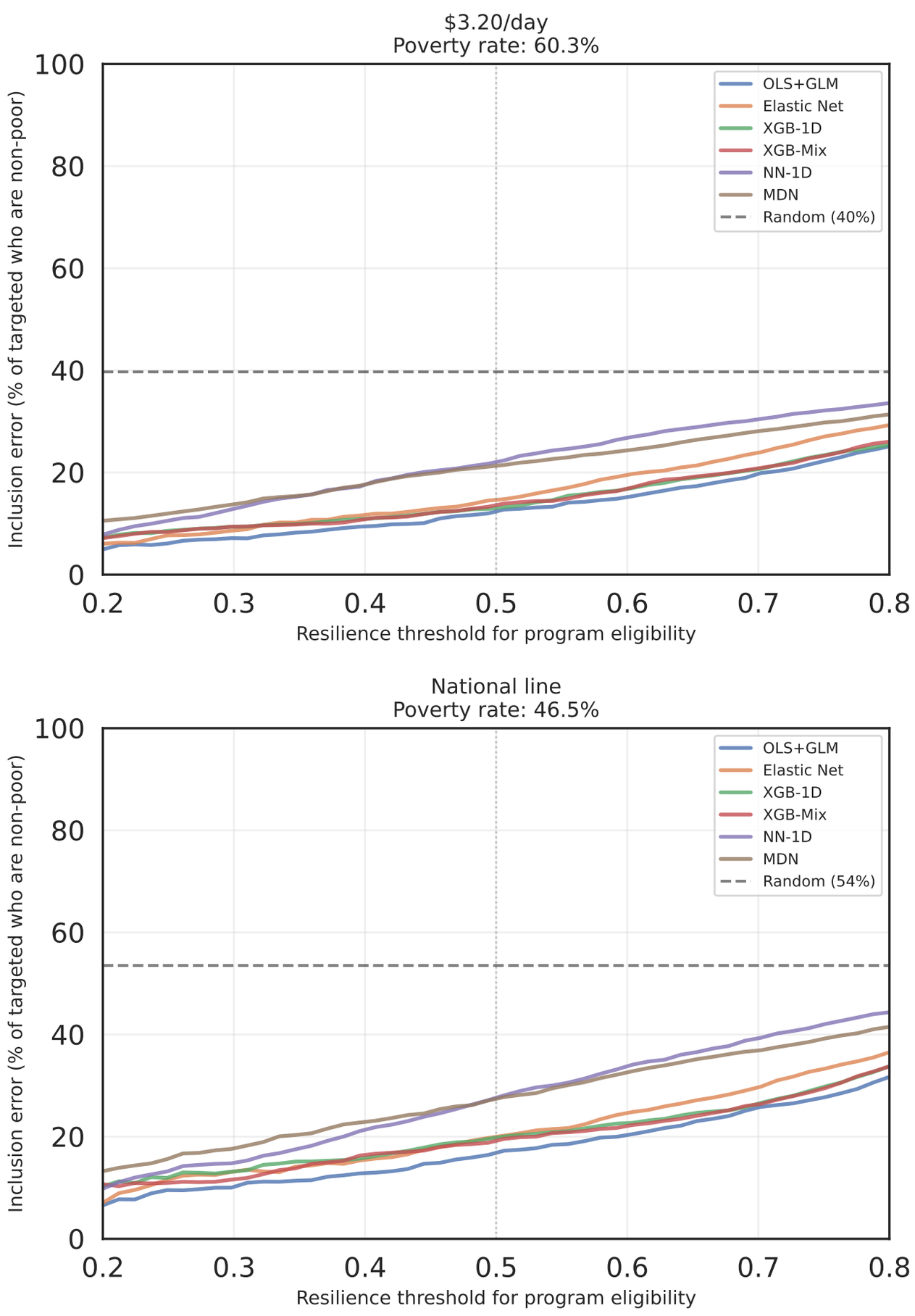

$1.90/day
Poverty rate: 30.7%
Exclusion error (% of poor excluded)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random targeting


$3.20/day
Poverty rate: 60.3%
Exclusion error (% of poor excluded)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random targeting


$5.50/day
Poverty rate: 83.9%
Exclusion error (% of poor excluded)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random targeting

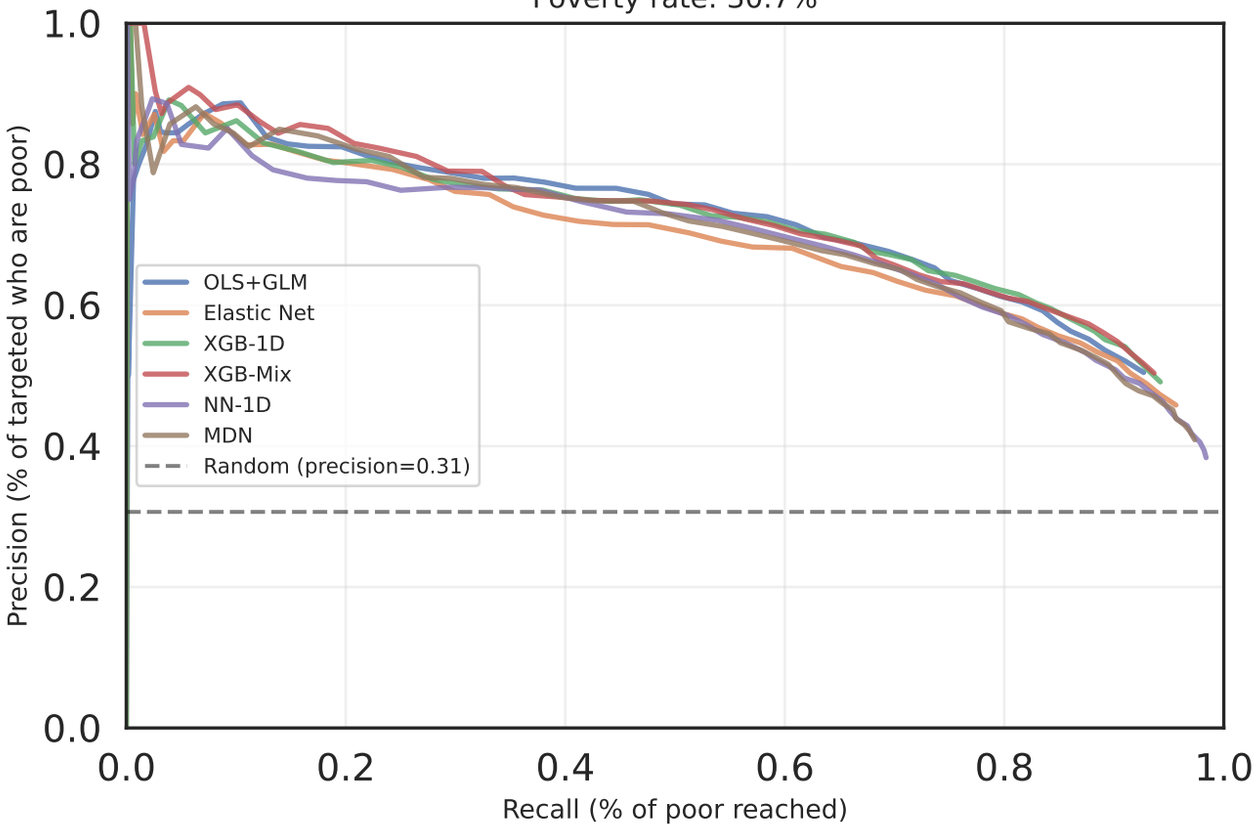

National line
Poverty rate: 46.5%
Exclusion error (% of poor excluded)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random targeting

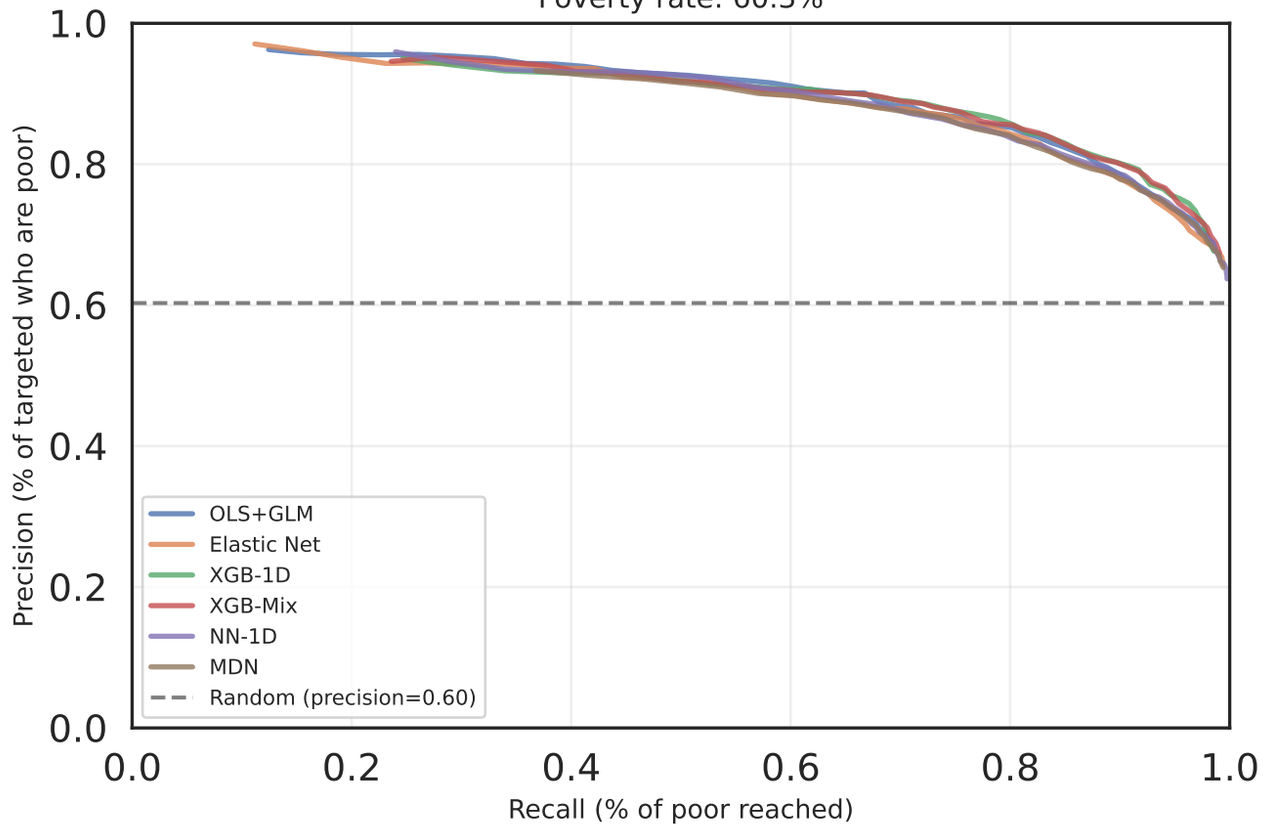

$1.90/day
Poverty rate: 30.7%
Inclusion error (% of targeted who are non-poor)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (69%)

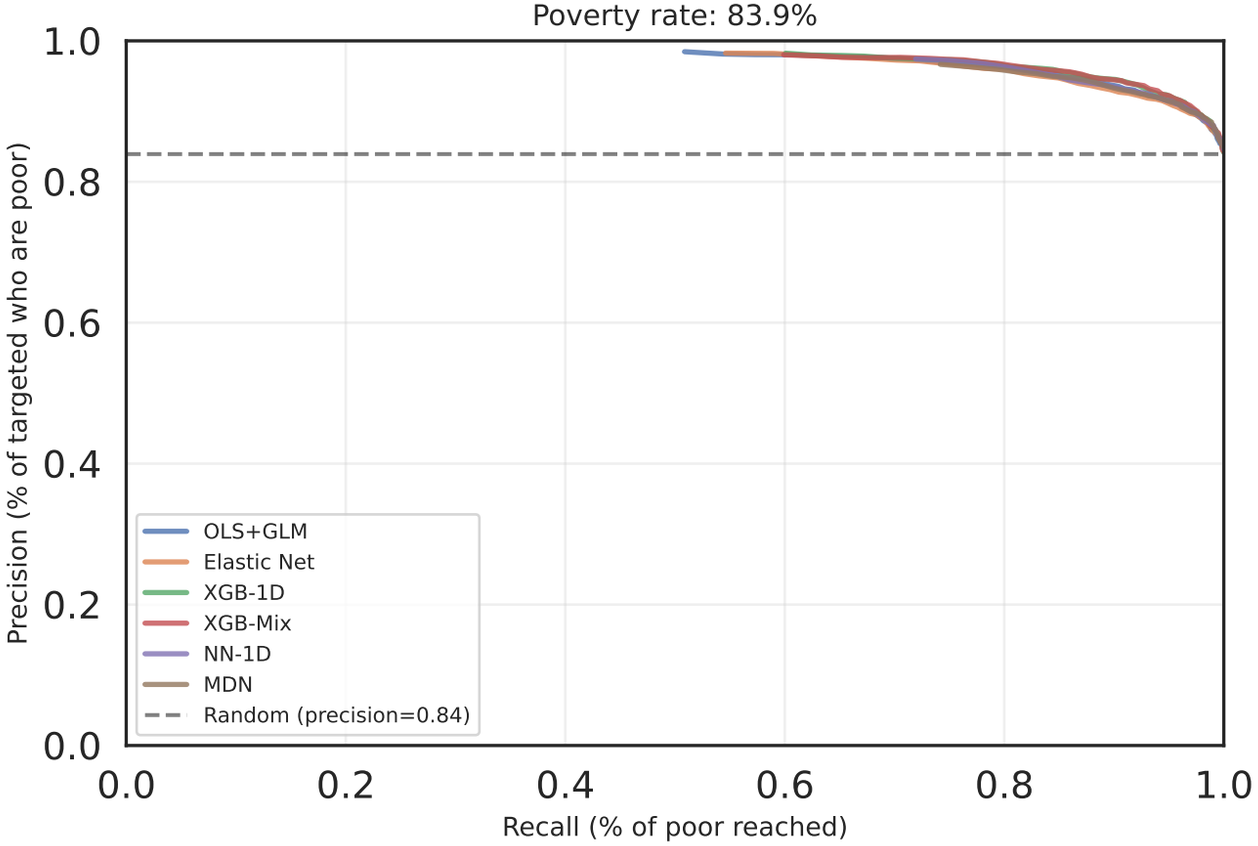

$3.20/day
Poverty rate: 60.3%
Inclusion error (% of targeted who are non-poor)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (40%)

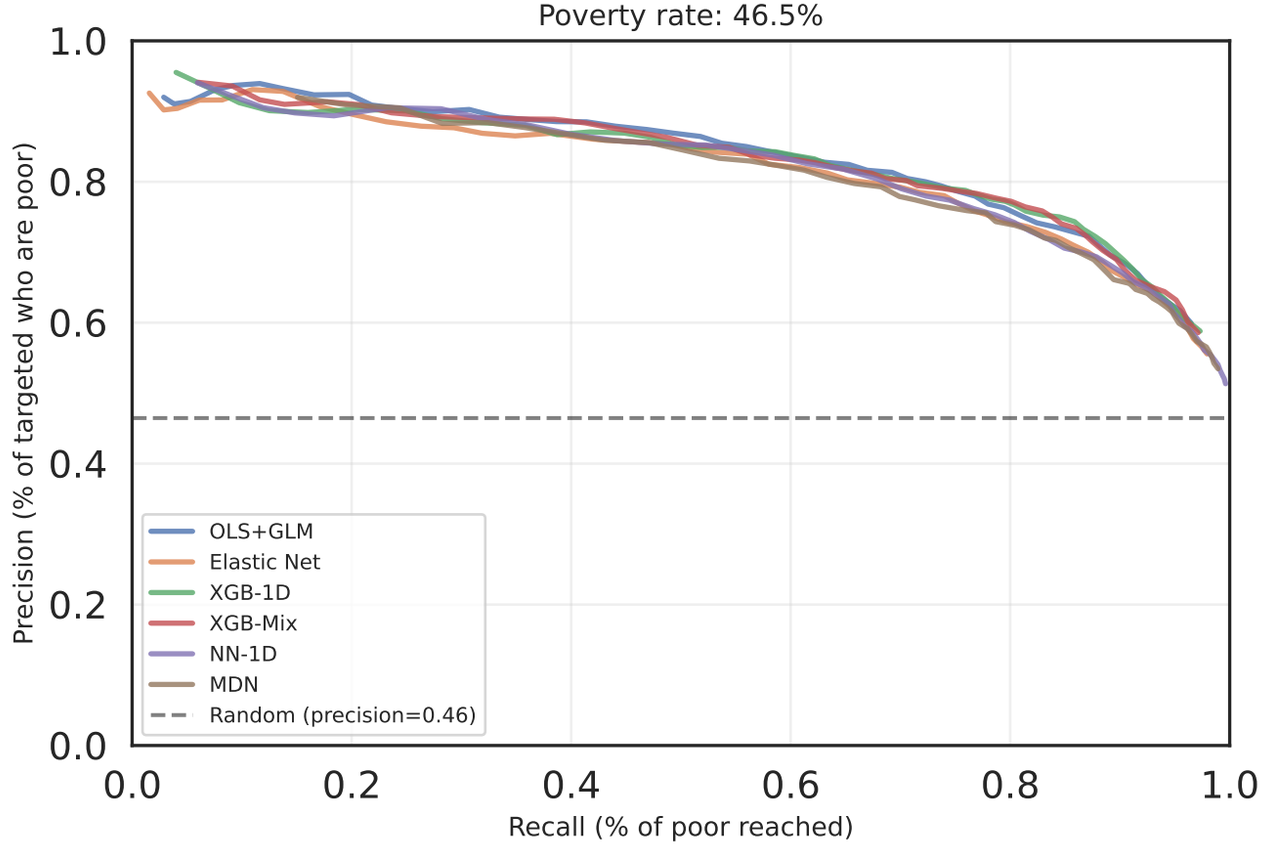
$5.50/day
Poverty rate: 83.9%
Inclusion error (% of targeted who are non-poor)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (16%)

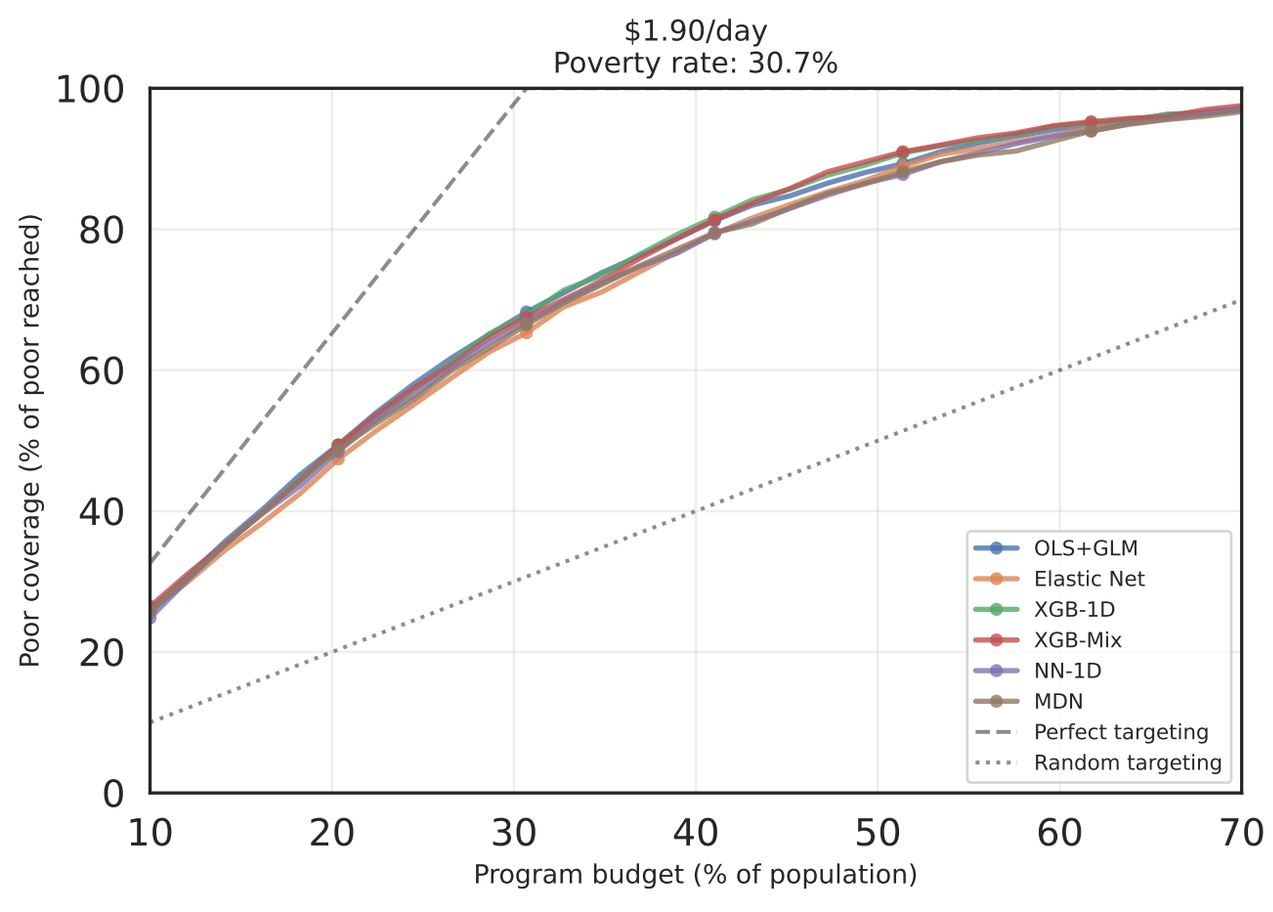
National line
Poverty rate: 46.5%
Inclusion error (% of targeted who are non-poor)
Resilience threshold for program eligibility
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (54%)

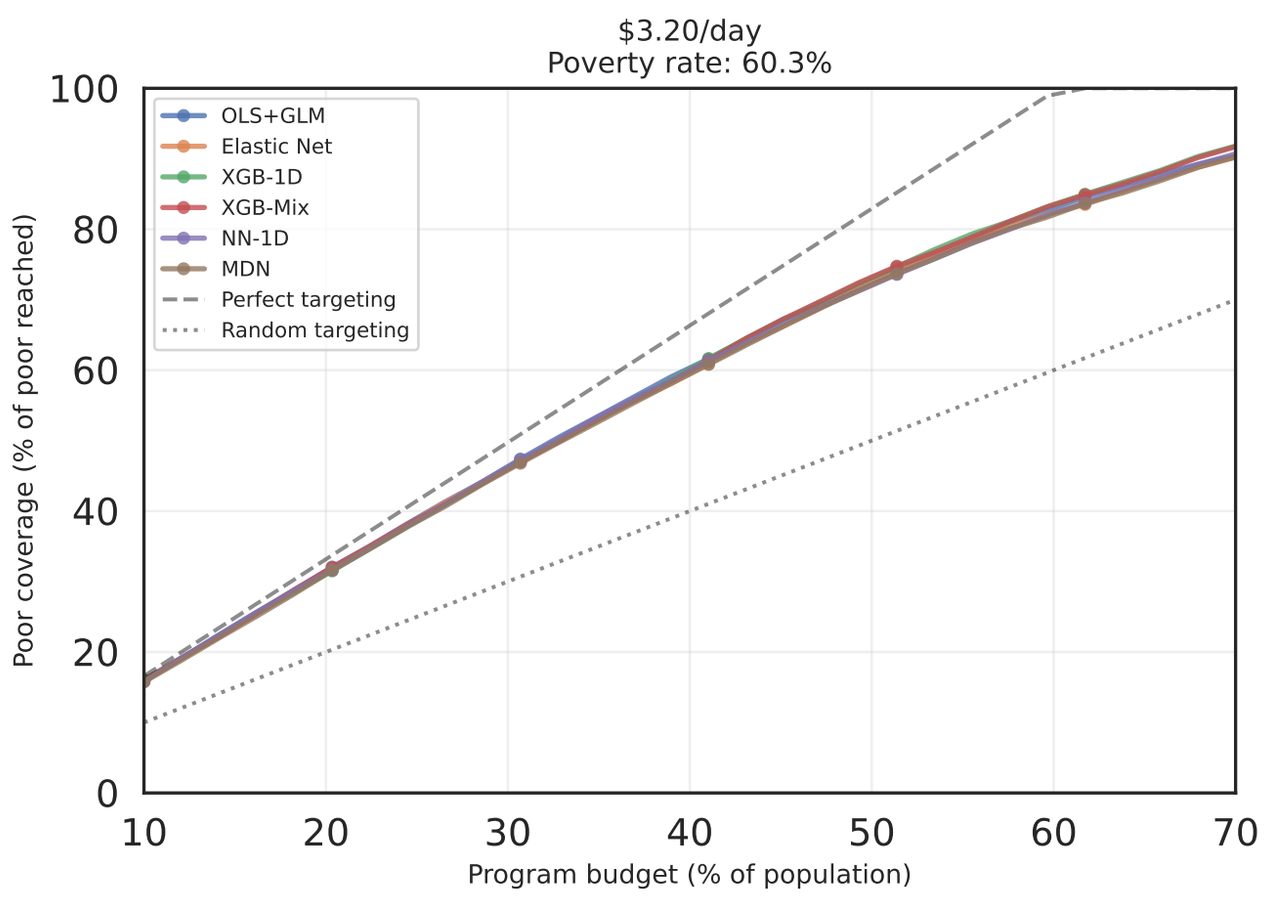

$1.90/day
Poverty rate: 30.7%
Precision (% of targeted who are poor)
Recall (% of poor reached)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (precision=0.31)

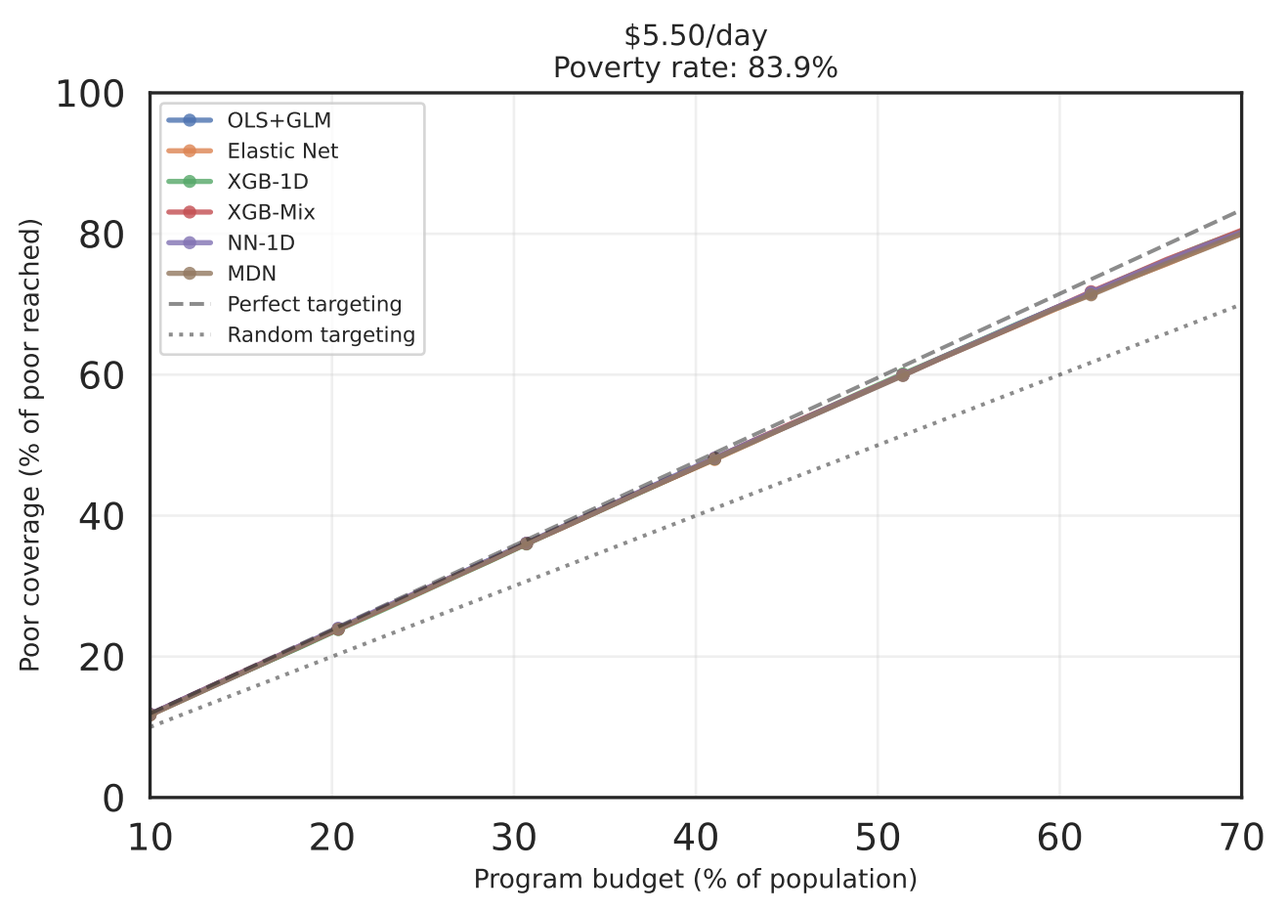

$3.20/day
Poverty rate: 60.3%
Precision (% of targeted who are poor)
Recall (% of poor reached)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (precision=0.60)

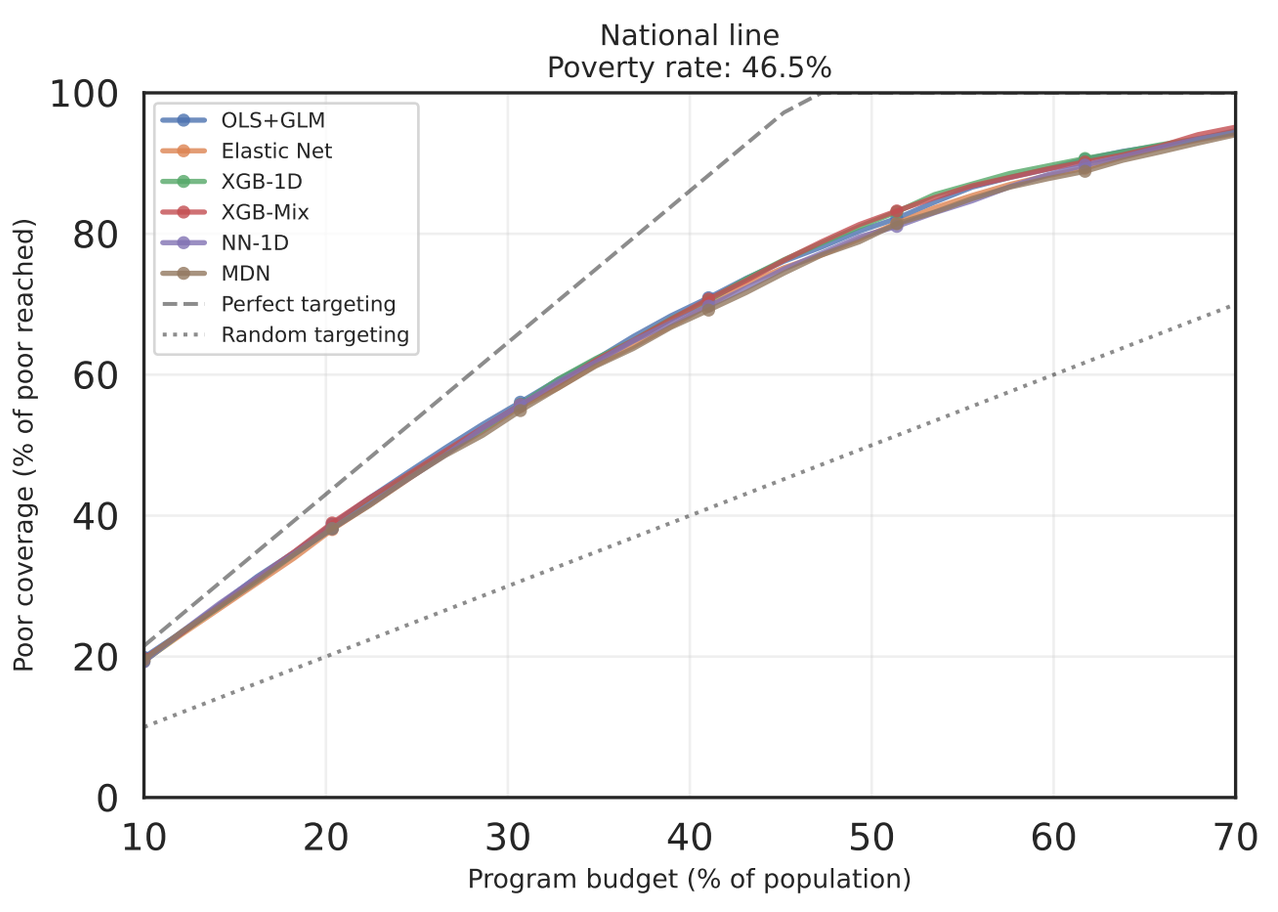

$5.50/day
Poverty rate: 83.9%
Precision (% of targeted who are poor)
Recall (% of poor reached)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (precision=0.84)

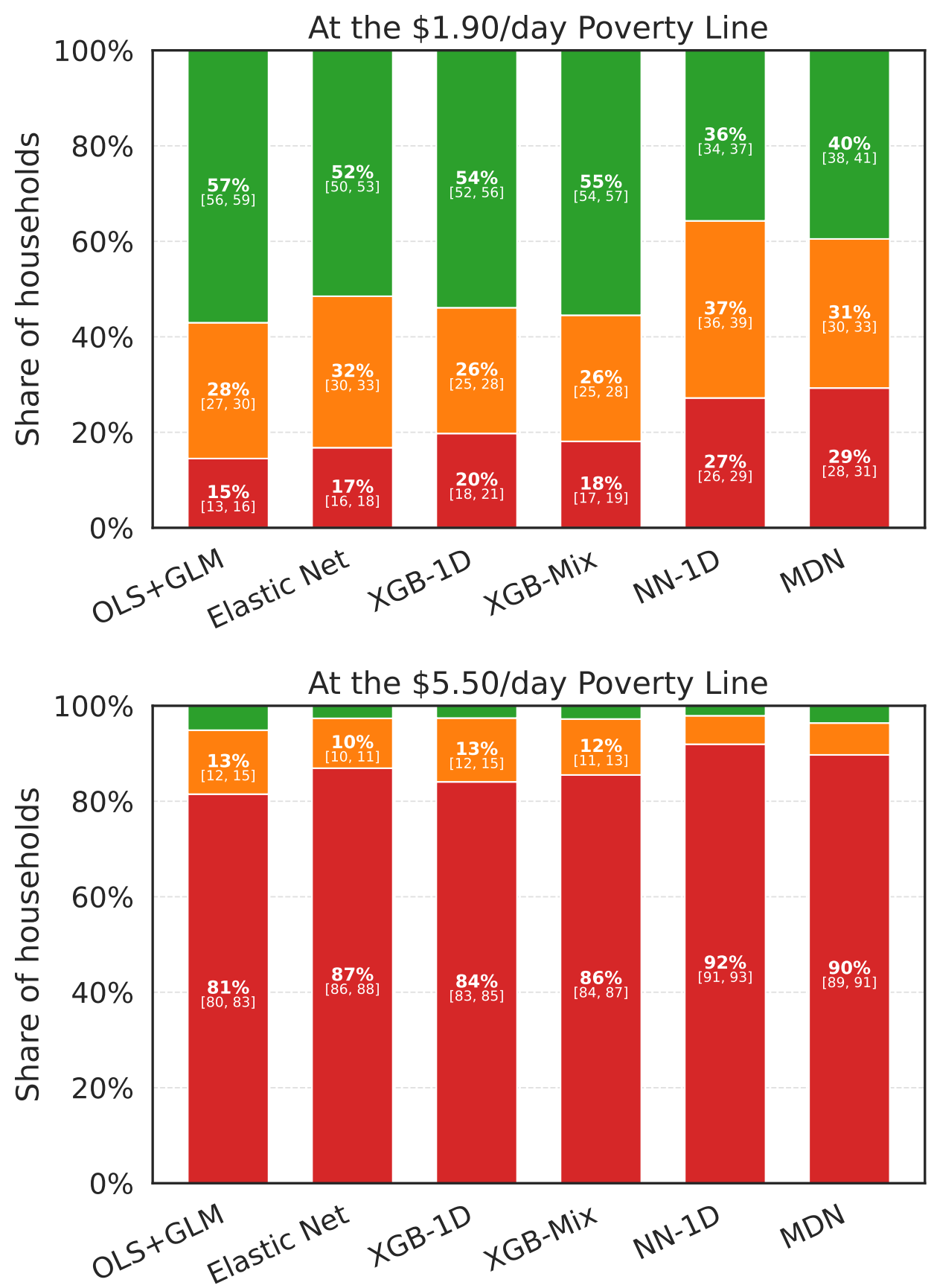

National line
Poverty rate: 46.5%
Precision (% of targeted who are poor)
Recall (% of poor reached)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Random (precision=0.46)

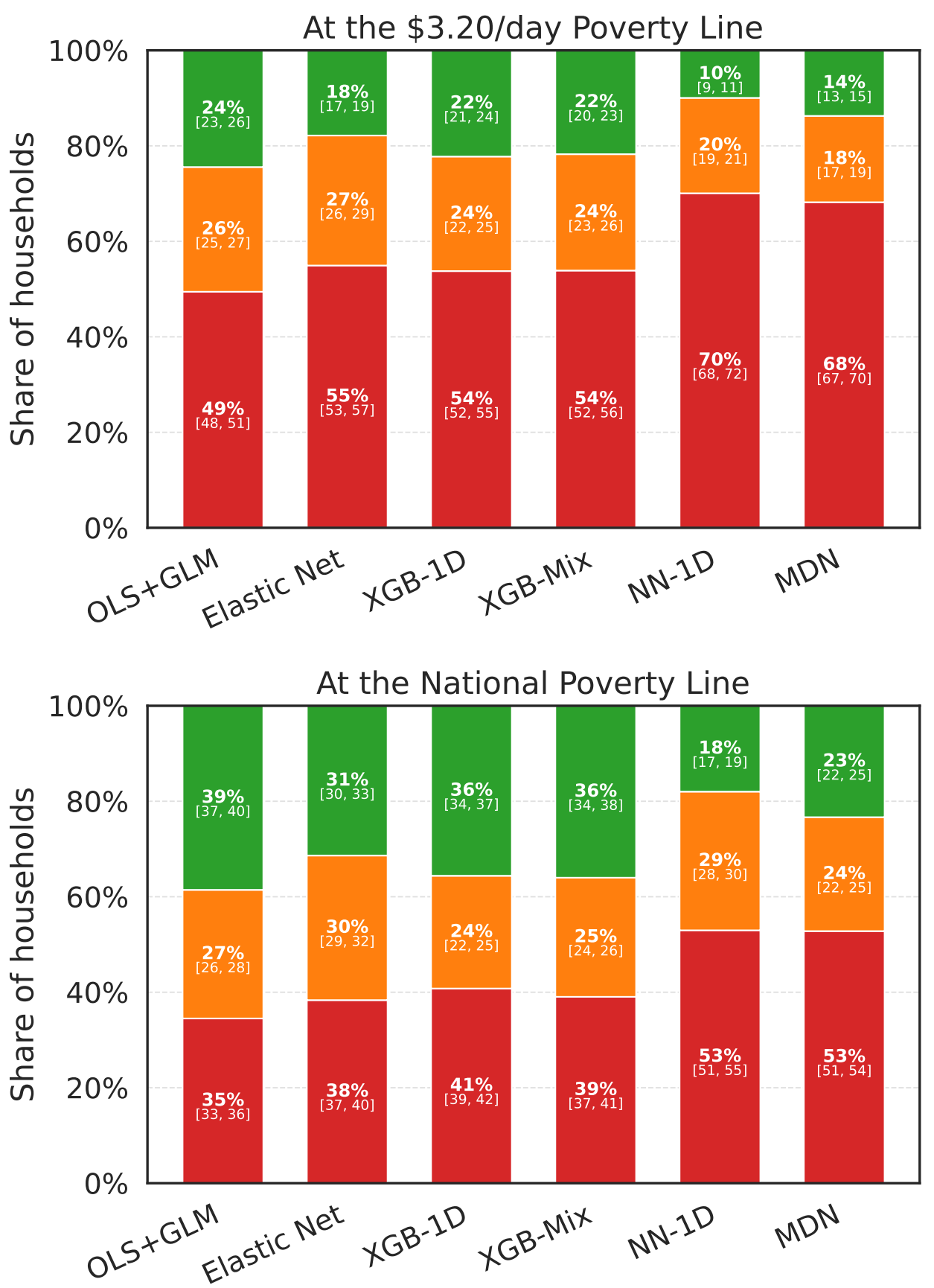

$1.90/day
Poverty rate: 30.7%
Poor coverage (% of poor reached)
Program budget (% of population)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Random targeting


$3.20/day
Poverty rate: 60.3%
Poor coverage (% of poor reached)
Program budget (% of population)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Random targeting


$5.50/day
Poverty rate: 83.9%
Poor coverage (% of poor reached)
Program budget (% of population)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Random targeting


National line
Poverty rate: 46.5%
Poor coverage (% of poor reached)
Program budget (% of population)
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN
Perfect targeting
Random targeting

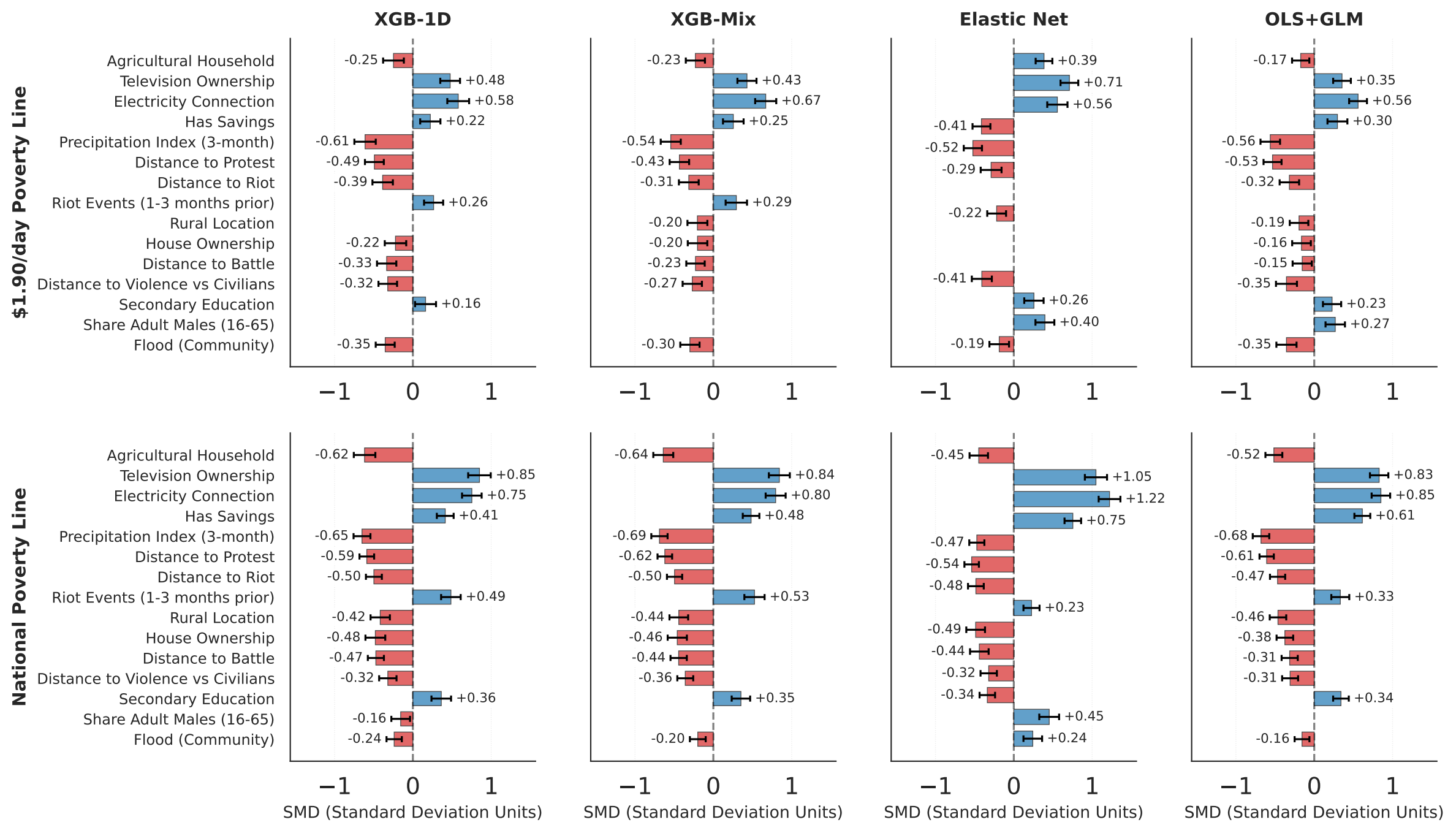

At the $1.90/day Poverty Line
Share of households
100%
80%
60%
40%
20%
0%
57% [56, 59]
28% [27, 30]
15% [13, 16]
52% [50, 53]
32% [30, 33]
17% [16, 18]
54% [52, 56]
26% [25, 28]
20% [18, 21]
55% [54, 57]
26% [25, 28]
18% [17, 19]
36% [34, 37]
37% [36, 39]
27% [26, 29]
40% [38, 41]
31% [30, 33]
29% [28, 31]
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN


At the $3.20/day Poverty Line
Share of households
100%
80%
60%
40%
20%
0%
24% [23, 26]
26% [25, 27]
49% [48, 51]
18% [17, 19]
27% [26, 29]
55% [53, 57]
22% [21, 24]
24% [22, 25]
54% [52, 55]
22% [20, 23]
24% [23, 26]
54% [52, 56]
10% [9, 11]
20% [19, 21]
70% [68, 72]
14% [13, 15]
18% [17, 19]
68% [67, 70]
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN

At the $5.50/day Poverty Line
Share of households
100%
80%
60%
40%
20%
0%
13% [12, 15]
81% [80, 83]
10% [10, 11]
87% [86, 88]
13% [12, 15]
84% [83, 85]
12% [11, 13]
86% [84, 87]
92% [91, 93]
90% [89, 91]
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN

At the National Poverty Line
Share of households
100%
80%
60%
40%
20%
0%
39% [37, 40]
27% [26, 28]
35% [33, 36]
31% [30, 33]
30% [29, 32]
38% [37, 40]
36% [34, 37]
24% [22, 25]
41% [39, 42]
36% [34, 38]
25% [24, 26]
39% [37, 41]
18% [17, 19]
29% [28, 30]
53% [51, 55]
23% [22, 25]
24% [22, 25]
53% [51, 54]
OLS+GLM
Elastic Net
XGB-1D
XGB-Mix
NN-1D
MDN

Non-resilient (<0.5)
Resilient (0.5-0.8)
Most-resilient (>=0.8)


XGB-1D
XGB-Mix
Elastic Net
OLS+GLM
$1.90/day Poverty Line
National Poverty Line
Agricultural Household
Television Ownership
Electricity Connection
Has Savings
Precipitation Index (3-month)
Distance to Protest
Distance to Riot
Riot Events (1-3 months prior)
Rural Location
House Ownership
Distance to Battle
Distance to Violence vs Civilians
Secondary Education
Share Adult Males (16-65)
Flood (Community)
−1
0
1
SMD (Standard Deviation Units)

$1.90/day
$3.20/day
$5.50/day
National
Effort Difference: Model - MDN
(Negative = Less Effort)
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
−0.12
−0.14
−0.16
To Resilient (rho=0.5)
To Most-resilient (rho=0.8)
-0.076
-0.091
-0.073
-0.103
-0.039
-0.016
-0.100
-0.110
-0.109
-0.142
-0.060
-0.096
-0.066
-0.102
-0.026
-0.008
-0.101
-0.116
-0.116
-0.154
-0.049
-0.060
-0.048
-0.056
-0.003
+0.004
-0.059
-0.061
-0.080
-0.097
-0.064
-0.098
-0.069
-0.108
-0.036
-0.015
-0.108
-0.125
-0.124
-0.160
XGB-1D
XGB-Mix
NN-1D
Elastic Net
OLS+GLM